AGR 101

FARMING BASED LIVELIHOOD SYSTEMS

Lecture-Wise Theory Study Material

B.Sc. (Hons) Agriculture - 2025 Syllabus

First Year - First Semester

Tamil Nadu Agricultural University

Complete coverage of all 34 lectures with definitions, comparison tables, worked examples, text diagrams and exam-focus question banks.

Compiled for examination preparation. Figures marked as standard ranges are indicative and should be verified against current official sources before quoting in an examination.

Contents

No.	Title	Page
1	1. Status of Agriculture in India, Different States and Tamil Nadu	3
2	2. Farmers' Income - Trends, Challenges and Policy Initiatives	7
3	3. Challenges and Opportunities for Enhancing Farmers' Income	10
4	4. Livelihood - Definition, Concept and Patterns	13
5	5. Livelihood Systems Indicators for Analysis	16
6	6. Agricultural Livelihood Systems - Definition and Components	19
7	7. Approaches to Agricultural Livelihood Systems	22
8	8. Challenges and Future Perspectives in Agricultural Livelihood Systems	25
9	9. Introduction to Farming Systems and Farming-Based Livelihood Systems	28
10	10. Types of Farming Systems, Characteristics and Farm Type Assessment	30
11	11. Prevalent Farming Systems in India and Their Role in Livelihoods	33
12	12. Component - Cropping Systems	36
13	13. Component - Dairy Farming and Small Livestock	39
14	14. Component - Horticulture	42
15	15. Component - Agroforestry	45
16	16. Component - Fish Farming and Integrated Models	48
17	17. Mid Semester Examination - Revision and Model Paper	51
18	18. Component - Small, Medium and Large Farm-Based Enterprises	54
19	19. Value Chains in Farming-Based Livelihood Systems	57
20	20. Secondary Enterprises and Their Role	59
21	21. Challenges and Future Prospects of Farming-Based Livelihood Systems	62
22	22. Factors Affecting Integration of Enterprises	64
23	23. Feasibility of Farming Systems Across Agro-Climatic Zones	67
24	24. Future Directions in Integrated Farming Systems	70
25	25. Overview of Commercial Farming-Based Livelihood Models in India	73
26	26. NABARD's Initiatives in Rural Livelihoods and Agriculture	76
27	27. Climate-Resilient Farming, NGOs and Government Programmes	79
28	28. Risks in Farming and Success Factors	82
29	29. Central and State Government Schemes for Farming-Based Livelihoods	85
30	30. Role of Public and Private Organizations	88
31	31. Circular Economy and Zero-Waste Farming Systems	90
32	32. Green Economy and Sustainable Agriculture	93
33	33. Digitalization and Its Impact on Farming-Based Livelihood Enterprises	96
34	34. Changing Lifestyles and Their Impact on Farming-Based Livelihood Enterprises	99
35	Glossary of Key Terms	102
36	Abbreviations and Acronyms	107
37	References and Further Reading	108
38	Appendix A - Quick Revision Tables	110
39	Appendix B - Model Question Papers	113
40	Appendix C - Practice Numericals with Solutions	115
41	Appendix D - Summary of Schemes and Institutions at a Glance	117
42	Appendix E - Quick Reference Tables for Revision	119
43	Appendix F - Case Studies, Exam Tips and Syllabus Map	121

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CHAPTER

1. Status of Agriculture in India, Different States and Tamil Nadu

1.1 Historical significance of Indian agriculture

Agriculture is the oldest and the largest livelihood activity in India. The story of Indian civilisation is built on farming: the Indus Valley settlements grew wheat and barley, the Vedic period developed plough-based cultivation and irrigation, and the Sangam age in the Tamil region was organised around the wetlands of the Cauvery delta.

- **Neolithic farming (about 7000-3000 BCE)** began in the north-west, in the Baluchistan and Indus regions, with wheat, barley and later rice.
- **Sangam period (about 300 BCE - 300 CE)** in present-day Tamil Nadu recognised five eco-zones called *thinai* - *kurinji* (hills), *mullai* (forest), *marutham* (wetland), *neithal* (coastal) and *paalai* (dry land). Each zone had its own crops and livelihood.
- **Grand Anicut (Kallanai)** was built across the Cauvery by the Chola king Karikala, one of the oldest functioning water-diversion structures in the world. It still irrigates parts of the Thanjavur delta.
- Agriculture supplied the revenue base of successive empires, supported village self-sufficiency, and created India's famous crop diversity.

1.2 Role of agriculture in India's economy

Agriculture remains the backbone of the rural economy even though its share in national output has fallen. The three roles are output (GDP), employment and foreign exchange.

Indicator	Approximate position	Trend
Share in Gross Value Added (GVA)	16-20 per cent	Slowly falling with industrial growth
Share in employment	42-48 per cent	Falling slowly, still the largest employer
Share in exports	10-13 per cent	Rising; agri-exports crossed USD 45 billion
Population dependent on agriculture	About 55-65 per cent of rural households	Declining but dominant
Foodgrain production	About 300-355 million tonnes	Record highs in recent years
Net sown area	About 140 million hectares	Nearly stable
Gross cropped area	About 200-220 million hectares	Slowly rising
Cropping intensity	About 140-145 per cent	Rising

The gap between the 16-20 per cent GDP share and the 45 per cent employment share is the central economic problem of Indian agriculture: too many people depend on too little value addition. Raising productivity per worker is the key to raising rural incomes.

1.3 Major crops and cropping patterns

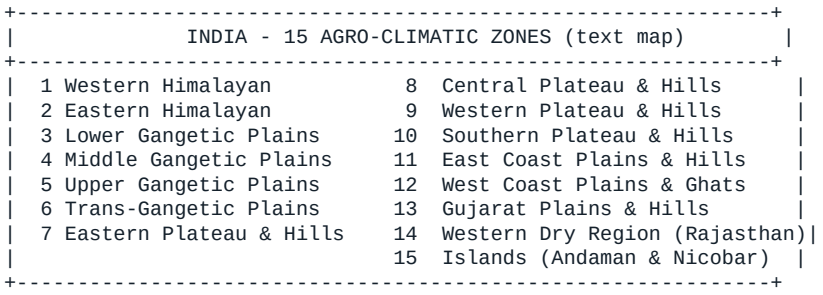
Indian crops are grouped by use.

- **Foodgrains:** cereals (rice, wheat, maize, sorghum/*jowar*, pearl millet/*bajra*, finger millet/*ragi*, small millets) and pulses (gram, pigeon pea/*tur*, black gram, green gram, lentil).
- **Oilseeds:** groundnut, mustard-rapeseed, soybean, sunflower, sesame, safflower, niger, castor.
- **Fibres:** cotton, jute, mesta.
- **Plantation and commercial:** sugarcane, tea, coffee, rubber, coconut, tobacco.
- **Horticulture:** fruits, vegetables, spices, flowers, medicinal and aromatic plants.

Tamil Nadu's principal crops are rice (the delta districts), sugarcane, cotton, groundnut, coconut, banana, mango, tapioca, pulses (black gram, green gram) and millets. The state is a leading producer of banana, coconut, tapioca, flowers and several spices.

1.4 Agro-climatic zones of India

The Planning Commission divided India into **15 agro-climatic zones** based on soil, climate, rainfall and cropping pattern. Tamil Nadu is further divided into **7 agro-climatic zones** by TNAU.



Tamil Nadu - 7 agro-climatic zones (TNAU):

1. North Eastern Zone
2. North Western Zone
3. Western Zone
4. Cauvery Delta Zone
5. Southern Zone
6. High Rainfall Zone
7. Hill and High Altitude Zone

1.5 Productivity comparison across states

Productivity (yield per hectare) varies widely because of irrigation, soil, technology and institutional support.

State	Strength	Notable crop performance
Punjab, Haryana	Irrigated wheat-rice	Highest wheat and rice yields in India
Uttar Pradesh	Largest producer	Top in sugarcane, wheat, foodgrains
West Bengal	Rice, potato	Highest rice output in several years
Andhra Pradesh, Telangana	Rice, cotton, chilli	High value crops, strong irrigation
Maharashtra	Cotton, sugarcane, soybean, horticulture	Largest in pulses, cotton, onion
Gujarat	Groundnut, cotton, milk	Leading groundnut and milk producer
Tamil Nadu	Rice, banana, coconut, flowers	High horticulture yields; delta rice
Karnataka	Ragi, coffee, silk	Largest ragi and coarse cereal state

Tamil Nadu's average rice yield in the Cauvery delta is among the highest in the country, helped by canal irrigation and the Grand Anicut system. However, average farm size in Tamil Nadu is small, so **income per farm**, not yield alone, is the binding constraint.

1.6 Challenges in Indian agriculture

- **Fragmentation:** about 86 per cent of holdings are small or marginal (below 2 hectares); average holding is roughly 1.0-1.1 hectares.
- **Monsoon dependence:** a large share of cropped area is still rainfed.
- **Low productivity per worker:** too many workers share a modest output.
- **Input cost inflation:** fertiliser, diesel, labour and seed costs rise faster than output prices.
- **Market access and price volatility:** weak storage, transport and price information.
- **Land and water degradation:** salinity, waterlogging, declining groundwater tables.
- **Climate risk:** droughts, floods, cyclones and heat stress, especially on the east coast and in Tamil Nadu.

1.7 Tamil Nadu - a snapshot

- Agriculture and allied sectors contribute roughly 10-13 per cent of the state's Gross State Domestic Product (GSDP).
- About 40-45 per cent of the state's workforce depends on agriculture and allied activities.
- The **Cauvery delta** (Thanjavur, Thiruvarur, Nagapattinam, Mayiladuthurai, parts of Trichy) is the rice bowl of Tamil Nadu.
- **Thanjavur** is famous for paddy, the Grand Anicut and the Samba and Thaladi rice seasons.
- The state has a strong **horticulture** base - banana, coconut, mango, flowers (jasmine in Madurai and Dindigul), and spices.
- Tamil Nadu leads in **millets revival** and in **integrated farming system** research led by TNAU.

Exam tip: The mismatch between agriculture's small GDP share and its large employment share is the single most repeated exam point. Learn to explain the *structural transformation* argument: as an economy develops, output shifts to industry and services while labour shifts out of agriculture more slowly.

Landholding structure and agrarian change

The structure of landholding explains why Indian agriculture, despite large national output, leaves many households poor.

- About 86 per cent of operational holdings are small or marginal (below 2 hectares).
- The average operational holding is roughly 1.0-1.1 hectares and continues to shrink with inheritance.
- Small and marginal holdings together operate a large share of the total cultivated area, so their welfare decides rural prosperity.
- Tenancy and sharecropping persist informally in many states, weakening the tenant's incentive to invest.
- Land fragmentation raises the cost of every operation - ploughing, irrigation, harvesting and transport.

Agrarian change in Tamil Nadu: the state has one of the smallest average holding sizes in India. Combined with high input costs, this makes collective action through FPOs and integrated farming the practical route to viability. Consolidation through voluntary pooling, group farming and custom-hiring of machinery helps smallholders act at a larger scale without losing ownership.

1.8 Exam focus

Key points

- Agriculture contributes about 16-20 per cent of India's GVA but employs 42-48 per cent of the workforce.
- India is divided into 15 agro-climatic zones; Tamil Nadu has 7 agro-climatic zones.
- About 86 per cent of Indian holdings are small or marginal, with an average size near 1 hectare.
- Tamil Nadu's Cauvery delta (Thanjavur region) is the state's rice bowl.
- Cropping intensity in India is about 140-145 per cent.
- The core problem is low income per worker, not low national output.
- Historical structures such as the Grand Anicut (Kallanai) still support delta irrigation.

Short-answer questions with model answers

Q1. State the importance of agriculture in India's economy. Model answer: Agriculture provides livelihood to about 45 per cent of the workforce, contributes roughly 16-20 per cent of national GVA, supplies raw material to industry, earns foreign exchange through agri-exports, and ensures national food security. It is also the largest source of rural employment and the base of the rural market for manufactured goods.

Q2. What is an agro-climatic zone? Model answer: An agro-climatic zone is a region with a distinct combination of climate, soil, rainfall, topography and cropping pattern. India is divided into 15 such zones to plan region-specific crop and technology recommendations.

Q3. Give two features of Tamil Nadu agriculture. Model answer: (i) The Cauvery delta is a major rice-growing tract with canal irrigation from the Grand Anicut. (ii) The state has a strong horticulture base, leading in banana, coconut, flowers and spices, and has small average farm size.

Long-answer questions with model answer outlines

Q1. Describe the status of agriculture in India with reference to GDP, employment and major crops. Outline: (i) Introduction - historical role; (ii) GDP and employment shares and the structural gap; (iii) major crop groups - foodgrains, oilseeds, fibres, plantation, horticulture; (iv) key production figures and cropping intensity; (v) regional concentration (Punjab, UP, Maharashtra, Tamil Nadu); (vi) challenges; (vii) conclusion - need for higher per-worker productivity.

Q2. Discuss the agro-climatic zones of India and Tamil Nadu and their importance in crop planning. Outline: (i) Definition and basis of zoning; (ii) the 15 Indian zones with brief features; (iii) the 7 Tamil Nadu zones; (iv) how zoning guides crop choice, technology transfer and risk management; (v) role of TNAU and KVKs; (vi) conclusion.

CHAPTER

2. Farmers' Income - Trends, Challenges and Policy Initiatives

2.1 Why farmers' income matters

National food security is achieved when the country grows enough food, but farm household welfare is achieved only when farmers **earn enough**. India's policy debate has therefore shifted from production alone to *income*. The Government of India constituted the **National Commission on Farmers** (Swaminathan Commission) and later the **Committee on Doubling Farmers' Income** (Dalwai Committee, 2016) to examine farm income.

2.2 Current trends in farmers' income

- Average income per farm household is low and highly unequal across regions.
- A large share of farm households have income below the level needed for a decent living.
- Income sources are increasingly diversified: crops, livestock, wages, remittances and non-farm enterprise.
- The **Dalwai Committee (2016-17)** estimated that a meaningful doubling of income needed an annual growth of about 10-12 per cent, far above the historical trend.
- Real income growth has been modest and uneven; irrigated and high-value crop regions do better than rainfed regions.

Note on figures: exact income levels change every survey round (NSSO Situation Assessment Survey, NABARD All India Rural Financial Inclusion Survey). In an examination, state the trend and the range rather than a single unverified number.

2.3 Challenges to farm income

- **Low profitability:** thin margins between cost of cultivation and price received.
- **High and rising input costs:** fertiliser, pesticide, seed, diesel, hired labour.
- **Small and fragmented holdings:** poor economies of scale.
- **Weak market access:** long supply chains, many intermediaries, poor price discovery.
- **Price volatility and distress sales:** lack of storage and credit forces immediate sale.
- **Monsoon and climate risk:** yield losses from drought, flood and pest attack.
- **Indebtedness:** dependence on informal credit at high interest.
- **Low value addition:** farmers sell raw produce rather than processed products.

2.4 Role of Minimum Support Price (MSP)

The MSP is the price at which the government promises to buy notified crops from farmers if market prices fall below it.

- **Recommended by** the Commission for Agricultural Costs and Prices (CACP).
- **Announced by** the Ministry of Agriculture and Farmers' Welfare before the sowing season.
- **Cost concepts:** the CACP computes A2 (paid-out cost), A2+FL (paid-out cost plus imputed family labour), and C2 (comprehensive cost including imputed rent and interest). The government has announced MSP at 1.5 times A2+FL for many crops.
- **Procurement:** effective mainly for wheat and paddy through the Food Corporation of India and state agencies; coverage is weaker for pulses, oilseeds and horticulture.
- **Limitations:** not all crops are procured, not all farmers can access procurement centres, and MSP does not directly help perishable horticulture crops.

2.5 Government schemes for income support

Scheme / Programme	Purpose
PM-KISAN	Direct income support of Rs. 6,000 per year in three instalments to eligible landholding farmers
Pradhan Mantri Fasal Bima Yojana (PMFBY)	Crop insurance against yield loss from natural calamities
Kisan Credit Card (KCC)	Institutional credit for crop and allied needs at concessional interest
e-NAM	Online national market platform for transparent price discovery
PM Kisan Maan Dhan Yojana	Old-age pension for small and marginal farmers
Soil Health Card	Soil-test based fertiliser recommendation to cut cost
PM Krishi Sinchayee Yojana	Irrigation - "Har Khet Ko Pani" and "More Crop Per Drop"
National Mission on Edible Oils	Reduce import dependence, support oilseed income
Rashtriya Krishi Vikas Yojana (RKVY)	State-level agri-development projects
Agriculture Infrastructure Fund	Post-harvest and community asset financing

2.6 Doubling Farmers' Income - the strategy

The Dalwai Committee's strategy had three pillars: **increase productivity**, **increase price realisation**, and **reduce cost**. Specific levers:

1. **Irrigation and water use efficiency** to raise and stabilise yields.
2. **Crop diversification** into high-value crops and horticulture.
3. **Livestock and fisheries integration** for round-the-year income.
4. **Post-harvest and value addition** to capture more of the consumer rupee.
5. **Market reform and e-NAM** for better prices.
6. **Risk management** through insurance and climate-resilient practices.
7. **Farmer Producer Organizations (FPOs)** for collective bargaining.

2.7 Worked example - a simple farm income calculation

A smallholder in the Cauvery delta cultivates 1 hectare and keeps a milch animal.

Enterprise	Gross income (Rs.)	Cost (Rs.)	Net income (Rs.)
Paddy (1 ha, 5 t at Rs. 2,000/quintal)	100,000	55,000	45,000
Dairy (1 cow, 8 litres/day, 250 days)	60,000	38,000	22,000
Backyard poultry (50 birds, 2 batches)	30,000	22,000	8,000
Total	**190,000**	**115,000**	**75,000**

Observation: the crop alone gives Rs. 45,000 net. Adding dairy and poultry raises net income to Rs. 75,000 - a rise of about 67 per cent - purely through **integration**, not through more land. This is the practical logic of farming-based livelihood systems.

Note on units: one quintal = 100 kg; one tonne = 10 quintals. A 5 tonne paddy yield is about 50 quintals per hectare, a realistic delta figure.

Measuring farm income and its sources

Understanding income requires knowing how it is measured and where it comes from.

Income = gross returns - total cost. Gross returns include the value of the main product and by-products. Total cost includes paid-out costs (seed, fertiliser, labour, irrigation) and imputed costs (family labour, rent, interest on own capital).

Sources of farm household income:

- Crop production (main and by-products).
- Livestock - milk, meat, eggs, manure.
- Wages from farm and non-farm work.
- Remittances from family members.
- Small business and handicrafts.
- Government transfers such as PM-KISAN.

Measurement methods: cost-of-cultivation surveys (CACP), the NSSO Situation Assessment Survey of farmers, and NABARD's rural financial inclusion surveys. Because income is uneven across seasons, an annual view is more reliable than a single-season figure.

2.8 Exam focus

Key points

- Farm policy has moved from production maximisation to income maximisation.
- MSP is recommended by CACP and announced by the Ministry of Agriculture.
- MSP is most effective for wheat and paddy procurement; weaker for other crops.
- PM-KISAN gives Rs. 6,000 per year income support to eligible farmers.
- The Dalwai Committee (2016) framed the doubling-of-income strategy.
- Low profitability, high input cost and weak market access are the main income constraints.
- Integration and value addition raise income without extra land.

Short-answer questions with model answers

Q1. What is MSP and who recommends it? Model answer: MSP (Minimum Support Price) is the assured price at which the government buys notified crops if market prices fall below it. It is recommended by the Commission for Agricultural Costs and Prices (CACP) and announced by the Ministry of Agriculture and Farmers' Welfare.

Q2. Name three constraints to raising farm income. Model answer: (i) Low profitability due to thin margins; (ii) high input costs; (iii) weak market access and price volatility. (Other acceptable answers: fragmentation, indebtedness, climate risk.)

Q3. What is PM-KISAN? Model answer: PM-KISAN is a central scheme giving direct income support of Rs. 6,000 per year in three equal instalments to eligible landholding farmer families, transferred to their bank accounts.

Long-answer questions with model answer outlines

Q1. Explain the challenges faced in raising farmers' income and the policy measures to address them. Outline: (i) Introduction - why income matters; (ii) challenges - profitability, input cost, fragmentation, market access, risk, debt; (iii) MSP and its role and limits; (iv) schemes - PM-KISAN, PMFBY, KCC, e-NAM, Soil Health Card; (v) Dalwai Committee strategy; (vi) conclusion.

Q2. Discuss the strategy for doubling farmers' income. Outline: (i) Background and target; (ii) three pillars - productivity, price realisation, cost reduction; (iii) levers - irrigation, diversification, livestock, value addition, markets, risk management, FPOs; (iv) role of states and Tamil Nadu initiatives; (v) challenges in implementation; (vi) conclusion.

CHAPTER

3. Challenges and Opportunities for Enhancing Farmers' Income

3.1 Framing the problem

Farm income can be raised only by increasing **gross returns**, decreasing **cost**, or both. The challenges below depress returns or raise cost; the opportunities below lift returns or cut cost. This two-sided view is the correct analytical frame.

3.2 Challenges

- **Climate change:** erratic monsoon, higher temperatures, heat stress at flowering, new pest and disease pressure, cyclones on the east coast and in Tamil Nadu.
- **Rising input costs:** fertiliser, pesticide, fuel and labour; wage rates rise faster than produce prices in many regions.
- **Fragmented landholdings:** small plots prevent mechanisation and block economies of scale; boundary disputes and poor water sharing follow.
- **Market access problems:** distant markets, weak roads and cold chains, many intermediaries, price information asymmetry.
- **Water stress:** declining groundwater, competition for canal water, salinity and waterlogging in delta and coastal tracts.
- **Post-harvest losses:** roughly 5-15 per cent in cereals and higher in fruits and vegetables because of poor storage and handling.
- **Credit and risk:** informal borrowing at high interest and inadequate insurance coverage.
- **Knowledge and extension gaps:** slow transfer of improved technology to small farms.

3.3 Opportunities

- **Crop diversification:** shifting from low-value cereals to high-value pulses, oilseeds, fruits, vegetables and spices.
- **Organic and natural farming:** premium prices, lower input cost, healthier soils; Tamil Nadu promotes organic certification and natural farming.
- **Digital agriculture:** mobile advisories, weather-based insurance, e-market platforms and precision tools.
- **Agro-processing:** converting raw produce into flour, oil, pulp, juice or packaged goods to capture more value.
- **Livestock and fisheries:** steady daily income and quick returns.
- **Integrated farming systems:** multiple enterprises on the same land for income stability.
- **Value chains and FPOs:** collective aggregation, grading, branding and direct marketing.
- **Agri-tourism and eco-tourism:** additional income from farm visits, farm stays and experience packages.

3.4 Comparison table - challenge and matching opportunity

Challenge	Opportunity / remedy	Example
Low crop price	Value addition, direct marketing	Banana pulp unit, farm-gate sale
High input cost	Organic inputs, soil health management	Vermicompost, green manure
Fragmentation	FPO collective farming	FPO machinery bank in Thanjavur
Market distance	e-NAM, e-commerce, cold chain	Online sale of mango and flowers
Climate risk	Climate-resilient varieties, NICRA practices	Drought-tolerant paddy, millet
Water stress	Micro-irrigation, IFS water budgeting	Drip with fertigation
Post-harvest loss	Storage, grading, packaging	Onion storage, cool chain

3.5 Worked example - value from diversification

A farmer with 1 hectare currently grows paddy with a net income of Rs. 45,000. Consider diversifying 0.4 hectare to vegetables and 0.2 hectare to a fruit orchard while retaining 0.4 hectare under paddy.

Land use	Area (ha)	Net income/ha (Rs.)	Net income (Rs.)
Paddy	0.4	45,000	18,000
Vegetables	0.4	90,000	36,000
Fruit (mature)	0.2	1,20,000	24,000
Total	**1.0**	-	**78,000**

Diversification raises net income from Rs. 45,000 to Rs. 78,000 - about 73 per cent higher - while reducing the risk of a single-crop failure.

Institutional support for income enhancement

Income grows faster when the household is linked to the right institutions.

- **Credit institutions:** Kisan Credit Card, cooperative banks and SHGs supply working capital at low cost.
- **Market institutions:** regulated markets, e-NAM and FPOs improve price discovery.
- **Insurance institutions:** PMFBY and livestock insurance absorb shocks.
- **Knowledge institutions:** KVKs, TNAU and the Uzhavan app supply technology and advice.
- **Collective institutions:** FPOs and SHGs give smallholders bargaining power.

The lesson is that technology alone does not raise income; it must be combined with credit, insurance and market linkage. A farmer with an improved variety but no credit, no insurance and no buyer remains vulnerable. Tamil Nadu's combination of KVK demonstrations, Uzhavan advisories and FPO marketing illustrates how these institutions work together.

3.6 Exam focus

Key points

- Income rises by increasing returns or decreasing cost.
- Climate change, input cost, fragmentation and market access are the main challenges.
- Diversification, organic farming, digital agriculture and agro-processing are the main opportunities.
- Diversification both raises income and spreads risk.
- Post-harvest losses of 5-15 per cent in cereals are a major avoidable income loss.
- FPOs and value chains help smallholders capture more of the consumer rupee.

Short-answer questions with model answers

Q1. How does crop diversification help farmers? Model answer: Diversification shifts land from low-value to high-value crops such as vegetables, fruits and spices, raising net income per hectare and reducing the risk of total loss from a single crop or single market.

Q2. List any three opportunities for enhancing farm income. Model answer: (i) Organic and natural farming; (ii) digital agriculture and e-marketing; (iii) agro-processing and value addition.

Q3. Why are fragmented holdings a challenge? Model answer: Small plots prevent the use of machinery, raise the per-unit cost of inputs and operations, and make it difficult to achieve economies of scale in production and marketing.

Long-answer questions with model answer outlines

Q1. Discuss the challenges and opportunities for enhancing farmers' income in India. Outline: (i) Introduction - income frame; (ii) challenges - climate, cost, fragmentation, market, water, post-harvest, credit; (iii) opportunities - diversification, organic, digital, processing, livestock, IFS, value chains, agri-tourism; (iv) a pairing table of challenge and remedy; (v) conclusion.

Q2. Explain how diversification and value addition can raise farm income. Outline: (i) Concept of diversification; (ii) high-value crops and examples; (iii) value addition - processing, grading, branding; (iv) worked numeric example; (v) role of FPOs and e-commerce; (vi) conclusion.

CHAPTER

4. Livelihood - Definition, Concept and Patterns

4.1 Definition of livelihood

A **livelihood** is the way a person or household makes a living. It comprises the **capabilities, assets (resources) and activities** required for a means of living. A livelihood is **sustainable** when it can cope with and recover from stresses and shocks and maintain or enhance its capabilities and assets, both now and in the future, without undermining the natural resource base.

The widely used definition comes from Chambers and Conway (1992) and is operationalised by the UK Department for International Development (DFID) Sustainable Livelihoods Framework.

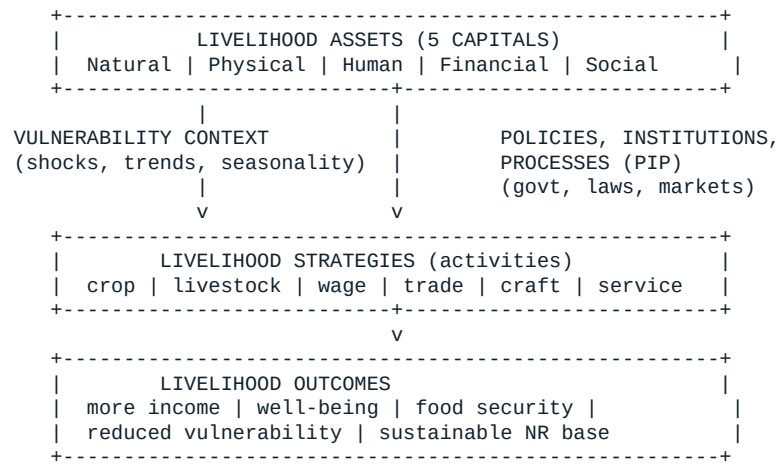
4.2 The three building blocks

- **Assets / resources:** the capital a household can draw on. These are the five capitals - natural, physical, human, financial and social.
- **Activities:** what people actually do - crop farming, livestock rearing, wage labour, trade, service, craft.
- **Capabilities:** skills, knowledge, health and the ability to convert assets into a living.

4.3 The five capitals

Capital	Meaning	Rural examples	Urban examples
Natural	Natural resource base	Land, water, forest, soil, climate	Access to water, green space
Physical	Produced assets	Irrigation, tools, tractor, house, roads	Transport, machinery, housing
Human	Skills, health, education	Farm knowledge, health, literacy	Education, job skills
Financial	Money and credit	Savings, crop loan, insurance, remittances	Salary, bank credit, savings
Social	Networks and institutions	SHGs, FPOs, caste/kinship networks, trust	Unions, associations, contacts

4.4 Sustainable Livelihood Framework (text diagram)



The framework shows that a household's assets, shaped by its vulnerability context and the policy environment, determine the strategies it can follow, which in turn produce outcomes.

4.5 Rural livelihood patterns

- **Farm-based:** crop cultivation, livestock, fisheries, agroforestry.

- **Farm plus non-farm:** part-time farming with wage labour, trade or craft.
- **Non-farm:** artisanship, services, transport, small business.
- **Seasonal migration:** moving to cities or other regions for work between crop seasons.
- **Common property dependence:** fuelwood, fodder, grazing and minor forest produce.

4.6 Urban livelihood patterns

- **Formal employment:** salaried jobs in industry, offices and services.
- **Informal employment:** street vending, construction labour, domestic work, gig work.
- **Self-employment and micro-enterprise:** shops, food stalls, repair services.
- **Home-based work:** tailoring, packing, food preparation.

4.7 Rural versus urban livelihood comparison

Feature	Rural livelihood	Urban livelihood
Main asset	Land and natural resources	Skills, education, networks
Main activity	Crop, livestock, fisheries	Industry, trade, services
Income pattern	Seasonal, harvest-linked	Regular, monthly or daily wage
Risk type	Monsoon, pest, price	Job loss, cost of living, market
Social capital	Strong kinship and village ties	Weaker, association-based
Dependence on nature	High	Low
Diversification	Multiple enterprises, migration	Multiple jobs, multiple income

4.8 Worked example - livelihood asset scoring

Score each capital from 1 (very weak) to 5 (very strong) for two households.

Capital	Farm household	Landless labour household
Natural	4	1
Physical	3	2
Human	3	3
Financial	2	2
Social	4	4
Total (max 25)	**16**	**12**

Interpretation: the farm household scores higher on natural and physical capital; both are equal on human and social capital. The landless household must strengthen financial and physical capital - through credit, skills and self-employment - to improve its livelihood outcome. This scoring method is the basis of livelihood indicator analysis.

Livelihood diversification and migration

Households rarely depend on a single activity. Diversification is the norm, not the exception.

- **On-farm diversification:** several crops and enterprises on the same holding.
- **Occupational diversification:** combining farming with trade, service or craft.
- **Seasonal migration:** moving to cities or other regions between crop seasons for wage work.
- **Permanent migration:** younger members leaving farming altogether.

Causes: small holdings, seasonal unemployment, low farm income, drought and lack of local jobs.

Consequences: remittances support the household, but migration also removes young labour from farming and can increase the workload on women and the elderly.

Policy response: creating local non-farm and agri-enterprise employment - processing units, dairy, poultry, agri-tourism - reduces forced migration and strengthens the rural livelihood base.

4.9 Exam focus

Key points

- Livelihood = capabilities + assets + activities for a means of living.
- A livelihood is sustainable when it withstands shocks without destroying the resource base.
- The five capitals are natural, physical, human, financial and social.
- The Sustainable Livelihood Framework links assets, vulnerability, institutions, strategies and outcomes.
- Rural livelihoods are land and nature based and seasonal; urban livelihoods are skill and wage based.
- Livelihood asset scoring is a practical method of analysis.

Short-answer questions with model answers

Q1. Define livelihood. Model answer: A livelihood comprises the capabilities, assets and activities required for a means of living. It is sustainable when it can cope with stresses and shocks and maintain or enhance its assets without undermining the natural resource base.

Q2. Name the five capitals of the sustainable livelihood framework. Model answer: Natural, physical, human, financial and social capital.

Q3. State two differences between rural and urban livelihoods. Model answer: (i) Rural livelihoods depend mainly on land and natural resources while urban livelihoods depend on skills and employment; (ii) rural income is seasonal while urban income is regular (monthly or daily wage).

Long-answer questions with model answer outlines

Q1. Explain the concept of livelihood and the sustainable livelihood framework. Outline: (i) Definition; (ii) the five capitals; (iii) vulnerability context and policies-institutions-processes; (iv) livelihood strategies; (v) livelihood outcomes; (vi) diagram; (vii) conclusion.

Q2. Compare rural and urban livelihood patterns with examples. Outline: (i) Meaning of livelihood patterns; (ii) rural patterns with examples; (iii) urban patterns with examples; (iv) comparison table; (v) implications for policy; (vi) conclusion.

CHAPTER

5. Livelihood Systems Indicators for Analysis

5.1 What is a livelihood indicator?

A **livelihood indicator** is a measurable variable that tells us how well a household or a community is making a living and whether that living is sustainable. Indicators convert complex realities into numbers or scores so that different households, villages or districts can be compared and monitored over time.

Good indicators are **specific, measurable, achievable, relevant and time-bound (SMART)**. A poor indicator is vague ("good farming"); a good indicator is precise ("net income per hectare per year in rupees").

5.2 Why indicators are needed

- To **diagnose** the weak points in a livelihood system before designing an intervention.
- To **compare** households, villages, districts and states on a common scale.
- To **monitor** change over time after a scheme or technology is introduced.
- To **evaluate** the impact of a project on income, employment and environment.
- To **target** support to the most vulnerable households.

5.3 The three pillars of livelihood indicators

Livelihood indicators are grouped into economic, social and environmental dimensions - the three pillars of sustainability.

Economic indicators

- **Income:** net farm income, total household income, income per capita.
- **Employment:** person-days of employment generated, share of family labour, off-farm employment.
- **Productivity:** yield per hectare, output per worker, milk yield per animal, water productivity.
- **Profitability:** benefit-cost ratio, net present value, gross margin.
- **Asset ownership:** land, livestock, machinery, savings.

Social indicators

- **Education:** literacy rate, school enrolment, years of schooling.
- **Health:** access to health care, nutrition status, infant mortality.
- **Gender equity:** women's participation in decision-making, women's asset ownership, wage parity.
- **Food security:** months of self-sufficiency, dietary diversity.
- **Social inclusion:** access of marginal groups to schemes and resources.
- **Institutional strength:** membership in SHGs, FPOs and cooperatives.

Environmental indicators

- **Resource sustainability:** soil organic carbon, groundwater level, soil erosion rate.
- **Biodiversity:** number of crops and varieties grown, tree cover.
- **Ecological balance:** use of renewable inputs, nutrient recycling.
- **Emission and waste:** residue burning, waste recycling, chemical load.
- **Water use efficiency:** litres of water per kilogram of produce.

5.4 Comprehensive indicator table

Dimension	Indicator	How measured
Economic	Net farm income	Total returns minus total cost per year
Economic	Benefit-cost ratio	Gross return divided by total cost
Economic	Employment generation	Person-days per hectare per season

Dimension	Indicator	How measured
Economic	Income diversification index	Share of non-crop income in total income
Social	Literacy rate	Percentage of adults who can read and write
Social	Women's participation	Percentage of decisions taken jointly
Social	Food self-sufficiency	Months of grain requirement met from own farm
Social	Institutional membership	Percentage of households in SHG or FPO
Environmental	Soil organic carbon	Percentage carbon in soil sample
Environmental	Groundwater trend	Change in water level over years
Environmental	Cropping diversity	Number of crops grown per year
Environmental	Recycling of residues	Percentage of residue composted, not burnt

5.5 Methods of assessment

- **Household survey:** structured questionnaire to collect income, assets, employment and consumption data.
- **Participatory Rural Appraisal (PRA):** villagers map resources, rank problems and score options in a group.
- **Focus group discussion:** a small group discusses a specific theme such as women's work or water use.
- **Transect walk:** a walk across the village to observe land use, crops and resources.
- **Seasonal calendar:** a chart showing labour, income and food availability month by month.
- **Scoring and ranking:** giving scores or ranks to capitals, enterprises or constraints.
- **Secondary data:** government statistics, census, NSSO and NABARD surveys.
- **Baseline and endline comparison:** measuring indicators before and after an intervention.

5.6 Worked example - livelihood indicator scoring

A village has three households. Each indicator is scored 1 (poor) to 5 (excellent). Scores are weighted by importance.

Indicator (weight)	HH A	HH B	HH C	Best possible	Weight
Net income (0.30)	4	2	3	5	0.30
Employment (0.15)	3	2	4	5	0.15
Education (0.15)	4	3	2	5	0.15
Health (0.10)	4	3	3	5	0.10
Gender equity (0.10)	3	4	2	5	0.10
Soil and water (0.20)	4	2	4	5	0.20

Weighted score = sum of (score x weight).

- HH A = (4x0.30) + (3x0.15) + (4x0.15) + (4x0.10) + (3x0.10) + (4x0.20)

= 1.20 + 0.45 + 0.60 + 0.40 + 0.30 + 0.80 = **3.75**

- HH B = (2x0.30) + (2x0.15) + (3x0.15) + (3x0.10) + (4x0.10) + (2x0.20)

= 0.60 + 0.30 + 0.45 + 0.30 + 0.40 + 0.40 = **2.45**

- HH C = (3x0.30) + (4x0.15) + (2x0.15) + (3x0.10) + (2x0.10) + (4x0.20)

= 0.90 + 0.60 + 0.30 + 0.30 + 0.20 + 0.80 = **3.10**

Interpretation: HH A has the strongest overall livelihood (3.75 out of 5), driven by income and soil-water management. HH B is the weakest (2.45), mainly on income and environment, and needs priority support. HH C is moderate but weak on gender equity and education.

5.7 Indicator scoring scale

Score	Meaning
1	Very poor / critical - urgent intervention
2	Poor - below acceptable
3	Moderate - acceptable but improvable
4	Good - above average
5	Excellent - model for others

Data sources for livelihood analysis

Reliable analysis needs reliable data. The main sources in India are:

- **Census of India:** population, literacy, occupation and housing.
- **NSSO surveys:** consumption expenditure, employment and farmer situation assessment.
- **Agricultural Census:** landholding size, distribution and tenancy.
- **NABARD surveys:** rural financial inclusion and household income.
- **State and district statistics:** crop area, production and yield.
- **Primary data:** household surveys, PRA and focus group discussions.

Good practice is to triangulate - combine secondary data with primary field data - because each source has gaps. For a village study, a transect walk and seasonal calendar add the local detail that official statistics miss.

5.8 Exam focus

Key points

- A livelihood indicator is a measurable variable of how well a household makes a living.
- Indicators are grouped as economic, social and environmental.
- Economic indicators include income, employment, productivity and B:C ratio.
- Social indicators include education, health, gender equity and food security.
- Environmental indicators include soil carbon, groundwater and recycling.
- Weighted scoring is a common assessment method.
- PRA and household surveys are standard tools of assessment.

Short-answer questions with model answers

Q1. Define a livelihood indicator and give two examples. Model answer: A livelihood indicator is a measurable variable used to assess the level and sustainability of a household's living. Examples: net farm income per hectare and benefit-cost ratio.

Q2. List the three dimensions of livelihood indicators. Model answer: Economic, social and environmental dimensions.

Q3. What is PRA? Model answer: Participatory Rural Appraisal is a set of participatory methods in which villagers themselves map resources, rank problems and score options, giving local knowledge a direct role in analysis and planning.

Long-answer questions with model answer outlines

Q1. Explain the economic, social and environmental indicators of livelihood analysis. Outline: (i) Meaning of indicators; (ii) economic indicators with examples; (iii) social indicators with examples; (iv) environmental indicators with examples; (v) indicator table; (vi) conclusion.

Q2. Describe the methods of livelihood assessment with a scoring example. Outline: (i) Need for assessment; (ii) methods - survey, PRA, focus group, transect, seasonal calendar, scoring; (iii) weighted scoring steps; (iv) worked example with three households; (v) interpretation and use in planning; (vi) conclusion.

CHAPTER

6. Agricultural Livelihood Systems - Definition and Components

6.1 Definition

An **Agricultural Livelihood System (ALS)** is the whole set of farm and farm-related activities, resources and relationships through which a farming household earns its living. It includes not only crop production but also livestock, fisheries, agroforestry, processing and marketing, together with the land, labour, capital and knowledge that support them.

A simple definition for examinations: *An agricultural livelihood system is the combination of crops, livestock and allied enterprises, supported by land, labour, capital and knowledge, through which a farm household secures food and income on a sustained basis.*

6.2 Why the "system" view matters

A single crop is not a livelihood. A livelihood is a **system** because its parts interact: manure from cattle feeds crops; crop residues feed cattle; trees shelter crops; processing raises the value of produce. Studying these links, rather than each enterprise alone, is the purpose of ALS analysis.

6.3 Major components of an ALS

Land

- The primary natural asset - its area, soil type, slope, drainage and irrigation status.
- Determines the choice of crops and enterprises.
- Fragmentation reduces efficiency; consolidation and collective farming help.

Labour

- Family labour and hired labour.
- Quality depends on skill, health and education.
- Labour peaks (transplanting, harvesting) and troughs shape enterprise choice.
- Off-farm employment competes with farm labour.

Capital

- **Physical capital:** tools, machinery, irrigation, buildings, livestock.
- **Financial capital:** savings, credit, insurance, remittances.
- **Working capital** for seed, fertiliser and wages is often the binding constraint.

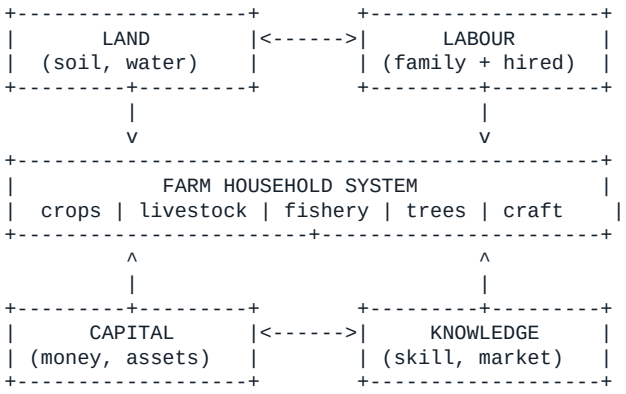
Knowledge

- Technical knowledge of crops, livestock and processing.
- Managerial knowledge of costs, markets and risk.
- Sources: KVKs, agricultural universities, extension, mobile advisories, peer farmers.
- Knowledge converts the other three components into income.

6.4 Components table

Component	What it includes	Role in livelihood	Limiting factor
Land	Area, soil, water, climate	Basis of crop and livestock	Fragmentation, degradation
Labour	Family and hired workers	Carries out all operations	Seasonality, migration
Capital	Physical, financial, working	Buys inputs, builds assets	Credit access, cost
Knowledge	Technical, managerial, market	Raises efficiency and value	Extension gaps, literacy

6.5 The four components as a system (text diagram)



All four components interact. A shortage in any one of them limits the whole system - this is the principle of the **limiting factor**.

6.6 ALS versus a single enterprise

Basis	Single enterprise	Agricultural livelihood system
Scope	One activity	Many linked activities
Income	Seasonal, single source	Continuous, multiple sources
Risk	High - one failure hurts	Spread across enterprises
Resource use	Often wasteful	By-products recycled
Labour use	Peaks and idle periods	Evenly distributed
Sustainability	Weak	Stronger

6.7 Worked example - identifying the limiting component

A farmer has 2 hectares of good land, adequate family labour and good knowledge of paddy cultivation but no access to credit and no savings. During the sowing season the farmer cannot buy seed and fertiliser on time.

Diagnosis: land, labour and knowledge are adequate; **capital is the limiting component**. The correct intervention is not more land or training but access to timely credit (Kisan Credit Card) and a small savings habit. Once capital is supplied, the same land, labour and knowledge can produce a much higher income. This shows why ALS analysis must identify the binding constraint before recommending a solution.

The four components in the Tamil Nadu context

- **Land:** small and fragmented, but often fertile; the delta's alluvial soils are among the best in India.
- **Labour:** family labour is abundant, but wage labour is costly and scarce at peak seasons.
- **Capital:** access to institutional credit is good through cooperatives and KCC; working capital remains the common constraint.
- **Knowledge:** strong extension support from TNAU, KVKs and the Uzhavan app; digital literacy is improving.

Because Tamil Nadu farms are small, the knowledge and capital components carry unusual weight: a small farm can raise income sharply through better knowledge and timely credit even without more land. This is why extension and credit reform have high returns in the state.

6.8 Exam focus

Key points

- An ALS is the whole set of farm and farm-related activities by which a household earns a living.
- The four major components are land, labour, capital and knowledge.
- Capital includes physical, financial and working capital.

- Knowledge converts the other components into income.
- The limiting-factor principle says the scarcest component controls the whole system.
- A system view spreads risk and recycles resources better than a single enterprise.

Short-answer questions with model answers

Q1. Define an agricultural livelihood system. Model answer: An agricultural livelihood system is the combination of crops, livestock and allied enterprises, supported by land, labour, capital and knowledge, through which a farm household secures food and income on a sustained basis.

Q2. Name the four major components of an ALS. Model answer: Land, labour, capital and knowledge.

Q3. What is the limiting-factor principle? Model answer: It states that the scarcest component of a system controls its overall performance; improving any other component will not help until the limiting one is addressed.

Long-answer questions with model answer outlines

Q1. Define ALS and explain its major components. Outline: (i) Definition; (ii) need for a system view; (iii) land; (iv) labour; (v) capital - physical, financial, working; (vi) knowledge; (vii) interaction diagram; (viii) conclusion.

Q2. Distinguish between a single enterprise and an agricultural livelihood system. Outline: (i) Meaning of each; (ii) comparison table - scope, income, risk, resource use, labour, sustainability; (iii) why systems are preferred for smallholders; (iv) example of an integrated farm; (v) conclusion.

CHAPTER

7. Approaches to Agricultural Livelihood Systems

7.1 Traditional versus modern approaches

Farming-based livelihood systems have evolved from traditional, self-reliant, low-input practices to modern, market-oriented, technology-intensive ones. Neither is perfect; the best systems blend the strengths of both.

Basis	Traditional approach	Modern approach
Inputs	Local, organic, on-farm	Purchased, external, high-yielding
Objective	Subsistence and security	Market surplus and profit
Technology	Hand tools, animal power	Machinery, precision tools, ICT
Cropping	Mixed, diverse, risk-spreading	Specialised, uniform
Yield	Low but stable	High but input-dependent
Knowledge	Inherited, local	Scientific, formal
Sustainability	Generally resource-conserving	Can degrade soil and water
Market link	Local, direct	Long value chains

7.2 Integrated Farming System (IFS) approach

An **Integrated Farming System** is a combination of two or more enterprises on the same farm such that the waste or by-product of one becomes the input of another. Examples: crop plus dairy plus poultry; crop plus fishery plus duck.

Advantages:

- Increased and stabilised income from several enterprises.
- Recycling of nutrients and reduction in purchased inputs.
- Better use of family labour throughout the year.
- Reduced risk from failure of any single enterprise.
- Improved soil health through organic recycling.

7.3 Sustainable practices approach

- **Conservation agriculture:** minimum tillage, residue retention, crop rotation.
- **Organic and natural farming:** no synthetic chemicals, on-farm inputs.
- **Integrated nutrient management:** balanced blend of organic and inorganic sources.
- **Integrated pest management:** biological, cultural, mechanical and need-based chemical control.
- **Micro-irrigation:** drip and sprinkler for water efficiency.

7.4 Climate-resilient approach

- Drought- and flood-tolerant crop varieties.
- Crop diversification and adjustment of sowing dates.
- Rainwater harvesting and farm ponds.
- Agroforestry and shelterbelts.
- Weather-based crop insurance and advisories.
- The **National Innovations in Climate Resilient Agriculture (NICRA)** programme develops and demonstrates these practices through KVKs.

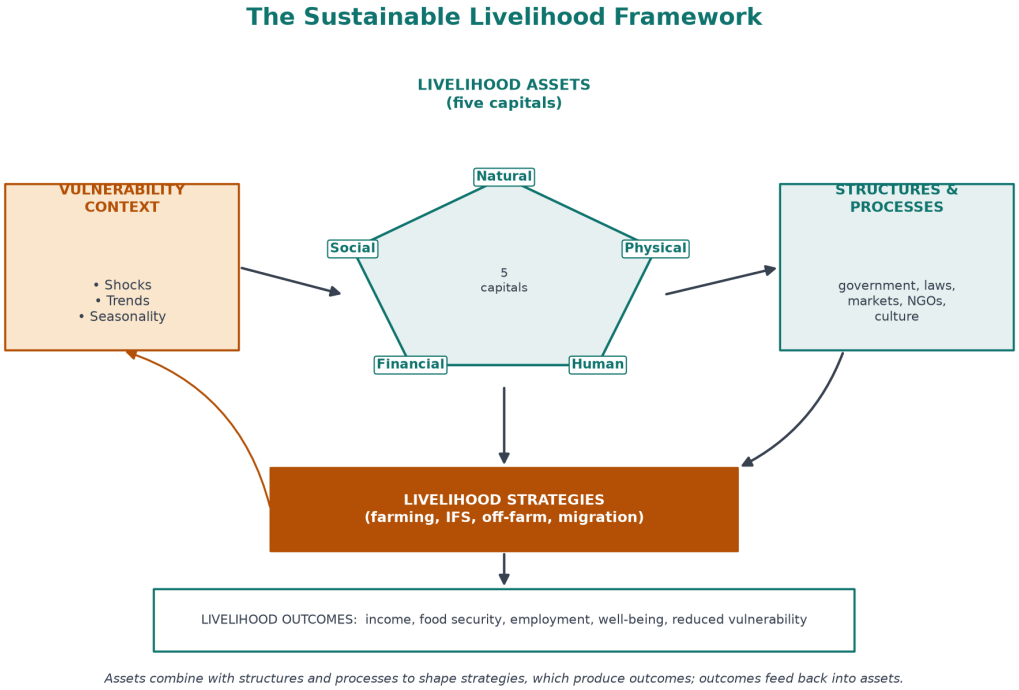


Figure 7.1 The sustainable livelihood framework

The sustainable livelihood framework — the five capitals (natural, physical, human, financial, social) shaped by the vulnerability context, structures and processes.

7.5 The Sustainable Livelihood Framework applied to farming



The framework is applied by (1) listing the household's five capitals, (2) identifying the shocks and the policy environment, (3) choosing livelihood strategies that fit the capitals, and (4) measuring outcomes against indicators, then repeating the cycle.

7.6 Choosing an approach - decision guide

1. **Assess resources:** how much land, water, labour and capital is available?
2. **Assess risk:** is the region drought-prone, flood-prone or stable?
3. **Assess market:** is there a buyer for the produce?
4. **Assess knowledge:** does the household have the skills for the chosen enterprise?
5. **Combine enterprises** so that by-products recycle and risk spreads.
6. **Start small, scale up** after learning.

7.7 Worked example - designing an IFS for a 1 hectare wetland farm

A farmer in the Cauvery delta has 1 hectare of irrigated wetland. Design a simple IFS.

Enterprise	Allocation	Function in the system
Paddy (main crop)	0.70 ha	Staple food and main cash
Vegetables (bund and border)	0.05 ha	Daily cash, home nutrition
Coconut or fruit on bunds	60 trees	Long-term income, shade
Dairy (2 cows)	Shed near house	Manure for crops, daily milk income
Poultry (50 birds)	Shed	Meat and eggs, insect control
Vermicompost unit	Small shed	Converts dung to manure
Farm pond (for fish and water)	0.05 ha	Fish, irrigation, recharge

Recycling: cow dung goes to the vermicompost unit and the farm pond; compost fertilises paddy and vegetables; paddy straw feeds the cows; pond silt is spread on the field. Purchased fertiliser is reduced, cost falls and income rises.

Participatory approaches in livelihood planning

Livelihood planning works best when farmers themselves identify problems and solutions.

- **Participatory Rural Appraisal (PRA):** villagers map resources, rank problems and score options.
- **Farmer field schools:** learning by observing and experimenting in the field.
- **Participatory technology development:** farmers help select and refine technologies.
- **Group planning:** SHGs and FPOs prepare collective plans.

Advantages: local knowledge is used, ownership is built, and solutions fit local conditions. **Limitations:** participatory processes take time, and group decisions can be dominated by influential members unless facilitation is careful. The best plans combine scientific knowledge from TNAU and KVKs with the farmer's own experience.

7.8 Exam focus

Key points

- Traditional systems are low-input, diverse and self-reliant; modern systems are market-oriented and input-intensive.
- IFS combines enterprises so the waste of one feeds another.
- IFS raises income, spreads risk and recycles nutrients.
- Sustainable practices include conservation agriculture, organic farming, IPM and micro-irrigation.
- NICRA develops climate-resilient practices demonstrated through KVKs.
- The Sustainable Livelihood Framework links assets, institutions, strategies and outcomes.

Short-answer questions with model answers

Q1. What is an integrated farming system? Model answer: An integrated farming system is a combination of two or more enterprises on the same farm in which the waste or by-product of one enterprise becomes the input of another, raising income and recycling resources.

Q2. Give two advantages of IFS. Model answer: (i) It stabilises and increases income through several enterprises; (ii) it recycles nutrients and reduces purchased inputs, lowering cost.

Q3. What is conservation agriculture? Model answer: Conservation agriculture is a farming approach based on minimum soil disturbance, retention of crop residues on the surface and crop rotation, to protect soil health and moisture.

Long-answer questions with model answer outlines

Q1. Compare traditional and modern approaches to farming-based livelihood systems. Outline: (i) Meaning of each approach; (ii) comparison table; (iii) strengths and weaknesses of each; (iv) the case for a blended approach; (v) conclusion.

Q2. Explain the integrated farming system approach with a design example. Outline: (i) Definition of IFS; (ii) principles of integration and recycling; (iii) advantages; (iv) design steps; (v) worked example for one hectare; (vi) role in climate resilience; (vii) conclusion.

CHAPTER

8. Challenges and Future Perspectives in Agricultural Livelihood Systems

8.1 Challenges

- **Resource constraints:** shrinking land, declining groundwater, degrading soil, rising labour cost.
- **Climate change:** erratic rainfall, heat waves, floods and cyclones.
- **Market volatility:** uncertain prices, weak linkages, dominance of intermediaries.
- **Knowledge and extension gaps:** slow adoption of new technology by smallholders.
- **Fragmentation:** small plots limiting mechanisation and investment.
- **Credit and risk:** inadequate and costly credit, low insurance penetration.
- **Youth out-migration:** declining interest of young people in farming.
- **Post-harvest losses:** poor storage, transport and processing.

8.2 Future perspectives

Digital agriculture

- Mobile advisories and weather alerts.
- Remote sensing and drones for crop health and spraying.
- Sensors and Internet of Things (IoT) for soil moisture and irrigation.
- Data analytics for yield prediction and precision input use.
- Digital market platforms for price discovery and direct selling.

Agro-ecotourism

- Farm stays, farm visits and guided tours.
- Income from visitors, food, and sale of farm products.
- Educates urban consumers and builds brand loyalty.
- Tamil Nadu has strong potential near hill stations and heritage delta sites.

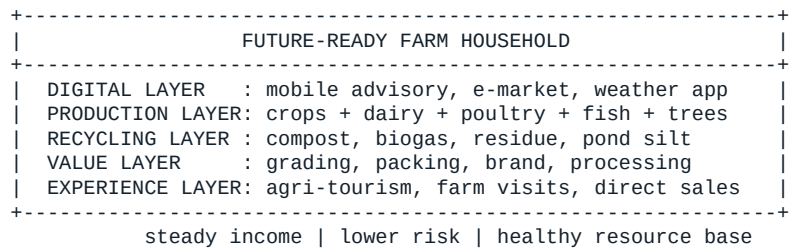
Value-chain integration

- Farmer Producer Organizations for aggregation.
- Grading, packaging, branding and certification.
- Contract farming and direct supply to retailers and exporters.
- Cold chains and processing units to cut losses and add value.

8.3 Challenge and future-response table

Challenge	Future response	Example
Resource constraint	Precision use, IFS recycling	Drip with fertigation, vermicompost
Climate change	Climate-resilient varieties and practices	NICRA demonstrations via KVK
Market volatility	Value chains, e-NAM, FPO	FPO selling graded produce online
Knowledge gap	Digital extension, farmer field schools	Uzhavan app in Tamil Nadu
Youth migration	Agri-entrepreneurship, agri-tourism	Farm-stay and processing units
Post-harvest loss	Cold chain, processing	Banana pulp and chips units

8.4 A future-ready farm - text diagram



8.5 Worked example - impact of digital adoption

A farmer sells paddy through a village trader at a lower price. After joining an FPO that lists on an electronic market platform, the farmer sells at a higher price.

Channel	Quantity (quintals)	Price per quintal (Rs.)	Gross return (Rs.)
Village trader	50	1,850	92,500
e-NAM / FPO	50	2,050	1,02,500
Gain	-	**200**	**10,000**

A price gain of Rs. 200 per quintal on 50 quintals adds Rs. 10,000 - about 11 per cent more - with no change in yield. Digital market access converts the same produce into higher income.

Digital extension services

Digital tools are changing how advice reaches farmers.

- **Mobile advisories:** crop and weather messages in the local language.
- **Apps:** Uzhavan in Tamil Nadu for advisory, subsidy and market information.
- **Call centres and helplines:** quick answers to specific queries.
- **Video and social media:** demonstrations and peer learning.
- **Remote sensing:** crop health and drought monitoring from satellite data.

Benefits: speed, low cost, wide reach and timeliness. **Challenges:** connectivity gaps, low digital literacy among older farmers, and the need for content in Tamil and other local languages. Digital extension complements, but does not replace, face-to-face visits by extension staff and KVK scientists.

8.6 Exam focus

Key points

- The main challenges are resource limits, climate change, market volatility and knowledge gaps.
- Digital agriculture, agro-ecotourism and value-chain integration are the key future directions.
- Digital tools include mobile advisories, drones, sensors and e-markets.
- Agro-tourism adds income from farm visits and farm stays.
- Value chains raise income by grading, processing and branding.
- Future-ready farms combine digital, production, recycling and value layers.

Short-answer questions with model answers

Q1. Name three future directions in agricultural livelihood systems. Model answer: Digital agriculture, agro-ecotourism, and value-chain integration.

Q2. How does digital agriculture help farmers? Model answer: It provides timely weather and advisory information, enables precision use of inputs through sensors and drones, and opens digital markets for better price discovery and direct selling.

Q3. What is agro-ecotourism? Model answer: Agro-ecotourism is a form of tourism in which visitors experience farming and rural life on a working farm, giving the farmer extra income from stays, tours and the

sale of farm products.

Long-answer questions with model answer outlines

Q1. Discuss the challenges and future perspectives of agricultural livelihood systems. Outline: (i) Introduction; (ii) challenges - resources, climate, market, knowledge, fragmentation, credit; (iii) future perspectives - digital agriculture, agro-tourism, value chains; (iv) challenge-response table; (v) conclusion.

Q2. Explain how digital agriculture and value chains can transform farm livelihoods. Outline: (i) Meaning of digital agriculture; (ii) tools - advisory, drones, sensors, analytics, e-market; (iii) value-chain concept and steps; (iv) worked example of price gain; (v) challenges of adoption; (vi) conclusion.

CHAPTER

9. Introduction to Farming Systems and Farming-Based Livelihood Systems

9.1 Definition of a farming system

A **farming system** is the combination of farm enterprises - crops, livestock, fishery, forestry and related activities - together with the resources and the management practices that a farm household uses to produce food and income. It is an organised arrangement of soil, water, plants, animals, labour, capital and technology on a holding.

A farming system is defined by three things: **what is produced** (enterprises), **how it is produced** (technology and practices) and **for whom** (household and market).

9.2 Definition of a farming-based livelihood system

A **farming-based livelihood system** is a farming system viewed through the lens of the household's living. It asks not only what the farm produces but also how the farm's activities, assets and products sustain the family's food, income, employment and security over time. In short: a farming system describes the farm; a farming-based livelihood system describes the farm *as a means of living*.

9.3 Components of a farming system

- **Enterprises:** crop, dairy, goat, poultry, piggery, fishery, apiculture, sericulture, mushroom, agroforestry, processing.
- **Resources:** land, water, labour, capital, livestock, equipment.
- **Technology:** varieties, practices, machinery, irrigation, information.
- **Management:** planning, resource allocation, marketing, risk handling.
- **Household:** the family that supplies labour and takes decisions.

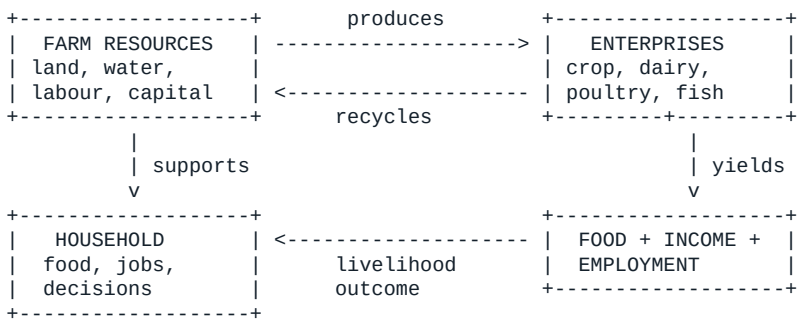
9.4 Relationship with rural livelihoods

Farming systems are the principal livelihood base of rural India. They provide:

- **Food** for the household and the market.
- **Income** from the sale of surplus and by-products.
- **Employment** for family and hired labour.
- **Assets** such as livestock, trees and equipment that build wealth.
- **Security** through diverse enterprises that reduce risk.

When a farming system is well integrated, it becomes a strong livelihood system; when it is weak or single-enterprise, the household is vulnerable and often forced to migrate.

9.5 Farming system as a livelihood - text diagram



9.6 Characteristics of a good farming system

- **Diversity:** several enterprises to spread risk.

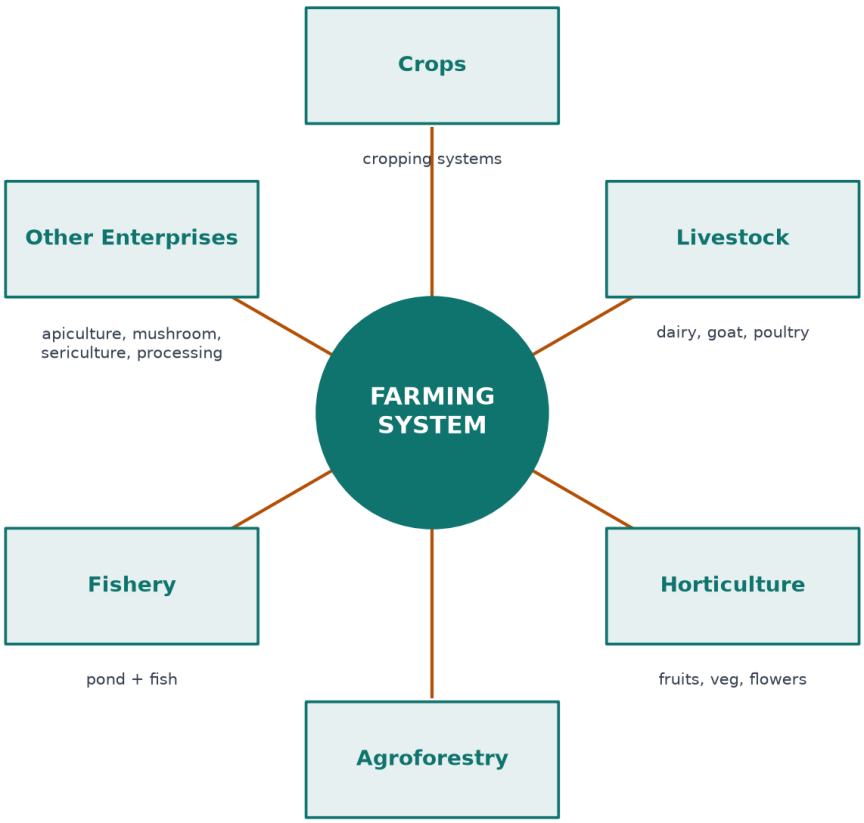


Figure 9.1 Components of a farming system

The main components of a farming system — crops, livestock, horticulture, agroforestry, fishery and other enterprises arranged around the farm.

- **Complementarity:** enterprises support each other (manure, fodder, residue).
- **Profitability:** positive net return over full cost.
- **Stability:** income through the year and across years.
- **Sustainability:** protects soil, water and biodiversity.
- **Acceptability:** fits the household's skills, labour and culture.

Systems thinking in farm planning

A farming system is more than a list of enterprises; it is a set of interactions.

- **Flows:** nutrients, water, labour, capital and information move between enterprises.
- **Feedback:** high milk yield needs good fodder, which needs manure, which needs livestock.
- **Bottlenecks:** the limiting factor - often capital or water - controls the whole system.
- **Emergent properties:** a well-integrated farm is more stable than the sum of its enterprises.

Planning rule: identify the limiting factor first, then design the enterprise mix around it. A farm with abundant land but scarce water should not add water-hungry enterprises; a farm with abundant labour but little capital should choose low-investment, labour-intensive enterprises such as mushroom or backyard poultry.

9.7 Exam focus

Key points

- A farming system is the combination of enterprises, resources and practices on a holding.
- A farming-based livelihood system views the farm as a means of living.
- Components are enterprises, resources, technology, management and the household.
- Farming systems provide food, income, employment, assets and security.
- Good systems are diverse, complementary, profitable, stable and sustainable.

Short-answer questions with model answers

Q1. Define a farming system. Model answer: A farming system is the combination of farm enterprises together with the resources and management practices by which a farm household produces food and income.

Q2. How is a farming system related to rural livelihoods? Model answer: Farming systems are the main livelihood base of rural households, providing food, income, employment, assets and security; a well-integrated system gives a strong, stable livelihood.

Q3. State two characteristics of a good farming system. Model answer: (i) Diversity of enterprises to spread risk; (ii) complementarity, where the by-product of one enterprise feeds another.

Long-answer questions with model answer outlines

Q1. Explain the concept of a farming system and its relationship with rural livelihoods. Outline: (i) Definition; (ii) components; (iii) role in rural livelihoods - food, income, employment, assets, security; (iv) characteristics of a good system; (v) diagram; (vi) conclusion.

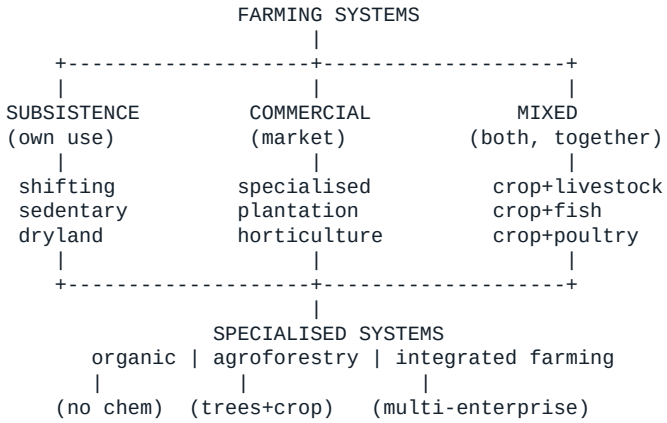
Q2. Distinguish between a farming system and a farming-based livelihood system. Outline: (i) Meaning of each; (ii) focus of each - production versus living; (iii) components of each; (iv) why the livelihood view matters for policy; (v) example; (vi) conclusion.

CHAPTER

10. Types of Farming Systems, Characteristics and Farm Type Assessment

10.1 Classification of farming systems

Farming systems are classified by the purpose of production, the degree of specialisation and the resources used. The classification tree below organises the main types.



10.2 Subsistence farming

- Produces mainly for the family's own consumption.
- Small area, low external inputs, simple tools.
- Low surplus, low cash income.
- Examples: shifting cultivation, subsistence paddy in rainfed tracts.

10.3 Commercial farming

- Produces mainly for sale and profit.
- Uses high inputs, improved technology and hired labour.
- Examples: sugarcane, cotton, horticulture, commercial dairy and poultry.

10.4 Mixed farming

- Combines crop production with livestock.
- Crop residues feed animals; manure feeds crops.
- Spreads income and risk; common on Indian smallholdings.

10.5 Specialised systems

- **Organic farming:** no synthetic chemicals; on-farm inputs; certified produce.
- **Agroforestry:** trees with crops or livestock on the same land.
- **Integrated Farming System:** several enterprises linked by recycling.

10.6 Comparison table of farming types

Basis	Subsistence	Commercial	Mixed	Specialised
Purpose	Home use	Market profit	Both	Depends
Farm size	Small	Medium to large	Small to medium	Varies
Inputs	Low	High	Moderate	Low (organic)
Technology	Traditional	Modern	Mixed	Specialised
Risk	High	High	Moderate	Depends

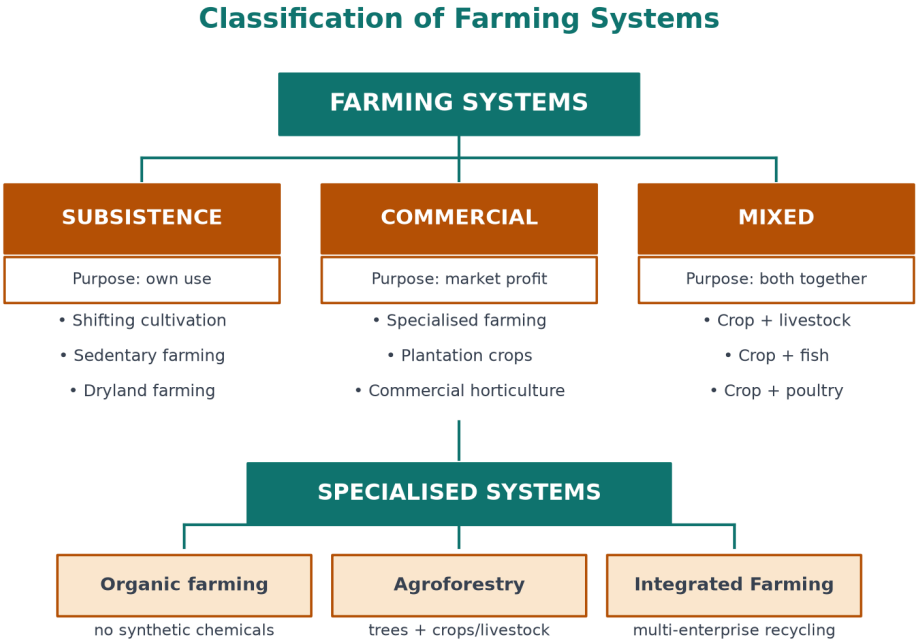


Figure 10.1 Classification of farming systems

Classification tree of farming systems — subsistence, commercial and mixed types, and specialised systems (organic, agroforestry, integrated farming).

Basis	Subsistence	Commercial	Mixed	Specialised
Examples	Shifting, rainfed	Sugarcane, poultry	Crop + dairy	Organic, agroforestry

10.7 Farm type assessment

Farm typology is the grouping of farms into types that share similar resources, enterprises and constraints, so that recommendations can be targeted. Assessment uses:

- **Resource criteria:** land area, irrigation, livestock number.
- **Enterprise criteria:** main crop, main animal, main off-farm activity.
- **Socio-economic criteria:** income level, caste, tenancy, credit access.
- **Methods:** farm survey, PRA, secondary data, cluster analysis.

Common Indian farm typology:

1. Marginal (below 1 ha) - mostly subsistence plus wage work.
2. Small (1-2 ha) - mixed crop and livestock.
3. Semi-medium (2-4 ha) - crop with some commercialisation.
4. Medium (4-10 ha) - commercial crop plus machinery.
5. Large (above 10 ha) - commercial and mechanised.

10.8 Worked example - classifying a farm

A household has 1.4 hectares of irrigated land, one pair of bullocks, two cows, grows paddy and vegetables and sells surplus milk.

- Land class: **small farmer** (1-2 ha).
- Purpose: partly subsistence (food) and partly market (milk, vegetables) - **mixed farming**.
- Enterprises: crop + dairy - **crop-livestock integrated**.
- Recommendation: strengthen the dairy-crop nutrient cycle, add a small poultry unit and a vermicompost unit, and join an FPO for vegetable marketing.

Farm size, tenancy and efficiency

Farm size and tenure shape what a household can do.

- **Marginal and small farms** dominate in number; they need collective action and high-value enterprises.
- **Medium and large farms** can mechanise and commercialise but must manage labour and risk.
- **Tenancy** (renting land) is widespread informally; insecure tenure discourages investment in soil and water.
- **Efficiency** in Indian agriculture is often higher on small farms per hectare (the "inverse size-yield relationship"), but lower per worker.

The policy implication is twofold: raise per-hectare productivity on small farms through technology and integration, and raise per-worker income through diversification, value addition and off-farm employment.

10.9 Exam focus

Key points

- Farming systems are classified by purpose, specialisation and resources.
- Subsistence farms produce for home use; commercial farms for the market.
- Mixed farming combines crops and livestock and is common in India.
- Specialised systems include organic, agroforestry and IFS.
- Farm typology groups farms for targeted recommendations.
- Indian farm classes run from marginal (below 1 ha) to large (above 10 ha).

Short-answer questions with model answers

Q1. Define farm typology. Model answer: Farm typology is the grouping of farms into types that share similar resources, enterprises and constraints so that recommendations and support can be targeted.

Q2. Give two differences between subsistence and commercial farming. Model answer: (i) Subsistence farming produces mainly for home use while commercial farming produces for the market; (ii) subsistence farming uses low inputs and traditional technology while commercial farming uses high inputs and modern technology.

Q3. Name any three specialised farming systems. Model answer: Organic farming, agroforestry and integrated farming system.

Long-answer questions with model answer outlines

Q1. Classify farming systems and describe their characteristics. Outline: (i) Basis of classification; (ii) subsistence farming; (iii) commercial farming; (iv) mixed farming; (v) specialised systems; (vi) comparison table; (vii) conclusion.

Q2. Explain farm type assessment and its importance. Outline: (i) Meaning of farm typology; (ii) criteria - resources, enterprises, socio-economic; (iii) methods; (iv) Indian farm classes; (v) worked example; (vi) use in extension and policy; (vii) conclusion.

CHAPTER

11. Prevalent Farming Systems in India and Their Role in Livelihoods

11.1 Overview

Indian farming systems are as diverse as its agro-climatic zones. The main groups are crop-based, livestock-based and horticulture-based systems, with fisheries and agroforestry in specific regions. Most smallholders follow a **mixed crop-livestock** system, which is the single most prevalent type.

11.2 Crop-based systems

- **Rice-wheat system:** Punjab, Haryana, western Uttar Pradesh; high irrigation, high output, but concerns over groundwater and soil health.
- **Rice-rice system:** Cauvery delta of Tamil Nadu, coastal Andhra, West Bengal; two or three crops a year.
- **Millet-pulse system:** Karnataka, Tamil Nadu drylands, Maharashtra; rainfed, drought-tolerant, low input.
- **Cotton-groundnut system:** Gujarat, Maharashtra, Telangana.
- **Maize-wheat / maize-pulse:** central India and the Deccan.

11.3 Livestock-based systems

- **Dairy:** Punjab, Haryana, Gujarat, Rajasthan, Tamil Nadu; daily income, manure for crops.
- **Goat rearing:** dryland and semi-arid regions; low investment, quick returns, suited to landless households.
- **Poultry:** commercial broiler and layer belts around cities; backyard poultry for households.
- **Piggery:** North-East India, Kerala and parts of Tamil Nadu; important for tribal livelihoods.

11.4 Horticulture-based systems

- **Fruit belts:** mango (Andhra, UP, Maharashtra), banana (Tamil Nadu), grape (Maharashtra), apple (Himachal).
- **Vegetable belts:** around all major cities; high value per hectare.
- **Spice systems:** chilli (Andhra), turmeric (Tamil Nadu, Telangana), pepper and cardamom (Kerala, Karnataka).
- **Floriculture:** jasmine (Tamil Nadu), rose and marigold near cities.

11.5 Role in livelihoods - comparison table

System	Main regions	Livelihood role	Key risk
Rice-wheat	Punjab, Haryana, UP	Food, income, employment	Groundwater, soil health
Rice-rice (delta)	Cauvery delta, TN	Staple, cash, farm labour	Water availability
Millet-pulse	Drylands of TN, KA, MH	Food security, fodder	Drought
Cotton-groundnut	GJ, MH, TS	Cash income	Pest, price
Dairy	PB, HR, GJ, TN	Daily income, manure	Feed cost, disease
Goat and poultry	Drylands, all India	Quick income, landless	Disease, market
Horticulture	TN, AP, MH	High value, exports	Perishability, price
Piggery	NE India, Kerala, TN	Tribal livelihood	Disease, market

11.6 Tamil Nadu farming systems

- **Cauvery delta (Thanjavur, Thiruvarur, Nagapattinam):** rice-rice-pulse; the state's rice bowl.
- **Western zone (Coimbatore, Erode):** cotton, sugarcane, coconut, turmeric.
- **Southern drylands (Virudhunagar, Ramanathapuram):** millets, pulses, goat, cotton.

Agro-Climatic Zones of India and Representative Crops

1. Western Himalayan	Wheat, maize, apple, potato
2. Eastern Himalayan	Rice, maize, tea, horticulture
3. Lower Gangetic Plains	Rice, jute, potato
4. Middle Gangetic Plains	Rice, wheat, sugarcane
5. Upper Gangetic Plains	Wheat, rice, sugarcane
6. Trans-Gangetic Plains	Wheat, rice, cotton
7. Eastern Plateau & Hills	Rice, pulses, oilseeds
8. Central Plateau & Hills	Wheat, soybean, pulses
9. Western Plateau & Hills	Cotton, sorghum, sugarcane
10. Southern Plateau & Hills	Millets, pulses, oilseeds
11. East Coast Plains & Hills	Rice, groundnut, coconut
12. West Coast Plains & Ghats	Rice, coconut, spices
13. Gujarat Plains & Hills	Groundnut, cotton, wheat
14. Western Dry Region	Bajra, pulses, livestock
15. Islands	Rice, coconut, spices

15 agro-climatic zones (Planning Commission), based on soil, climate, rainfall and cropping pattern.

Figure 11.1 Agro-climatic zones of India

The 15 agro-climatic zones of India with representative crops for each zone.

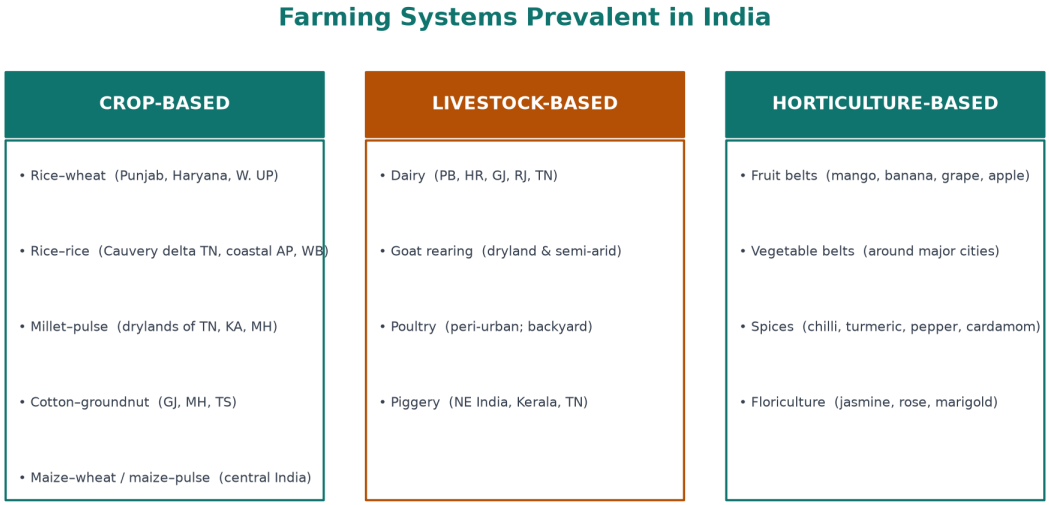


Figure 11.2 Farming systems prevalent in India

The three broad groups of farming systems prevalent in India — crop-based, livestock-based and horticulture-based — with example systems.

- **Hilly zone (Nilgiris):** tea, vegetables, potato, flowers.
- **Coastal zone:** coconut, cashew, fisheries, rice.

11.7 Worked example - income mix of a mixed system

A smallholder in the southern drylands of Tamil Nadu has 1.5 hectares rainfed plus five goats.

Enterprise	Net income (Rs./year)	Share of total
Millets and pulses	22,000	34 per cent
Cotton	15,000	23 per cent
Goat rearing	18,000	28 per cent
Wage labour (off-farm)	10,000	15 per cent
Total	**65,000**	**100 per cent**

No single source exceeds 34 per cent, so the household is resilient: a crop failure in one season is cushioned by goats and wages. This is the practical strength of a mixed system in a risky dryland environment.

Regional specialisation and comparative advantage

Each region tends to specialise in what it does best - its comparative advantage.

- **Punjab and Haryana:** irrigated wheat and rice, high yields, but stressed groundwater.
- **Cauvery delta, Tamil Nadu:** rice, with a strong canal system and high yields.
- **Gujarat:** groundnut, cotton and milk.
- **Maharashtra:** cotton, sugarcane, horticulture.
- **Karnataka and Tamil Nadu drylands:** millets, pulses and livestock.

Why specialisation matters: it raises efficiency and marketability, but it also raises risk if the specialised crop fails or its price crashes. The remedy is to specialise in the main enterprise while keeping secondary enterprises for stability - the logic of integrated farming within a specialised region.

11.8 Exam focus

Key points

- Indian farming systems are crop-based, livestock-based and horticulture-based.
- The mixed crop-livestock system is the most prevalent.
- Rice-wheat dominates the north-west; rice-rice dominates the Cauvery delta.
- Millet-pulse systems suit rainfed drylands.
- Dairy, goat and poultry provide income to landless and marginal households.
- Horticulture gives high value per hectare but carries price and perishability risk.

Short-answer questions with model answers

Q1. Name the most prevalent farming system in India. Model answer: The mixed crop-livestock system, in which crop production is combined with cattle or other livestock, is the most prevalent farming system in India.

Q2. Which farming system is typical of the Cauvery delta? Model answer: The rice-rice system (with a pulse crop in some rotations) is typical of the Cauvery delta of Tamil Nadu.

Q3. Why is goat rearing suited to landless households? Model answer: Goat rearing needs low investment, can be done on common land and crop residues, gives quick returns and requires little land, so it suits landless and marginal households.

Long-answer questions with model answer outlines

Q1. Describe the prevalent farming systems of India and their role in livelihoods. Outline: (i) Overview and diversity; (ii) crop-based systems; (iii) livestock-based systems; (iv) horticulture-based systems; (v) role in livelihoods table; (vi) Tamil Nadu examples; (vii) conclusion.

Q2. Explain the crop-livestock mixed farming system and its importance for smallholders. Outline: (i) Meaning of mixed farming; (ii) nutrient and income cycle; (iii) income and risk benefits; (iv) worked example; (v) role in drylands; (vi) conclusion.

CHAPTER

12. Component - Cropping Systems

12.1 Definition

A **cropping system** is the pattern of crops and their sequence and management on a given piece of land over time. It includes what is grown, in what order, in what association and with what practices. A **cropping pattern** is the proportion of area under different crops at a point of time.

12.2 Monocropping

- Growing a single crop on the same land year after year.
- **Advantages:** simple management, suited to mechanisation, market specialisation.
- **Disadvantages:** nutrient depletion, build-up of pests and diseases, high risk if the crop fails.
- Example: continuous sugarcane or continuous rice.

12.3 Mixed cropping

- Growing two or more crops together in the same field without a definite row arrangement.
- **Advantages:** risk spreading, better use of space, fodder and food together.
- **Disadvantages:** difficult to manage, harvest and market separately.
- Example: maize with cowpea; sorghum with pigeon pea.

12.4 Intercropping

- Growing two or more crops in **definite rows** on the same field at the same time.
- Types: **row intercropping**, **strip intercropping**, **relay intercropping**.
- **Advantages:** higher total yield, better use of light and nutrients, insurance against failure, legume benefit.
- Example: cotton with black gram; maize with cowpea; coconut with banana.

12.5 Crop rotation

- Growing different crops in a planned sequence on the same land.
- **Advantages:** breaks pest and disease cycles, restores fertility (with legumes), balances nutrient demand, controls weeds.
- Example: rice - pulse - rice; cotton - sorghum - groundnut; paddy - black gram.

12.6 Comparison of cropping systems

System	Arrangement	Main advantage	Main limitation
Monocropping	Single crop, repeated	Simple, mechanisable	Nutrient loss, pest build-up
Mixed cropping	No definite rows	Risk spreading	Hard to manage
Intercropping	Definite rows	Higher total yield	Complex management
Crop rotation	Sequence over seasons	Fertility, pest break	Planning needed

12.7 Factors influencing cropping patterns

- **Physical:** soil type, rainfall, temperature, irrigation, topography.
- **Economic:** price, market demand, cost of cultivation, MSP.
- **Technical:** availability of seed, machinery and knowledge.
- **Social:** food habits, tradition, tenancy.
- **Institutional:** irrigation policy, procurement, insurance.
- **Biological:** pest and disease pressure, crop duration.

12.8 Region-specific examples

Region	Cropping pattern	Reason
Cauvery delta, TN	Rice-rice-pulse	Assured canal irrigation
Punjab	Rice-wheat	Irrigation, MSP, mechanisation
Tamil Nadu drylands	Millet-pulse-cotton	Low rainfall, risk spreading
West Bengal	Rice-potato-rice	Moisture, market for potato
Maharashtra	Cotton-soybean	Black soil, market
Nilgiris, TN	Tea and vegetables	Cool climate, hills

12.9 Cropping intensity and multiple cropping

- **Cropping intensity = (Gross cropped area / Net sown area) x 100.**
- Higher intensity means more crops per year on the same land.
- **Multiple cropping** raises land productivity and income.

Worked example: A farm has a net sown area of 1 hectare and grows paddy on 1 hectare, a pulse on 0.9 hectare and a vegetable on 0.4 hectare in one year.

Gross cropped area = 1.0 + 0.9 + 0.4 = 2.3 hectares.

Cropping intensity = (2.3 / 1.0) x 100 = **230 per cent**.

This means the farmer harvests on average 2.3 crops per year from each hectare - a high intensity typical of the delta.

Sustainable cropping practices

Cropping systems can be managed to protect soil and water while raising output.

- **Legume inclusion:** a pulse in the rotation fixes nitrogen and breaks pest cycles.
- **Crop residue management:** retain or compost residues instead of burning them.
- **Balanced fertilisation:** soil-test based, with organic sources.
- **Integrated pest management:** reduce chemical load and resistance.
- **Micro-irrigation and mulching:** save water and suppress weeds.
- **Diversified rotation:** avoids nutrient depletion from continuous monocropping.

Example: a delta rotation of rice - black gram - rice uses the pulse to restore nitrogen and improve soil, reducing the fertiliser needed for the next rice crop while adding a second income.

12.10 Exam focus

Key points

- A cropping system is the pattern, sequence and management of crops on a piece of land.
- Monocropping is simple but depletes nutrients and builds up pests.
- Mixed cropping has no definite rows; intercropping has definite rows.
- Crop rotation breaks pest cycles and restores fertility.
- Cropping intensity = gross cropped area divided by net sown area, times 100.
- Factors influencing cropping patterns are physical, economic, technical, social and institutional.

Short-answer questions with model answers

Q1. Define cropping intensity. Model answer: Cropping intensity is the ratio of gross cropped area to net sown area, expressed as a percentage; it indicates how many crops are grown per year on the same land.

Q2. Differentiate intercropping from mixed cropping. Model answer: In intercropping two or more crops are grown in definite rows; in mixed cropping the crops are grown together without a definite row arrangement.

Q3. Give two advantages of crop rotation. Model answer: (i) It breaks the cycle of pests and diseases; (ii) it restores soil fertility, especially when a legume is included.

Long-answer questions with model answer outlines

Q1. Explain the different cropping systems with examples. Outline: (i) Definition; (ii) monocropping; (iii) mixed cropping; (iv) intercropping and its types; (v) crop rotation; (vi) comparison table; (vii) examples; (viii) conclusion.

Q2. Discuss the factors influencing cropping patterns in India. Outline: (i) Meaning of cropping pattern; (ii) physical factors; (iii) economic factors; (iv) technical and institutional factors; (v) social factors; (vi) region-specific examples; (vii) conclusion.

CHAPTER

13. Component - Dairy Farming and Small Livestock

13.1 Dairy farming

Dairy farming is the rearing of milch animals - cows and buffaloes - for milk production. It provides daily income, manure for crops and a buffer against crop failure. India is the world's largest milk producer.

- **Milch breeds of cattle:** Sahiwal, Gir, Red Sindhi, Tharparkar (indigenous); crossbreds of Holstein Friesian and Jersey with local cattle.
- **Buffalo breeds:** Murrah, Surti, Jaffarabadi, Mehsana.
- **Milk yield:** indigenous cows typically 2-8 litres per day; crossbred cows 8-20 litres per day; buffaloes 5-12 litres per day, depending on breed and management.
- **Tamil Nadu:** Kangayam and Umblachery are important local cattle breeds; dairy is a major allied enterprise.

13.2 Fodder management

- **Green fodder:** hybrid napier, CO fodder grasses, maize, lucerne, berseem.
- **Dry fodder:** paddy straw, sorghum stover, hay.
- **Concentrates:** oil cakes, bran, mineral mixture.
- **Balanced ration:** green fodder + dry fodder + concentrate + mineral mixture + water.
- **Fodder from IFS:** crop residues, tree leaves, bund grasses reduce the cost of feed.

13.3 Small livestock

Goat rearing

- Low investment, quick returns, suited to landless households.
- Breeds: Tellicherry, Boer (meat); Jamunapari, Sirohi (dual).
- Feeds on shrubs, crop residues and common land.

Piggery

- Important in North-East India, Kerala and tribal belts.
- Breeds: Large White Yorkshire, Landrace, Duroc; indigenous Ghungroo.
- Converts kitchen waste and crop residues into meat.

Poultry

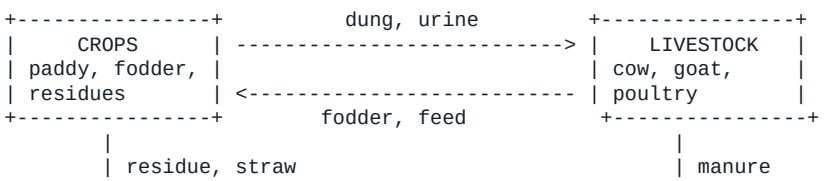
- **Backyard poultry:** 10-50 birds, low cost, eggs and meat for the family and local sale.
- **Commercial:** broilers (meat) and layers (eggs).
- Breeds: Aseel, Kadaknath (indigenous); Vanaraja, Gramapriya (improved backyard).
- Quick returns; suited to women's self-help groups.

Duck rearing

- Suited to wetlands and water bodies; feeds on aquatic plants and snails.
- Important in Kerala, West Bengal, Assam and coastal Tamil Nadu.
- Integrates well with rice fields and fish ponds.

13.4 Integration of livestock with crop farming

Livestock and crops support each other:



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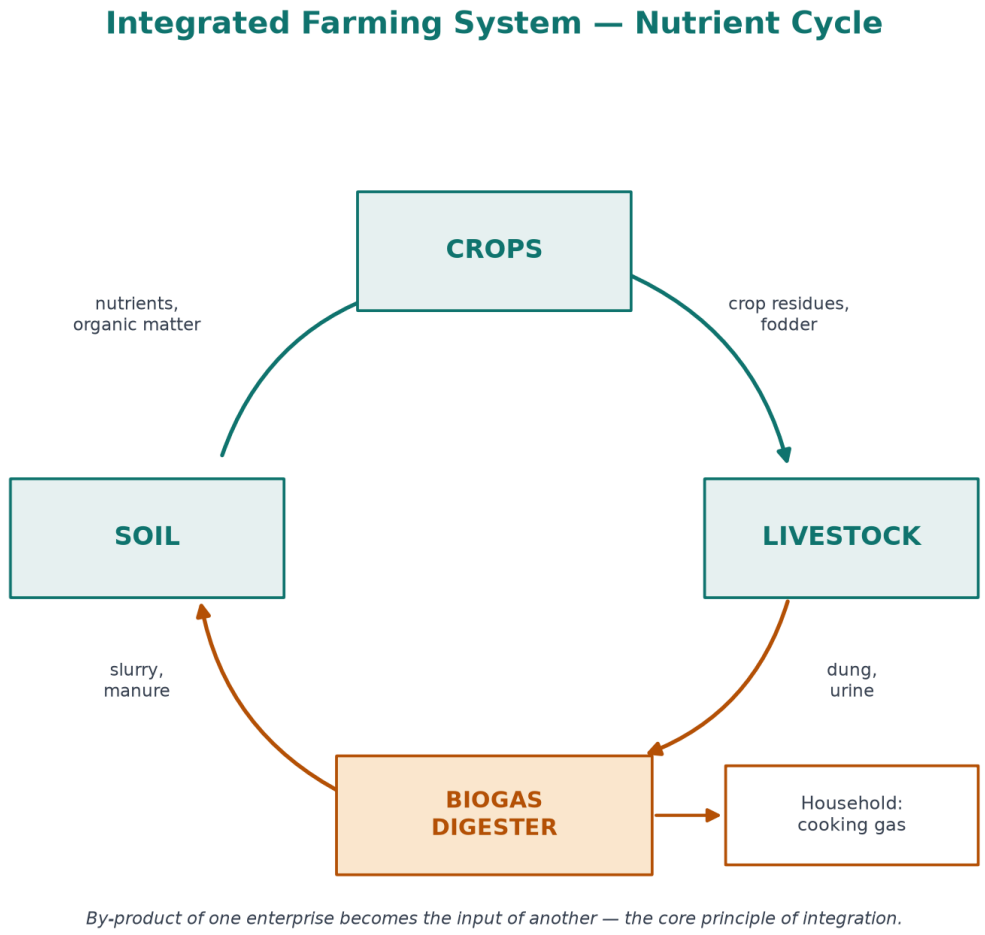
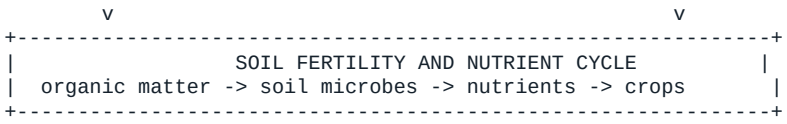


Figure 13.1 Integrated Farming System nutrient cycle

Nutrient cycle in an integrated farming system — crop residues feed livestock, dung feeds a biogas digester, and slurry and manure return nutrients to the soil and crops.



- Crop residues and fodder feed animals.
- Dung and urine become manure and biogas.
- Manure returns nutrients to the soil, reducing fertiliser cost.
- Draught animals provide power for ploughing.

13.5 Benefit-cost analysis of a dairy unit (worked example)

A farmer keeps two crossbred cows. Consider one year.

Item	Quantity / rate	Amount (Rs.)
A. Costs		
Cost of two cows (amortised over 5 years)	80,000 / 5	16,000
Green fodder	2 cows x 365 days	18,000
Concentrate feed	3 kg/day x 2 x 365 x Rs. 25	54,750
Labour (family valued)	-	24,000
Veterinary and breeding	-	4,000
Miscellaneous	-	3,250
Total cost		**1,20,000**
B. Returns		
Milk (2 cows x 8 L x 250 days = 4,000 L at Rs. 35)	4,000 L	1,40,000
Sale of manure	-	8,000
Value of calves (one per cow per 18 months)	-	10,000
Total returns		**1,58,000**
C. Net return	1,58,000 - 1,20,000	**38,000**
D. Benefit-cost ratio	1,58,000 / 1,20,000	**1.32**

Interpretation: for every rupee spent, the farmer gets Rs. 1.32 back. The unit gives a net return of Rs. 38,000 plus manure worth Rs. 8,000 for the crop enterprise. When manure substitutes purchased fertiliser, the effective return is higher still.

13.6 Poultry unit - quick benefit-cost example

Item	Amount (Rs.)
Cost of 100 day-old chicks	3,500
Feed for 6 weeks	22,000
Medicine, litter, labour	4,500
Total cost	**30,000**
Sale of 95 birds at Rs. 90 per kg, 1.6 kg each	43,700
Net return	**13,700**
Benefit-cost ratio	**1.46**

Livestock health and productivity

The profitability of livestock depends as much on health and breeding as on feed.

- **Veterinary care:** regular vaccination against foot-and-mouth disease, haemorrhagic septicaemia and other diseases.
- **Breeding:** artificial insemination with improved or crossbred semen raises milk yield.
- **Housing:** clean, ventilated sheds reduce heat stress and disease.
- **Nutrition:** balanced ration with mineral mixture raises yield and fertility.
- **Record keeping:** milk yield, feed and health records guide management decisions.

Tamil Nadu support: the Animal Husbandry Department, cooperatives and milk producers' unions provide breeding, veterinary and marketing services. For a smallholder, the dairy enterprise succeeds only when health, breeding and feed are managed together with a reliable milk buyer.

13.7 Exam focus

Key points

- Dairy farming gives daily income, manure and a buffer against crop failure.
- Indigenous cows yield less but are hardy; crossbreds yield more but need better feed.
- Balanced ration = green fodder + dry fodder + concentrate + mineral mixture.
- Goat and backyard poultry suit landless households and women's groups.
- Livestock and crops form a nutrient cycle - residues to animals, manure to soil.
- A dairy unit with a B:C ratio above 1 is profitable.

Short-answer questions with model answers

Q1. What is a balanced ration? Model answer: A balanced ration is a feed mixture that supplies all nutrients in the correct proportion for maintenance and production, typically combining green fodder, dry fodder, concentrate and a mineral mixture with clean water.

Q2. How do livestock and crops support each other? Model answer: Crop residues and fodder feed livestock; livestock provide dung and urine that become manure and biogas, returning nutrients to the soil and reducing fertiliser cost, and draught animals provide farm power.

Q3. Name two backyard poultry breeds. Model answer: Vanaraja and Gramapriya are improved backyard poultry breeds (Aseel and Kadaknath are indigenous examples).

Long-answer questions with model answer outlines

Q1. Explain dairy farming and the integration of livestock with crop farming. Outline: (i) Meaning of dairy farming; (ii) breeds of cattle and buffalo; (iii) fodder and balanced ration; (iv) nutrient cycle between crops and livestock; (v) benefit-cost example; (vi) conclusion.

Q2. Describe small livestock enterprises and their role in rural livelihoods. Outline: (i) Meaning of small livestock; (ii) goat rearing; (iii) piggery; (iv) poultry and duck rearing; (v) role for landless and women; (vi) benefit-cost example; (vii) conclusion.

CHAPTER

14. Component - Horticulture

14.1 Definition and scope

Horticulture is the science and art of cultivating fruits, vegetables, flowers, spices, plantation crops and medicinal and aromatic plants. It is the most **income-dense** branch of agriculture: a hectare of vegetables or flowers can earn several times the income of a hectare of cereals.

Horticulture is divided into:

- **Pomology:** fruit crops.
- **Olericulture:** vegetable crops.
- **Floriculture:** flowers and ornamental plants.
- **Spices and plantation crops:** chilli, turmeric, pepper, cardamom, coconut.
- **Medicinal and aromatic plants:** medicinal herbs and essential-oil crops.

14.2 Importance in livelihoods

- **High value per hectare** - raises farm income faster than field crops.
- **Labour-intensive** - generates rural employment, especially for women.
- **Nutrition security** - supplies vitamins and minerals.
- **Short duration** - vegetables give quick returns.
- **Export earnings** - fruits, spices and flowers earn foreign exchange.
- **Diversification and risk reduction** - spreads income across crops.

14.3 Important horticultural crops

Crop	Leading states	Tamil Nadu position
Banana	Tamil Nadu, Maharashtra, Gujarat	Tamil Nadu is a leading producer
Mango	Uttar Pradesh, Andhra Pradesh, Karnataka	Krishnagiri, Salem, Dharmapuri
Coconut	Kerala, Karnataka, Tamil Nadu	Coimbatore, Thanjavur, Pollachi
Tomato	Andhra Pradesh, Karnataka, Madhya Pradesh	Coimbatore, Dharmapuri
Onion	Maharashtra, Madhya Pradesh, Karnataka	Perambalur, Tiruchirappalli
Chilli	Andhra Pradesh, Telangana, Karnataka	Ramanathapuram, Virudhunagar
Turmeric	Telangana, Tamil Nadu, Maharashtra	Erode, Coimbatore
Jasmine	Tamil Nadu, Karnataka	Madurai, Dindigul
Tapioca	Tamil Nadu, Kerala	Salem, Namakkal

14.4 High-value horticulture

Floriculture

- **Loose flowers:** jasmine, marigold, chrysanthemum, crossandra.
- **Cut flowers:** rose, carnation, gerbera, orchid, gladiolus.
- **Tamil Nadu** is a leading producer of jasmine (Madurai, Dindigul) and other loose flowers.
- Income is high but flowers are highly perishable; cold chain and quick marketing are essential.

Medicinal and aromatic plants

- **Medicinal:** ashwagandha, senna, coleus, aloe vera, glory lily, sarpagandha.

- **Aromatic:** lemongrass, palmarosa, citronella, vetiver, eucalyptus.
- Tamil Nadu has a strong base in coleus, senna and glory lily for export.
- **Advantages:** low water need, high value, suited to drylands; contract farming with buyers is common.

14.5 Horticulture-based livelihood table

Enterprise	Investment	Return profile	Suitability
Vegetables	Moderate	Quick, 3-4 months	Irrigated, near market
Banana	Moderate-high	11-14 months	Irrigated, delta
Mango orchard	High, long-term	After 4-5 years	Rainfed and dryland
Jasmine	Moderate	Continuous, daily	Near cities
Medicinal plants	Low-moderate	6-12 months	Dryland, contract
Cut flowers	High	Weekly harvest	Protected, near city

14.6 Worked example - income from one hectare of horticulture

A farmer grows vegetables on 0.5 hectare and maintains a young mango orchard on 0.5 hectare.

Crop	Area (ha)	Gross income (Rs.)	Cost (Rs.)	Net (Rs.)
Vegetables (tomato and brinjal)	0.5	1,20,000	55,000	65,000
Mango (young, intercropped)	0.5	20,000	10,000	10,000
Total	**1.0**	**1,40,000**	**65,000**	**75,000**

Net income of Rs. 75,000 from one hectare compares with about Rs. 45,000 from paddy alone - a clear gain, but with higher labour and market risk. Intercropping the young orchard with a pulse or vegetable keeps cash flowing until the trees bear fruit.

Post-harvest management of horticultural produce

Horticultural produce is perishable, so post-harvest care decides profit.

- **Harvesting:** at the right maturity, with care to avoid bruising.
- **Grading and sorting:** uniform size and quality fetch higher prices.
- **Pre-cooling and cold chain:** removes field heat and extends shelf life.
- **Packaging:** protects produce and enables branding.
- **Storage:** cool storage for fruits and vegetables reduces glut-season losses.
- **Processing:** converts surplus into pulp, chips, pickles and dried products.

Loss figures: post-harvest losses in fruits and vegetables are typically higher than in cereals, often in the range of 10-20 per cent or more because of poor handling. Cutting these losses is one of the cheapest ways to raise horticultural income.

14.7 Exam focus

Key points

- Horticulture covers fruits, vegetables, flowers, spices and medicinal plants.
- It is income-dense and labour-intensive, generating employment for women.
- Banana, mango, coconut, chilli and turmeric are important in Tamil Nadu.
- Floriculture gives high but perishable income; cold chain is essential.
- Medicinal and aromatic plants suit drylands and contract farming.
- Horticulture raises income but carries higher market and perishability risk.

Short-answer questions with model answers

Q1. Define horticulture and name its main branches. Model answer: Horticulture is the science and art of cultivating fruits, vegetables, flowers, spices and medicinal and aromatic plants. Its main branches are pomology, olericulture, floriculture and the cultivation of spices and medicinal plants.

Q2. Why is horticulture important for livelihoods? Model answer: Horticulture gives high value per hectare, generates labour employment, provides nutrition and export earnings, and diversifies income, raising farm income faster than field crops.

Q3. Give two advantages of medicinal plants for dryland farmers. Model answer: (i) Low water requirement, suited to drylands; (ii) high value with contract farming, giving assured income.

Long-answer questions with model answer outlines

Q1. Explain the importance of horticulture in rural livelihoods. Outline: (i) Definition and branches; (ii) importance - income, employment, nutrition, export, diversification; (iii) important crops and states; (iv) worked income example; (v) risks; (vi) conclusion.

Q2. Describe high-value horticulture with reference to floriculture and medicinal plants. Outline: (i) Meaning of high-value horticulture; (ii) floriculture - types, Tamil Nadu position, cold chain; (iii) medicinal and aromatic plants; (iv) advantages for drylands; (v) contract farming; (vi) conclusion.

CHAPTER

15. Component - Agroforestry

15.1 Definition

Agroforestry is a land-use system in which woody perennials (trees, shrubs, palms, bamboo) are grown on the same land as crops or livestock, in a deliberate and managed combination. It combines agriculture and forestry to get the benefits of both.

15.2 Types of agroforestry

- **Silviculture / silvopasture:** trees with pasture and livestock.
- **Agri-silviculture:** trees with crops (e.g. eucalyptus or casuarina with groundnut).
- **Agri-horticulture:** fruit trees with crops (e.g. mango or coconut with vegetables or pulses).
- **Agri-silvi-horticulture:** trees + fruit trees + crops together.
- **Silvi-pastoral:** trees with pasture for grazing.
- **Agro-silvi-pastoral:** trees + crops + pasture together.
- **Home gardens:** a mixture of trees, crops and animals around the homestead.

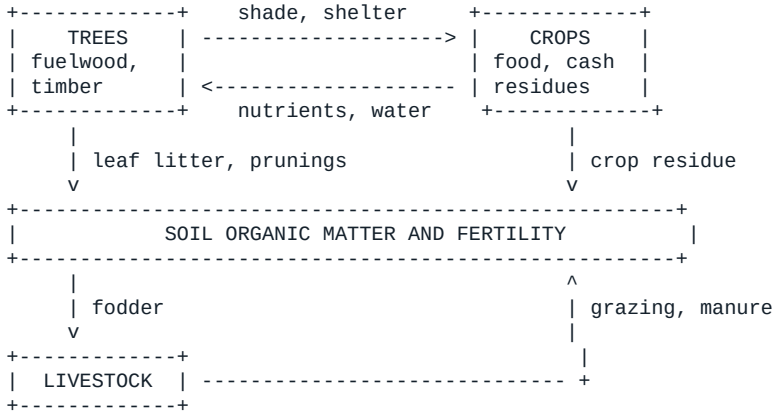
15.3 Classification table

Type	Combination	Example
Agri-silviculture	Trees + crops	Casuarina with groundnut
Agri-horticulture	Fruit trees + crops	Coconut with banana or pulses
Silvopasture	Trees + pasture + animals	Subabul with fodder and cattle
Agri-silvi-horticulture	Trees + fruits + crops	Neem + mango + cowpea
Home garden	Trees + crops + animals	Jackfruit, banana, poultry at home

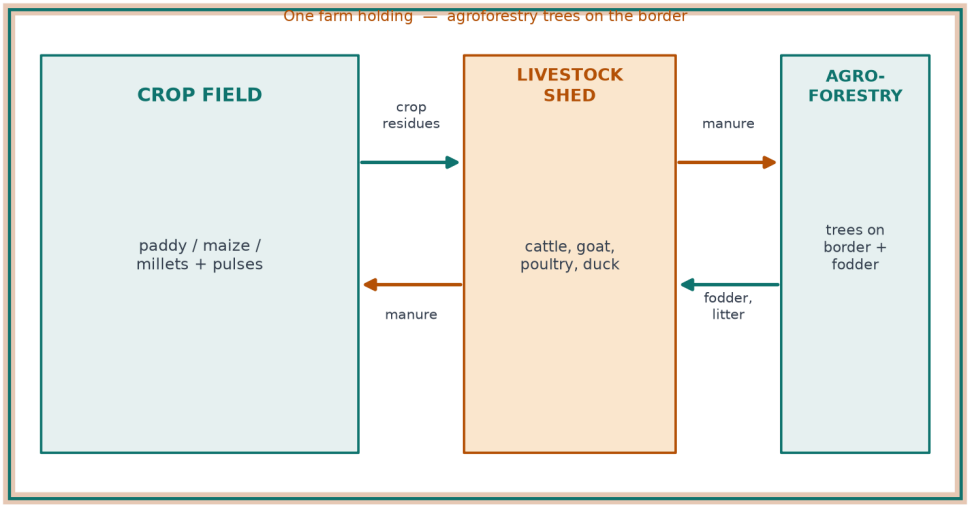
15.4 Benefits of agroforestry

- **Income diversification:** timber, fruit, fuelwood, fodder and crops together.
- **Soil conservation:** tree roots bind soil, reduce erosion.
- **Improved soil fertility:** leaf litter adds organic matter; leguminous trees fix nitrogen.
- **Microclimate:** shade reduces heat stress on crops and animals.
- **Fodder and fuelwood:** reduces pressure on forests.
- **Risk reduction:** trees act as a buffer when crops fail.
- **Carbon sequestration:** trees store carbon, helping climate goals.
- **Biodiversity:** supports birds, insects and soil life.

15.5 Nutrient and resource flow in agroforestry (text diagram)



Crop–Livestock–Tree Integration on One Holding



Crop residues feed livestock; manure feeds crops and trees; tree fodder and leaf litter return to the system.

All three components share the same land, labour, water and nutrients — the basis of a mixed farming system.

Figure 15.1 Crop–livestock–tree integration

Crop–livestock–tree integration on a single holding — the three components share the same land, water, labour and nutrients.

15.6 Suitable tree species for Tamil Nadu

Purpose	Species
Timber	Teak, casuarina, eucalyptus, melia
Fruit	Mango, coconut, sapota, jackfruit, guava
Fodder	Subabul, glyricidia, sesbania, moringa
Nitrogen fixing	Subabul, glyricidia, acacia, casuarina (with mycorrhiza)
Fuelwood	Casuarina, prosopis, acacia
Border and shelter	Casuarina, teak, silver oak

15.7 Worked example - agri-horticulture on one hectare

A farmer plants coconut at 7.5 m x 7.5 m (about 175 palms per hectare) and intercrops with banana for the first three years and then with a pulse or fodder.

Component	Annual net income (Rs.)
Coconut (175 palms, mature)	55,000
Intercrop (banana, years 1-3)	40,000
Fodder or pulse (after year 3)	15,000
Total (mature phase)	**70,000**

The coconut palm is productive for decades, so this system gives long-term stable income, and the intercrop keeps income flowing during the early years.

Tree-crop interaction and management

In agroforestry, trees and crops compete for light, water and nutrients; good management balances the competition.

- **Spacing:** wide spacing reduces shade and root competition.
- **Pruning:** regular pruning of tree branches reduces shade and supplies fodder and mulch.
- **Root management:** deep-rooted trees compete less with shallow-rooted crops.
- **Species choice:** nitrogen-fixing and light-crowned trees suit intercropping.
- **Rotation:** crops are grown in the early years before the tree canopy closes.

Example: in a coconut-based agri-horticulture system, banana is grown for the first few years and then replaced by shade-tolerant crops or fodder, matching the intercrop to the increasing shade as the palms grow.

15.8 Exam focus

Key points

- Agroforestry combines trees with crops or livestock on the same land.
- Main types are agri-silviculture, agri-horticulture, silvopasture and home gardens.
- Benefits include income diversification, soil conservation, fertility and microclimate.
- Leguminous trees fix nitrogen and improve soil.
- Agroforestry reduces risk and sequesters carbon.
- Coconut-based agri-horticulture is common in Tamil Nadu.

Short-answer questions with model answers

Q1. Define agroforestry. Model answer: Agroforestry is a land-use system in which woody perennials are deliberately grown on the same land as crops or livestock, combining agriculture and forestry.

Q2. Name any four types of agroforestry. Model answer: Agri-silviculture, agri-horticulture, silvopasture and home gardens.

Q3. State two benefits of agroforestry. Model answer: (i) It diversifies income through timber, fruit, fuelwood and fodder; (ii) it conserves soil and improves fertility through leaf litter and nitrogen fixation.

Long-answer questions with model answer outlines

Q1. Explain agroforestry, its types and benefits. Outline: (i) Definition; (ii) types with examples; (iii) benefits - income, soil, fertility, microclimate, fodder, risk, carbon; (iv) nutrient flow diagram; (v) suitable species; (vi) worked example; (vii) conclusion.

Q2. Discuss the role of agroforestry in sustainable farming-based livelihoods. Outline: (i) Meaning; (ii) resource recycling; (iii) income stability; (iv) environmental benefits; (v) Tamil Nadu examples; (vi) constraints - slow returns, tenure; (vii) conclusion.

CHAPTER

16. Component - Fish Farming and Integrated Models

16.1 Fish farming

Fish farming (aquaculture) is the rearing of fish in ponds, tanks and reservoirs for food and income. It is one of the fastest-growing allied enterprises and gives high returns per unit area.

- **Major species:** Indian major carps - rohu, catla, mrigal; and exotic carps - grass carp, silver carp, common carp.
- **Pond preparation:** lime, manuring, filling with water, stocking fingerlings.
- **Composite fish culture:** stocking several species together that feed at different levels of the pond, so all food is used.
- **Feeding:** supplementary feed of rice bran and oil cake.
- **Harvesting:** after about 10-12 months, or staggered for continuous supply.

16.2 Composite fish culture - feeding zones

```
Surface water  : Silver carp (phytoplankton feeder)
Mid water      : Catla (zooplankton), Rohu (column feeder)
Bottom         : Mrigal, Common carp (detritus and bottom feeder)
---------------------------------------------------------------
Three feeding niches -> one pond -> maximum yield per unit area
```

16.3 Integrated fish farming models

Integrated fish farming combines fish with other enterprises so that wastes feed the fish.

Duck-cum-fish

- Ducks are housed over or beside the pond.
- Duck droppings fertilise the pond, promoting plankton.
- Ducks eat insects and pests; reduce feed cost.
- About 200-300 ducks per hectare of pond.

Poultry-cum-fish

- Poultry droppings fertilise the pond.
- A poultry shed is built over the pond bund.
- Reduces fish feed cost and gives two incomes.

Dairy-cum-fish

- Cattle shed washings and dung enrich the pond.
- Cattle provide milk income; pond gets organic manure.
- Pond silt is later used on crop fields.

Piggery-cum-fish

- Pig dung fertilises the pond; pigs eat crop and kitchen waste.
- Popular in North-East India and Kerala.

Rice-fish culture

- Fish are reared in the standing water of rice fields.
- Fish eat insects and weeds; their waste fertilises the rice.
- Common in Tamil Nadu delta and Kerala.

16.4 Integrated model comparison table

Model	Waste used	Benefit	Caution
Duck-cum-fish	Duck droppings	Two incomes, low feed	Pond water quality
Poultry-cum-fish	Poultry litter	Low feed cost	Disease control
Dairy-cum-fish	Dung and washings	Manure, milk, fish	Organic load balance
Piggery-cum-fish	Pig dung	High fertility	Odour, hygiene
Rice-fish	Fish waste, insects	Weed and pest control	Pesticide use must stop

16.5 Economic and environmental benefits

- **Economic:** two or three incomes from one pond; lower feed cost; quick returns.
- **Environmental:** nutrients recycled, less pollution, less wild fish pressure.
- **Nutritional:** cheap protein for the household.
- **Employment:** pond management, feeding and harvesting give jobs.

16.6 Benefit-cost analysis of a one-hectare fish pond (worked example)

Item	Amount (Rs.)
A. Costs	
Pond preparation, lime, manure	15,000
Fingerlings (10,000 at Rs. 3)	30,000
Feed (rice bran, oil cake)	70,000
Labour	30,000
Miscellaneous	15,000
Total cost	**1,60,000**
B. Returns	
Fish yield 4,000 kg at Rs. 65/kg	2,60,000
Net return	**1,00,000**
Benefit-cost ratio	**1.63**

With a duck-cum-fish model, feed cost can fall by Rs. 15,000-20,000 because duck droppings fertilise the pond, raising the B:C ratio further.

Water quality management in fish ponds

Fish health and yield depend on pond water quality.

- **Dissolved oxygen:** maintain adequate levels; avoid overstocking and over-manuring.
- **pH:** keep near neutral to slightly alkaline; lime corrects acidity.
- **Manuring:** apply organic manure in controlled doses to promote plankton without depleting oxygen.
- **Water exchange:** periodic partial exchange removes wastes.
- **Avoiding chemicals:** stop pesticide use in adjoining fields to protect fish.

Integration caution: while duck or pig droppings fertilise the pond, excessive loading can deplete oxygen and kill fish. The correct management is a balanced stocking and manuring schedule, monitored regularly - the reason integration needs skill as well as resources.

16.7 Exam focus

Key points

- Fish farming rears fish in ponds for food and income.

- Composite fish culture stocks species that feed at different pond levels.
- Integrated models are duck-cum-fish, poultry-cum-fish, dairy-cum-fish and piggery-cum-fish.
- Integration lowers feed cost by using waste to fertilise the pond.
- Rice-fish culture controls weeds and pests in paddy fields.
- Fish farming gives high returns per unit area with a favourable B:C ratio.

Short-answer questions with model answers

Q1. What is composite fish culture? Model answer: Composite fish culture is the practice of stocking several fish species that feed at different levels of the pond - surface, column and bottom - so that all available food is used and total yield per unit area is maximised.

Q2. Name two integrated fish farming models. Model answer: Duck-cum-fish and dairy-cum-fish (also poultry-cum-fish and piggery-cum-fish).

Q3. State two benefits of integrated fish farming. Model answer: (i) It lowers feed cost because animal waste fertilises the pond; (ii) it gives two or more incomes from the same pond.

Long-answer questions with model answer outlines

Q1. Explain fish farming and the integrated fish farming models. Outline: (i) Meaning of fish farming; (ii) species and composite culture; (iii) integrated models - duck, poultry, dairy, piggery, rice-fish; (iv) economic and environmental benefits; (v) benefit-cost example; (vi) conclusion.

Q2. Discuss the economic and environmental benefits of integrated fish farming. Outline: (i) Concept of integration; (ii) nutrient recycling; (iii) economic benefits; (iv) environmental benefits; (v) worked B:C analysis; (vi) conclusion.

CHAPTER

17. Mid Semester Examination - Revision and Model Paper

17.1 Purpose of this lecture

This lecture is the mid-semester examination. It covers Lectures 1 to 16. The material below is a revision summary of the key concepts and a model question paper for practice.

17.2 Revision summary - key concepts

Lecture	Core concept to remember
1	Agriculture: 16-20 per cent of GVA, 42-48 per cent of employment; 15 agro-climatic zones; TN has 7
2	MSP by CACP; PM-KISAN; Dalwai Committee and doubling farmers' income
3	Challenges (cost, climate, market) and opportunities (diversification, digital, processing)
4	Livelihood = capabilities + assets + activities; five capitals; rural versus urban
5	Livelihood indicators - economic, social, environmental; weighted scoring
6	ALS components - land, labour, capital, knowledge; limiting factor
7	Traditional versus modern; IFS; Sustainable Livelihood Framework
8	Future - digital agriculture, agro-tourism, value chains
9	Farming system and farming-based livelihood system
10	Types - subsistence, commercial, mixed, specialised; farm typology
11	Prevalent systems - rice-wheat, millet-pulse, dairy, horticulture
12	Cropping systems - monocropping, mixed, intercropping, rotation
13	Dairy and small livestock; nutrient cycle; B:C analysis
14	Horticulture - income-dense, labour-intensive; floriculture
15	Agroforestry - types and benefits
16	Fish farming and integrated models

17.3 Model question paper

Time: 2 hours. Maximum marks: 50.

Section A - Answer all (5 x 2 = 10 marks).

1 Define agro-climatic zone.

2 What is MSP and who recommends it?

3 Name the five capitals of the sustainable livelihood framework.

4 Define cropping intensity.

5 What is composite fish culture?

Section B - Answer any four (4 x 5 = 20 marks).

1 Explain the role of agriculture in India's economy.

2 Describe the components of an agricultural livelihood system.

3 Compare traditional and modern approaches to farming-based livelihood systems.
4 Explain the types of agroforestry with examples.
5 Describe the integration of livestock with crop farming.

Section C - Answer any one (1 x 20 = 20 marks).

1 Discuss the challenges and opportunities for enhancing farmers' income in India, with a worked example.
2 Explain integrated farming systems and design one for a one-hectare wetland farm in the Cauvery delta.

17.4 Model answers to Section A

1 **Agro-climatic zone:** a region with a distinct combination of climate, soil, rainfall and cropping pattern; India has 15 such zones.
2 **MSP:** the assured price at which the government buys notified crops; recommended by the CACP and announced by the Ministry of Agriculture.
3 **Five capitals:** natural, physical, human, financial and social capital.
4 **Cropping intensity:** (gross cropped area / net sown area) x 100.
5 **Composite fish culture:** stocking several fish species that feed at different pond levels to maximise yield.

17.5 Answer-writing tips

- Begin each answer with a **definition** in one or two lines.
- Use **headings** and **bullet points**; examiners reward structure.
- Support every point with an **Indian or Tamil Nadu example**.
- Include a **table or diagram** where possible; it earns presentation marks.
- In numerical answers, **show the working** and state units.
- End with a one-line **conclusion**.

Revision strategy and time management

A structured revision plan improves performance in the mid-semester examination.

- **Divide the syllabus** into blocks of three to four lectures and revise each block before moving on.
- **Make one-page summaries** for each lecture: definitions, one table, one example.
- **Practise numericals** on cropping intensity, B:C ratio and value chain margin.
- **Memorise key figures** as ranges, not exact numbers, and note the source.
- **Write practice answers** within time limits to build speed.
- **Draw diagrams from memory** - the classification tree, the livelihood framework and the circular loop.
- **Use the glossary** to check that every key term can be defined in one or two lines.

17.6 Exam focus

Key points

- The mid-semester examination covers Lectures 1 to 16.
- Definitions, comparisons and worked examples carry the most marks.
- Always give a definition, a structure and an example.
- Tables and diagrams improve presentation marks.
- Numericals must show the working and units.

Short-answer questions with model answers

Q1. What is the difference between a farming system and a farming-based livelihood system? Model answer: A farming system describes the combination of enterprises and practices on a farm; a farming-based livelihood system views the same farm as a means of living, focusing on food, income, employment and security.

Q2. State the three dimensions of livelihood indicators. Model answer: Economic, social and environmental.

Q3. Define IFS. Model answer: An integrated farming system combines two or more enterprises on the same farm so that the waste or by-product of one becomes the input of another.

Long-answer questions with model answer outlines

Q1. Explain the sustainable livelihood framework with a diagram. Outline: (i) Definition of livelihood; (ii) five capitals; (iii) vulnerability context; (iv) institutions and policies; (v) strategies and outcomes; (vi) diagram; (vii) conclusion.

Q2. Describe any four components of farming-based livelihood systems with examples. Outline: (i) Cropping systems; (ii) dairy and livestock; (iii) horticulture; (iv) agroforestry or fisheries; (v) how they integrate; (vi) conclusion.

CHAPTER

18. Component - Small, Medium and Large Farm-Based Enterprises

18.1 Concept of farm-based enterprises

A **farm-based enterprise** is any income-generating activity carried out on or around the farm using its resources. Enterprises are grouped by the scale of investment, the number of workers and the level of technology into **small, medium and large** categories. Small enterprises need little capital and suit marginal farmers and women; medium and large enterprises need more capital and management skill.

18.2 Small enterprises

Small enterprises are low-investment, quick-return activities that use family labour and local resources.

Beekeeping (apiculture)

- Rearing honey bees in boxes for honey, beeswax and pollination.
- **Species:** Indian bee (Apis cerana indica), Italian bee (Apis mellifera), rock bee, little bee.
- **Advantages:** low investment, no land needed, pollinates crops and raises their yield, ready cash from honey.
- A box can yield roughly 5-15 kg of honey per year depending on flora and management.
- Tamil Nadu has beekeeping potential in coconut, mango and horticulture belts.

Mushroom cultivation

- Growing edible fungi on agricultural waste such as paddy straw.
- **Common types:** oyster mushroom, milky mushroom, button mushroom (needs cool climate).
- **Advantages:** short cycle (3-4 weeks), low space, high value, uses crop waste, suits women's groups.
- Oyster mushroom can be grown on 10-20 kg of wet straw per bed with modest investment.

Sericulture

- Rearing silkworms for silk; includes mulberry cultivation and cocoon production.
- **Advantages:** high value, employment for the family, short cycle, integrates with mulberry on small land.
- Tamil Nadu is a silk producer, with mulberry growing in Dharmapuri, Krishnagiri and Salem.
- **Steps:** mulberry cultivation, silkworm rearing, cocoon harvesting, reeling.

18.3 Medium enterprises

Medium enterprises need more capital, some hired labour and better management.

- **Agro-processing units:** rice mill, dal mill, oil expeller, flour mill, spice grinding.
- **Dairy processing:** small milk chilling, paneer or curd making.
- **Poultry and goatery at semi-commercial scale.**
- **Nursery raising:** production of vegetable and fruit seedlings.
- **Feed and fodder units:** production and sale of cattle or poultry feed.

18.4 Large enterprises

Large enterprises need substantial capital, land and hired labour, and operate on commercial lines.

- **Commercial dairy and integrated livestock farms.**
- **Large poultry (broiler and layer) units.**
- **Food processing plants:** fruit pulp, juice, chips, packaged foods.
- **Cold storage and warehouses.**
- **Commercial horticulture and plantation estates.**

18.5 Enterprises by size - comparison table

Size	Examples	Investment and labour	Suitability
Small	Beekeeping, mushroom, sericulture, backyard poultry	Low investment, family labour	Marginal farmers, women, landless
Medium	Agro-processing, nursery, semi-commercial dairy	Moderate investment, some hired labour	Small and semi-medium farmers
Large	Commercial dairy, poultry, food plant, cold storage	High investment, hired labour	Medium and large farmers, companies

18.6 Choosing an enterprise - decision factors

- **Capital available** and access to credit.
- **Land and water** available.
- **Labour** - family and hired.
- **Skill and knowledge** of the entrepreneur.
- **Market** - demand, price and distance.
- **Risk** - perishability and price volatility.
- **Support** - schemes, subsidies, training.

18.7 Worked example - comparing three enterprises for a woman's SHG

A ten-member women's self-help group has Rs. 50,000 and wants to start an enterprise.

Enterprise	Investment (Rs.)	Annual net income (Rs.)	Skill needed	Risk
Mushroom (50 beds)	25,000	40,000	Moderate	Moderate
Backyard poultry (200 birds)	30,000	35,000	Low	Moderate
Beekeeping (20 boxes)	40,000	45,000	Moderate	Low-moderate

Recommendation: mushroom cultivation gives a good return at lower investment and a short cycle, so it suits a group starting out. Beekeeping gives the highest return but needs more boxes and flora nearby. A group could combine mushroom and a small poultry unit to spread risk.

Enterprise planning and market survey

Before starting an enterprise, a farmer or group should plan and survey the market.

- **Idea generation:** list enterprises suited to local resources and demand.
- **Market survey:** find out who will buy, at what price, in what quantity and when.
- **Cost estimation:** investment, working capital and operating cost.
- **Return estimation:** expected yield, price and net income.
- **Risk assessment:** perishability, price volatility and competition.
- **Skill check:** whether the household has the required knowledge.
- **Support check:** available schemes, subsidy and training.

Simple planning tool: prepare a one-page budget showing cost, return and B:C ratio for the enterprise. If the B:C ratio is below 1 or the market is uncertain, the enterprise should be reconsidered before investment.

18.8 Exam focus

Key points

- Farm-based enterprises are grouped by scale as small, medium and large.
- Small enterprises (beekeeping, mushroom, sericulture) need low capital and suit smallholders and women.

- Medium enterprises include agro-processing and nursery raising.
- Large enterprises include commercial dairy, poultry, food plants and cold storage.
- Enterprise choice depends on capital, land, labour, skill, market and risk.
- Mushroom and sericulture suit women's self-help groups.

Short-answer questions with model answers

Q1. Name three small farm-based enterprises. Model answer: Beekeeping, mushroom cultivation and sericulture.

Q2. Why does mushroom cultivation suit women's self-help groups? Model answer: It needs little space and capital, has a short three to four-week cycle, uses crop waste as substrate, gives high value and can be done at home, so it suits women's groups.

Q3. Give two examples of medium enterprises. Model answer: A small agro-processing unit such as a dal mill, and a plant nursery raising vegetable seedlings.

Long-answer questions with model answer outlines

Q1. Describe the small, medium and large farm-based enterprises with examples. Outline: (i) Concept of enterprise scale; (ii) small enterprises - beekeeping, mushroom, sericulture; (iii) medium enterprises - processing, nursery; (iv) large enterprises - commercial dairy, food plants; (v) comparison table; (vi) worked example; (vii) conclusion.

Q2. Explain the factors to consider when selecting a farm-based enterprise. Outline: (i) Need for enterprise selection; (ii) capital; (iii) land and water; (iv) labour; (v) skill; (vi) market and risk; (vii) support and schemes; (viii) worked example; (ix) conclusion.

CHAPTER

19. Value Chains in Farming-Based Livelihood Systems

19.1 Definition

A **value chain** is the full sequence of activities and actors that take a product from the farm to the final consumer, adding value at each stage. It includes input supply, production, processing, packaging, storage, transport, marketing and finally consumption.

The **value added** at each stage is the difference between the value of the output and the value of the inputs used at that stage.

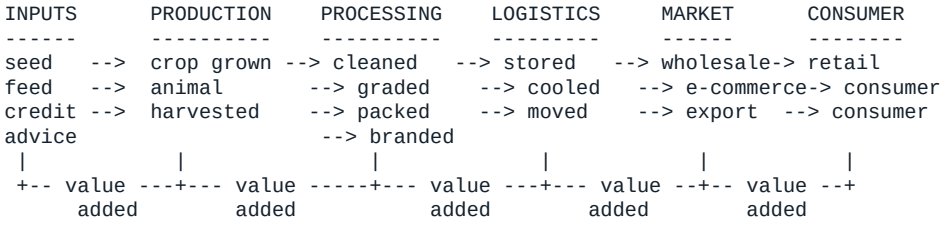
19.2 Importance of value chains

- **Higher income:** farmers capture more of the consumer rupee.
- **Reduced losses:** proper handling, storage and transport cut post-harvest loss.
- **Better quality:** grading and standardisation improve the product.
- **Market access:** links farmers to processors, retailers and exporters.
- **Employment:** processing and logistics create rural jobs.
- **Transparency:** price information flows better along a chain.
- **Traceability:** food safety and certification become possible.

19.3 Steps in value chain creation

1. **Input supply:** seed, fertiliser, feed, credit, advisory.
2. **Production:** growing the crop or rearing the animal.
3. **Primary processing:** cleaning, grading, drying, sorting.
4. **Secondary processing:** milling, pulping, packaging, branding.
5. **Storage and transport:** cold chain, warehouses, logistics.
6. **Marketing:** wholesale, retail, e-commerce, export.
7. **Consumption:** the final consumer.

19.4 Value chain flow (text diagram)



At each arrow, value is added. The longer and better-managed the chain, the more value the farmer can capture if the farmer or the FPO owns some of the later stages.

19.5 Value chain margin - worked example

Consider one quintal (100 kg) of tomatoes moving through the chain.

Stage	Price (Rs./quintal)	Cost added (Rs.)	Value added (Rs.)
Farm gate (farmer sells)	1,200	-	-
Trader and transport	1,500	200	100
Wholesaler	1,800	200	100
Retailer	2,400	300	300
Consumer pays	2,400	-	-

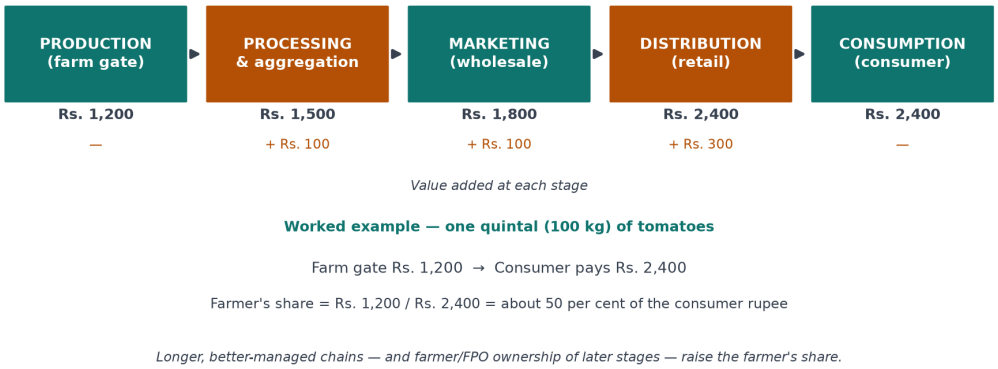


Figure 19.1 Value chain flow

Value chain flow from production to consumption, with the value added and the farmer’s share illustrated for one quintal of tomatoes.

Total consumer price = Rs. 2,400 per quintal. Farmer's share = (1,200 / 2,400) x 100 = 50 per cent. Total value added after the farm gate = Rs. 1,200 per quintal.

If the farmer or the FPO sells directly to a retailer or through an e-commerce platform at Rs. 2,000 per quintal, the farmer's share rises to about 83 per cent, and the farmer gains Rs. 800 per quintal. This is the core argument for value-chain participation by farmer groups.

19.6 Farmer Producer Organizations in the value chain

- An **FPO** is a collective of farmers that aggregates produce, buys inputs and markets in bulk.
- FPOs help smallholders reach scale, bargain better, access credit and own processing.
- Tamil Nadu has a growing number of FPOs in banana, mango, coconut, vegetables and millets.
- FPOs are promoted under the **10,000 FPOs scheme** and supported by NABARD and SFAC.

Value chain actors and governance

A value chain works only when its actors are coordinated.

- **Actors:** input suppliers, farmers, aggregators, processors, transporters, wholesalers, retailers and exporters.
- **Governance:** the actor that sets the rules and standards - often a large processor or retailer - is said to govern the chain.
- **Coordination mechanisms:** contracts, FPOs, cooperatives and e-platforms.
- **Information flow:** price, quality and demand information must reach the farmer.

Why governance matters: when farmers are weak, powerful buyers capture most of the value. FPOs and cooperatives shift bargaining power toward farmers. Transparent platforms such as e-NAM reduce information asymmetry, so the farmer is not forced to accept the first price offered.

19.7 Exam focus

Key points

- A value chain is the sequence of activities from input supply to the consumer.
- Value added is the difference between the output value and input value at a stage.
- Value chains raise income, cut losses and improve quality and market access.
- The steps are input supply, production, processing, logistics, marketing and consumption.
- In a typical vegetable chain the farmer's share of the consumer price is about half.
- FPOs help smallholders capture more of the value chain.

Short-answer questions with model answers

Q1. Define a value chain. Model answer: A value chain is the full sequence of activities and actors that take a product from the farm to the final consumer, adding value at each stage from input supply to consumption.

Q2. What is value addition? Model answer: Value addition is the increase in the value of a product at a stage of the chain, equal to the value of the output minus the value of the inputs used at that stage.

Q3. State two benefits of value chains for farmers. Model answer: (i) Farmers capture a larger share of the consumer price; (ii) post-harvest losses fall because of better handling, storage and transport.

Long-answer questions with model answer outlines

Q1. Explain value chains and the steps in value chain creation. Outline: (i) Definition; (ii) importance; (iii) steps - inputs, production, processing, logistics, marketing, consumption; (iv) value chain flow diagram; (v) margin example; (vi) role of FPOs; (vii) conclusion.

Q2. Discuss how value-chain participation raises farm income, with a worked example. Outline: (i) Meaning of the farmer's share; (ii) margin calculation; (iii) strategies - direct marketing, processing, branding; (iv) role of FPOs and e-commerce; (v) challenges; (vi) conclusion.

CHAPTER

20. Secondary Enterprises and Their Role

20.1 Definition

Secondary enterprises are income-generating activities based on the **processing of primary farm produce** or on allied non-farm skills. They are "secondary" because they add value to what the primary enterprise (crop or livestock) produces, rather than producing raw output.

20.2 Food processing enterprises

- **Cereal processing:** rice milling, flour milling, poha and vermicelli making.
- **Pulse processing:** dal milling, papad and vadam making.
- **Oil processing:** oil expelling from groundnut, coconut, sesame.
- **Fruit and vegetable processing:** pulp, juice, jam, squash, pickles, chips, dried products.
- **Spice processing:** grinding and packing of chilli, turmeric and coriander.
- **Millet processing:** millet flour, ready mixes, health foods.

20.3 Dairy product enterprises

- Curd, buttermilk, paneer, ghee, khoa and milk sweets.
- Value addition raises the return on every litre of milk.
- Small units can be run by SHGs and FPOs.

20.4 Handicrafts and allied enterprises

- Coir products from coconut husk (Tamil Nadu is a major coir state).
- Handloom and handicrafts, palm-leaf products, mats and baskets.
- Nursery and landscaping services.
- Vermicompost and organic manure production.

20.5 Role of secondary enterprises

- **Income diversification:** a second income stream beyond raw produce.
- **Employment:** processing and packing create jobs, especially for women.
- **Value addition:** raises the price realised per unit of produce.
- **Reduced waste:** uses by-products and surplus that would otherwise spoil.
- **Reduced migration:** local jobs reduce the need to move to cities.
- **Resilience:** income continues even when raw prices are low.

20.6 Secondary enterprises table

Enterprise	Raw material	Product	Key advantage
Rice and flour mill	Paddy, wheat	Rice, flour	Local demand
Oil expeller	Groundnut, coconut	Oil, cake	Cake feeds cattle
Fruit processing	Mango, banana	Pulp, chips	Uses surplus
Dairy products	Milk	Curd, paneer, ghee	High margin
Spice grinding	Chilli, turmeric	Powder	Branding
Coir unit	Coconut husk	Rope, mats	Uses waste
Vermicompost	Dung, waste	Manure	Cuts input cost

20.7 Worked example - value added by a small fruit-processing unit

A group buys 1,000 kg of mangoes at Rs. 30 per kg and makes pulp and chips.

Item	Amount (Rs.)
A. Costs	
Raw mango (1,000 kg at Rs. 30)	30,000
Sugar, preservative, packaging	12,000
Labour	6,000
Fuel and power	2,000
Total cost	**50,000**
B. Returns	
Pulp and chips from 1,000 kg, sold	78,000
Net value added	**28,000**

The same mangoes sold raw would have fetched Rs. 30,000. Processing them adds Rs. 28,000 of net value - about 93 per cent more - and creates local employment. During the glut season, when raw prices crash, processing is the farmer's best protection.

Quality standards and certification

Quality and certification open premium markets for secondary enterprises.

- **Grading:** sorting produce by size, colour and quality.
- **Standards:** food safety and hygiene norms for processed products.
- **Certification:** organic certification, Good Agricultural Practices and India Organic.
- **Labelling and branding:** builds trust and allows a higher price.
- **Traceability:** records the origin and handling of the product.

Benefits: access to modern retail, export markets and health-conscious consumers. **Costs:** certification and compliance need investment and record keeping. For small groups, collective certification through an FPO or SHG lowers the cost per member and is the practical route to premium markets.

20.8 Exam focus

Key points

- Secondary enterprises process primary produce to add value.
- Examples are food processing, dairy products and handicrafts.
- They diversify income and create rural employment, especially for women.
- They reduce waste by using surplus and by-products.
- Processing raises the return on the same raw produce.
- Coir and handicrafts are important in Tamil Nadu.

Short-answer questions with model answers

Q1. What are secondary enterprises? Model answer: Secondary enterprises are income-generating activities based on the processing of primary farm produce or allied skills, adding value to raw output rather than producing it.

Q2. State two roles of secondary enterprises in rural livelihoods. Model answer: (i) They diversify income by adding a second income stream; (ii) they create rural employment, especially for women.

Q3. Give two examples of dairy product enterprises. Model answer: Paneer making and ghee making (also curd, khoa and milk sweets).

Long-answer questions with model answer outlines

Q1. Explain secondary enterprises and their role in rural livelihoods. Outline: (i) Definition; (ii) food processing; (iii) dairy products; (iv) handicrafts and coir; (v) role - income, employment, value addition, waste reduction, reduced migration; (vi) worked example; (vii) conclusion.

Q2. Discuss value addition through food processing with an example. Outline: (i) Meaning of value addition; (ii) types of processing; (iii) economic logic; (iv) worked example of mango processing; (v) role of SHGs and FPOs; (vi) constraints - capital, quality, market; (vii) conclusion.

CHAPTER

21. Challenges and Future Prospects of Farming-Based Livelihood Systems

21.1 Challenges

- **Resource constraints:** shrinking and fragmented land, declining groundwater, degrading soil, rising labour cost.
- **Climate change:** erratic rainfall, heat waves, floods, cyclones and new pests.
- **Market access:** weak infrastructure, many intermediaries, volatile prices, poor price information.
- **Policy gaps:** uneven implementation, weak extension, inadequate credit and insurance.
- **Knowledge gap:** low awareness of modern technology among smallholders.
- **Post-harvest losses:** poor storage and handling.
- **Youth out-migration:** declining interest in farming.
- **Capital shortage:** weak working capital and high cost of credit.

21.2 Future prospects and innovations

Digital agriculture

- Mobile apps for advisory, weather and prices.
- Drones for spraying and crop monitoring.
- Sensors for soil moisture and automated irrigation.
- Data analytics for yield prediction and precision inputs.

Agro-tourism

- Farm stays, farm visits, guided tours and hands-on experiences.
- Additional income from visitors and direct sale of farm products.

Vertical farming

- Growing crops in stacked layers, often with hydroponics and controlled environment.
- Saves land and water; suited to peri-urban areas and leafy vegetables.
- High investment, so best for high-value crops.

Other directions

- **Protected cultivation:** polyhouse and shade-net for high-value crops.
- **Precision farming:** GPS-guided operations and variable-rate inputs.
- **Climate-smart agriculture:** resilient varieties and practices.
- **Value-chain integration and FPOs.**

21.3 Challenge and future prospect table

Challenge	Future prospect	Example
Resource constraint	Vertical and protected farming	Polyhouse vegetables
Climate change	Climate-smart agriculture	NICRA practices via KVK
Market access	Digital market platforms	e-NAM, e-commerce
Knowledge gap	Digital extension	Uzhavan app, SMS advisory
Post-harvest loss	Cold chain and processing	Fruit pulp units
Youth migration	Agri-entrepreneurship, agri-tourism	Farm-stay enterprises
Capital shortage	Microfinance, FPO credit	SHG bank linkage

21.4 Worked example - impact of protected cultivation

A farmer grows tomato in the open and then in a polyhouse on the same 0.2 hectare.

Method	Yield (tonnes)	Price (Rs./kg)	Gross income (Rs.)	Cost (Rs.)	Net (Rs.)
Open field	4	15	60,000	35,000	25,000
Polyhouse	9	22	1,98,000	1,10,000	88,000

Protected cultivation more than triples net income on the same land by raising yield and quality, though it needs higher investment. It suits a farmer with capital or access to a subsidy scheme.

Climate adaptation finance and insurance

Adapting to climate change costs money, and finance is now available for it.

- **Crop insurance:** PMFBY covers yield loss from natural calamities.
- **Weather-based (index) insurance:** pays out automatically when a weather trigger is crossed.
- **Livestock insurance:** covers animal death and, in some products, disease.
- **Subsidised inputs for adaptation:** tolerant seed, micro-irrigation and water harvesting support.
- **Climate finance:** national and international funds for resilient infrastructure.

Practical use: a farmer cannot prevent a drought, but can reduce its impact by combining a tolerant variety, protective irrigation, insurance and a diversified income. Adaptation is most effective when technical, financial and institutional measures work together.

21.5 Exam focus

Key points

- Main challenges are resource limits, climate change, market access and policy gaps.
- Digital agriculture, agro-tourism and vertical farming are key innovations.
- Protected cultivation raises yield and quality on the same land.
- Climate-smart agriculture builds resilience.
- Value chains and FPOs improve market access.
- Innovations need capital, skill and market linkage to succeed.

Short-answer questions with model answers

Q1. Name three future prospects of farming-based livelihood systems. Model answer: Digital agriculture, agro-tourism and vertical farming.

Q2. What is vertical farming? Model answer: Vertical farming is the growing of crops in stacked layers, often with hydroponics and a controlled environment, to save land and water and to suit high-value leafy vegetables in peri-urban areas.

Q3. How does protected cultivation help? Model answer: Protected cultivation in polyhouse or shade-net raises yield and quality by controlling temperature, pests and moisture, giving higher net income per unit of land.

Long-answer questions with model answer outlines

Q1. Discuss the challenges and future prospects of farming-based livelihood systems. Outline: (i) Introduction; (ii) challenges - resources, climate, market, policy, knowledge; (iii) future prospects - digital, agro-tourism, vertical farming, protected cultivation, precision farming; (iv) challenge-prospect table; (v) worked example; (vi) conclusion.

Q2. Explain the role of innovation in the future of farming-based livelihoods. Outline: (i) Meaning of innovation; (ii) digital agriculture; (iii) protected and vertical farming; (iv) agro-tourism; (v) climate-smart practices; (vi) constraints to adoption; (vii) conclusion.

CHAPTER

22. Factors Affecting Integration of Enterprises

22.1 Meaning of integration

Integration of enterprises means combining two or more farm enterprises so that they support each other through the recycling of resources and by-products. Successful integration depends on the resource base, the agro-climatic setting and socio-economic conditions.

22.2 Resource availability factors

- **Land:** area and type determine how many enterprises can be accommodated.
- **Water:** irrigation and pond availability decide crops, fish and livestock.
- **Labour:** family and hired labour supply and seasonal peaks.
- **Capital:** investment and working capital for each enterprise.
- **Feed and fodder:** local availability lowers livestock cost.

22.3 Agro-climatic factors

- **Soil:** type and fertility decide crops and tree species.
- **Rainfall:** amount and distribution decide rainfed versus irrigated systems.
- **Temperature:** decides crop and livestock suitability, and pest pressure.
- **Topography:** slope and drainage decide cropping and agroforestry design.
- **Climate risk:** flood, drought and cyclone frequency shape enterprise choice.

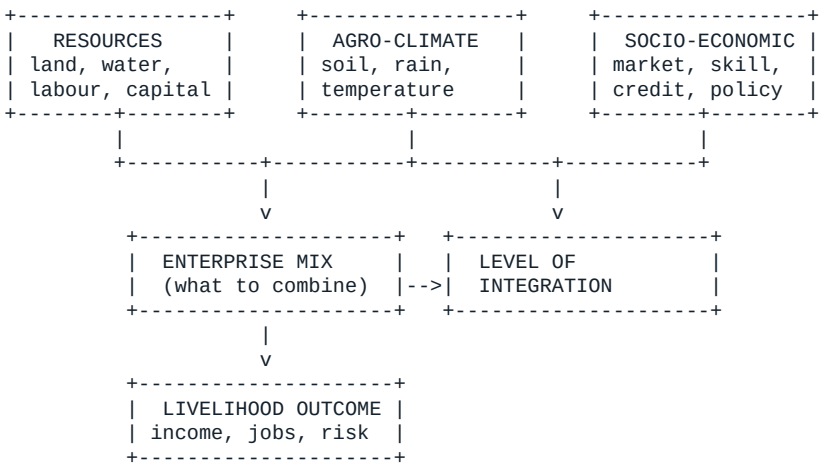
22.4 Socio-economic factors

- **Household needs:** food, cash and security priorities.
- **Skill and knowledge:** ability to manage each enterprise.
- **Market access:** demand and price for each product.
- **Credit and insurance:** access to finance and risk cover.
- **Social institutions:** SHGs, FPOs and cooperatives.
- **Tradition and culture:** food habits and customary practices.
- **Government policy:** subsidies and schemes.

22.5 Factors affecting integration - table

Factor group	Factor	Effect on integration	Example
Resource	Land area	Limits number of enterprises	Small farm keeps two only
Resource	Water	Enables fish and irrigation	Farm pond enables fish
Resource	Capital	Limits investment	Credit enables dairy
Agro-climatic	Rainfall	Decides rainfed or irrigated	Dryland keeps goat
Agro-climatic	Soil	Decides crops and trees	Black soil for cotton
Socio-economic	Market	Decides commercial choice	Milk route nearby
Socio-economic	Skill	Decides feasibility	Trained farmer keeps bees
Socio-economic	Institutions	Enables collective action	FPO markets produce

22.6 Interaction of factors (text diagram)



22.7 Worked example - matching enterprises to factors

A dryland farm in Ramanathapuram has 2 hectares of red soil, low rainfall, no canal water, a small farm pond and access to a goat market.

Enterprise	Suitability	Reason
Paddy	Poor	Needs assured water
Millets and pulses	Good	Drought-tolerant, suits red soil
Goat rearing	Good	Low water need, market nearby
Fish in farm pond	Moderate	Small pond, limited scale
Dairy (1-2 cows)	Moderate	Needs fodder and water

Recommended mix: millets and pulses + goat + a small dairy using pond water and crop residues. Paddy is rejected because water is the limiting factor. This shows how agro-climatic factors decide the enterprise mix.

Labour management in integrated farms

Integration changes the pattern of labour use, which must be planned.

- **Evening out peaks:** dairy and poultry need daily attention, filling the gaps between crop peaks.
- **Family versus hired labour:** family labour lowers cash cost but must be valued for true profit.
- **Skill matching:** livestock and fish need specific skills; training matters.
- **Women's role:** dairy, poultry, mushroom and processing often engage women, raising household income.
- **Labour-saving tools:** milking machines, chaff cutters and sprayers reduce drudgery.

Planning rule: design the enterprise mix so that labour demand is spread across the year. A farm with only paddy has idle labour between seasons; adding dairy and a processing activity uses that labour and raises income.

22.8 Exam focus

Key points

- Integration combines enterprises so they support each other.
- Resource factors are land, water, labour and capital.
- Agro-climatic factors are soil, rainfall, temperature and topography.
- Socio-economic factors are market, skill, credit, institutions and policy.
- The limiting factor decides which enterprise is feasible.
- Matching enterprises to the resource base is the key planning skill.

Short-answer questions with model answers

Q1. Name the three groups of factors affecting enterprise integration. Model answer: Resource availability, agro-climatic conditions and socio-economic factors.

Q2. How does water availability affect integration? Model answer: Water decides whether irrigated crops, fish farming and dairy are feasible; where water is scarce, drought-tolerant crops and goats are preferred over paddy and fish.

Q3. Give two socio-economic factors affecting integration. Model answer: Market access and the farmer's skill and knowledge (also credit, institutions, tradition and policy).

Long-answer questions with model answer outlines

Q1. Explain the factors affecting the integration of enterprises in a farming system. Outline: (i) Meaning of integration; (ii) resource factors; (iii) agro-climatic factors; (iv) socio-economic factors; (v) factors table; (vi) interaction diagram; (vii) worked example; (viii) conclusion.

Q2. Discuss how agro-climatic and socio-economic factors decide the enterprise mix. Outline: (i) Importance of matching; (ii) agro-climatic factors with examples; (iii) socio-economic factors with examples; (iv) worked dryland example; (v) role of extension in matching; (vi) conclusion.

CHAPTER

23. Feasibility of Farming Systems Across Agro-Climatic Zones

23.1 Concept of feasibility

Feasibility means whether a farming system can be successfully established and sustained in a given agro-climatic zone, considering its land, water, soil, climate, labour and market. A system that works in the wetland delta may fail on dryland. The same set of enterprises must therefore be re-designed for each zone.

23.2 Three broad land situations

- **Wetland:** low-lying, assured water, suited to paddy, fish and duck.
- **Garden land:** irrigated upland, suited to crops, horticulture, dairy and poultry.
- **Dryland:** rainfed, low rainfall, suited to millets, pulses, goat and agroforestry.

23.3 Wetland Integrated Farming System model

Situation: low-lying delta land with assured irrigation, as in the Cauvery delta (Thanjavur).

Model for one hectare:

Component	Area / number	Role
Paddy (main crop)	0.80 ha	Staple and cash
Fish in farm pond	0.05 ha	Protein and income
Duck rearing	50 birds	Eggs, meat, pond fertilisation
Dairy	2 cows	Milk and manure
Vegetables on bunds	0.05 ha	Daily cash and nutrition
Coconut on bunds	60 palms	Long-term income
Vermicompost unit	1 unit	Converts dung to manure

Feasibility: high, because water is assured, residues and dung recycle easily, and the pond supports fish and ducks. The delta's canal system and the Grand Anicut make this the classic IFS zone.

23.4 Garden land Integrated Farming System model

Situation: irrigated upland with a well or borewell, as in Coimbatore and Erode.

Model for one hectare:

Component	Area / number	Role
Maize or cotton or turmeric	0.60 ha	Main cash crop
Vegetables	0.15 ha	Frequent income
Fruit trees (mango or sapota)	0.20 ha	Long-term income
Dairy	2 cows	Milk, manure
Poultry	100 birds	Meat and eggs
Vermicompost	1 unit	Manure
Fodder on bunds	0.05 ha	Feed for cattle

Feasibility: high where borewell water is available; the mix of annual and perennial crops plus livestock spreads income and risk.

23.5 Dryland Integrated Farming System model

Situation: rainfed land with low, uncertain rainfall, as in Ramanathapuram or Virudhunagar.

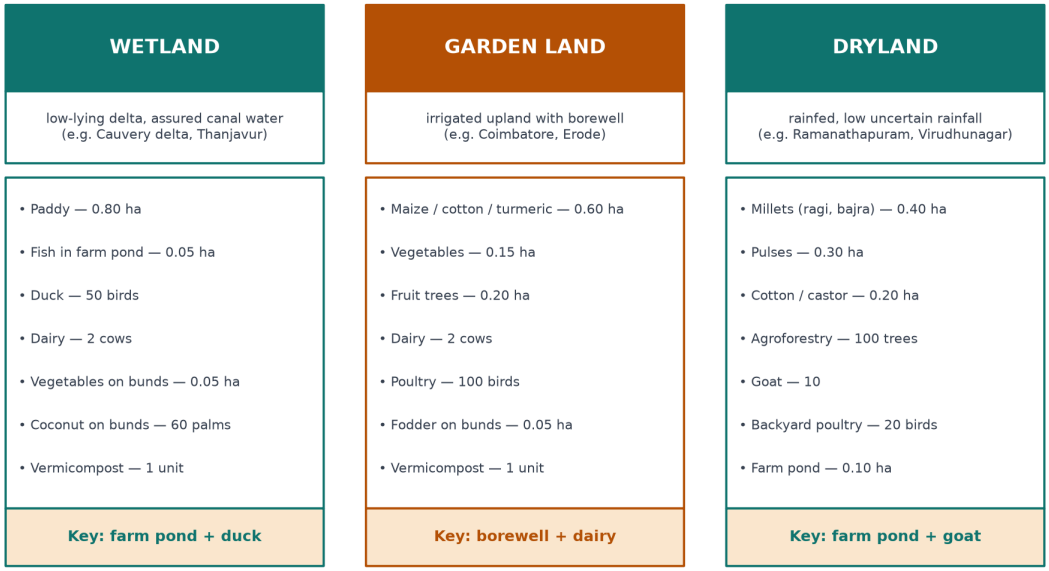


Figure 23.1 IFS models by agro-climatic zone

Integrated farming system models for the wetland, garden land and dryland situations, showing the enterprise mix for each zone.

Model for one hectare:

Component	Area / number	Role
Millets (ragi, bajra)	0.40 ha	Food and fodder
Pulses (black gram, green gram)	0.30 ha	Protein, soil fertility
Cotton or castor	0.20 ha	Cash
Agroforestry trees on borders	100 trees	Fuelwood, fodder, shelter
Goat rearing	10 goats	Quick income, low water
Backyard poultry	20 birds	Eggs, meat
Farm pond for rainwater	0.10 ha	Life-saving irrigation
Vermicompost	1 unit	Manure from waste

Feasibility: good if drought-tolerant crops, livestock and water harvesting are combined. The farm pond is the key, giving protective irrigation at critical stages.

23.6 Comparison of IFS models by agro-climatic zone

Feature	Wetland IFS	Garden land IFS	Dryland IFS
Water	Assured canal	Borewell	Rainfed and pond
Main crop	Paddy	Maize, cotton, turmeric	Millets, pulses
Livestock	Dairy, duck	Dairy, poultry	Goat, backyard poultry
Fisheries	Yes, farm pond	Limited	Small pond
Trees	Coconut on bunds	Mango, sapota	Border agroforestry
Risk	Flood	Water scarcity	Drought
Key component	Farm pond and duck	Borewell and dairy	Farm pond and goat

23.7 Worked example - IFS budgeting for one hectare (wetland)

Enterprise	Gross income (Rs.)	Net income (Rs.)
Paddy (0.80 ha)	80,000	36,000
Vegetables (0.05 ha)	15,000	8,000
Coconut (60 palms, mature)	20,000	14,000
Dairy (2 cows)	1,40,000	38,000
Duck (50 birds)	18,000	7,000
Fish (0.05 ha pond)	12,000	5,000
Total	**2,85,000**	**1,08,000**

Net income of Rs. 1,08,000 from one hectare compares with about Rs. 45,000 from paddy alone - a gain of about 140 per cent - because income comes from several enterprises and nutrients recycle to cut cost.

23.8 Worked example - IFS budgeting for one hectare (dryland)

Enterprise	Gross income (Rs.)	Net income (Rs.)
Millets (0.40 ha)	24,000	12,000
Pulses (0.30 ha)	18,000	10,000
Cotton (0.20 ha)	20,000	9,000
Goat (10 goats)	40,000	15,000

Enterprise	Gross income (Rs.)	Net income (Rs.)
Backyard poultry (20 birds)	10,000	4,000
Agroforestry (fuelwood, fodder)	8,000	5,000
Total	**1,20,000**	**55,000**

Net income of Rs. 55,000 from one hectare of dryland is a strong outcome for a rainfed farm, achieved mainly through livestock and agroforestry, which buffer crop failure.

Resource budgeting for an IFS

A good IFS plan budgets every key resource, not just money.

- **Land budget:** how much area for each enterprise, including bunds and pond.
- **Water budget:** total water available and requirement of each enterprise.
- **Labour budget:** person-days needed each month versus available.
- **Nutrient budget:** nutrients supplied by manure versus removed by crops.
- **Feed budget:** fodder and concentrate available versus required by livestock.
- **Cash budget:** monthly inflow and outflow to avoid a cash crunch.

Example check: if two cows need fodder that the farm cannot grow, the shortfall must be purchased, and this cost must appear in the budget. Resource budgeting prevents the common mistake of adding an enterprise that the farm's land, water or labour cannot support.

23.9 Exam focus

Key points

- Feasibility means whether a system can be sustained in a given agro-climatic zone.
- The three land situations are wetland, garden land and dryland.
- Wetland IFS suits the delta with paddy, fish, duck and dairy.
- Garden land IFS combines annual crops, horticulture, dairy and poultry.
- Dryland IFS relies on millets, pulses, goat, poultry, agroforestry and a farm pond.
- The farm pond and livestock are the key stabilizing components.

Short-answer questions with model answers

Q1. What is meant by feasibility of a farming system? Model answer: Feasibility is whether a farming system can be successfully established and sustained in a given agro-climatic zone, considering its land, water, soil, climate, labour and market.

Q2. Name the key component of a dryland IFS. Model answer: The farm pond for rainwater harvesting is the key component, providing life-saving irrigation, along with drought-tolerant crops and goat rearing.

Q3. Which IFS model suits the Cauvery delta and why? Model answer: The wetland IFS model suits the Cauvery delta because assured canal irrigation supports paddy, a farm pond supports fish and duck, and dairy provides manure, so enterprises recycle easily.

Long-answer questions with model answer outlines

Q1. Describe the wetland, garden land and dryland IFS models with budgets. Outline: (i) Meaning of feasibility; (ii) wetland model with components and budget; (iii) garden land model; (iv) dryland model with budget; (v) comparison table; (vi) conclusion.

Q2. Explain how farming systems are designed for different agro-climatic zones. Outline: (i) Land situations; (ii) role of water; (iii) wetland design; (iv) garden land design; (v) dryland design; (vi) matching enterprises to resources; (vii) conclusion.

CHAPTER

24. Future Directions in Integrated Farming Systems

24.1 Climate-smart agriculture

Climate-smart agriculture (CSA) is farming that (a) sustainably increases productivity and income, (b) builds resilience to climate change, and (c) reduces greenhouse gas emissions where possible. CSA is the main future direction for IFS.

Key practices:

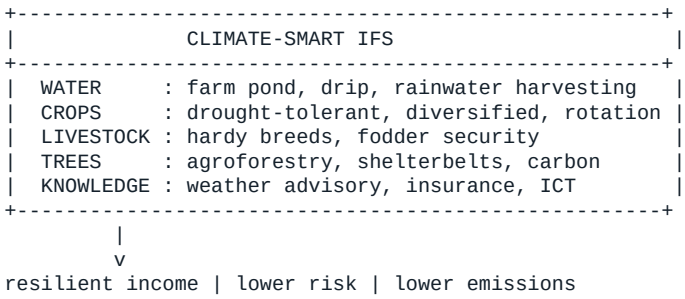
- **Climate-resilient varieties:** drought-, flood- and salinity-tolerant crops.
- **Water harvesting and micro-irrigation:** farm ponds, drip and sprinkler.
- **Conservation agriculture:** minimum tillage, residue retention, rotation.
- **Agroforestry and shelterbelts:** shade, wind protection, carbon storage.
- **Integrated nutrient and pest management:** lower chemical use.
- **Crop diversification and adjusted sowing dates.**
- **Livestock and fisheries integration** for income stability.
- **Weather-based advisories and insurance.**

24.2 Resilient farming systems

A **resilient farming system** can absorb shocks and still function. Resilience comes from:

- **Diversity** of crops and enterprises.
- **Redundancy** - more than one source of income and food.
- **Recycling** - nutrients and water reused within the farm.
- **Adaptability** - the ability to change crops or enterprises as conditions change.
- **Social capital** - SHGs, FPOs and community support.

24.3 Climate-smart IFS (text diagram)



24.4 Case studies

Case study 1 - Wetland IFS in the Cauvery delta (Tamil Nadu)

A farm household in Thanjavur district adopted a wetland IFS with paddy, a farm pond with fish, duck rearing, two cows and coconut on bunds. Over a few years the household reported a marked rise in net income and a steadier cash flow, because milk and eggs gave daily income while paddy gave seasonal income. Manure from the cattle and ducks reduced fertiliser purchases. The system withstood a poor paddy season because livestock income covered the shortfall.

Case study 2 - Dryland IFS in southern Tamil Nadu

A dryland household in Ramanathapuram adopted millets, pulses, a farm pond, ten goats and backyard poultry, with agroforestry on the borders. The farm pond gave protective irrigation at flowering, and the goats provided income during drought years. The system reduced the household's dependence on migration for wage work.

Case study 3 - NICRA demonstrations

Under the **National Innovations in Climate Resilient Agriculture (NICRA)** programme, KVKs across India demonstrate climate-resilient technologies - drought-tolerant varieties, water harvesting, and integrated farming - on farmers' fields. These demonstrations show that combining several enterprises with climate-resilient practices stabilises income in risky environments.

Note on figures: the exact income gains in case studies vary by year, location and management. In an examination, describe the direction and reasons for the gain rather than quoting a single unverified number.

24.5 Future directions table

Direction	What it means	Example
Climate-smart IFS	Resilient, low-emission integration	Farm pond with millets and goat
Digital IFS	ICT in planning and marketing	Mobile advisory and e-market
Precision IFS	Optimal use of inputs	Drip with fertigation
Circular IFS	Full recycling of waste	Biogas and compost
Market-linked IFS	Value chain integration	FPO with processing unit

Measuring the resilience of farming systems

Resilience can be assessed, not merely assumed.

- **Diversity index:** number and evenness of income sources.
- **Income stability:** year-to-year variation in net income.
- **Recovery time:** how quickly income returns to normal after a shock.
- **Resource trend:** whether soil and water are improving or degrading.
- **Institutional strength:** membership in SHGs, FPOs and insurance coverage.

A household that scores high on diversity, low on income variation, quick on recovery and positive on resource trend is resilient. Such scoring helps compare farms and target support to the least resilient, guiding the design of climate-smart farming systems.

24.6 Exam focus

Key points

- Climate-smart agriculture raises productivity, builds resilience and cuts emissions.
- Key practices are resilient varieties, water harvesting, conservation agriculture and agroforestry.
- Resilience comes from diversity, redundancy, recycling and adaptability.
- NICRA demonstrates climate-resilient technologies through KVKs.
- IFS case studies show more stable income from several enterprises.
- Future IFS will be climate-smart, digital, precise, circular and market-linked.

Short-answer questions with model answers

Q1. Define climate-smart agriculture. Model answer: Climate-smart agriculture is farming that sustainably increases productivity and income, builds resilience to climate change and reduces greenhouse gas emissions where possible.

Q2. What is NICRA? Model answer: NICRA (National Innovations in Climate Resilient Agriculture) is a programme that develops and demonstrates climate-resilient farming technologies on farmers' fields through Krishi Vigyan Kendras.

Q3. State two sources of resilience in a farming system. Model answer: (i) Diversity of crops and enterprises; (ii) recycling of nutrients and water within the farm.

Long-answer questions with model answer outlines

Q1. Explain climate-smart agriculture and its role in future integrated farming systems. Outline: (i) Definition and three goals; (ii) key practices; (iii) resilient farming systems; (iv) climate-smart IFS diagram; (v) NICRA and KVK role; (vi) case studies; (vii) conclusion.

Q2. Discuss future directions in integrated farming systems with case studies. Outline: (i) Introduction; (ii) climate-smart IFS; (iii) digital and precision IFS; (iv) circular and market-linked IFS; (v) case studies - wetland, dryland, NICRA; (vi) conclusion.

CHAPTER

25. Overview of Commercial Farming-Based Livelihood Models in India

25.1 Definition and scope

A **commercial farming-based livelihood model** is an organised system in which farming is carried out primarily for the market, with modern inputs, technology and management, to generate profit and employment. Such models may be run by individual farmers, farmer groups (FPOs), companies or cooperatives.

Scope: commercial models cover high-value crops, horticulture, dairy, poultry, fisheries, agro-processing, contract farming and agri-export.

25.2 Features of commercial models

- Market-oriented production, not subsistence.
- High investment and modern technology.
- Large scale or collective scale through FPOs.
- Emphasis on quality, grading and branding.
- Integration with value chains and processors.
- Formal credit, insurance and contracts.
- Employment of hired labour.

25.3 Examples of commercial models

- **Contract farming:** the farmer grows for a company that supplies inputs and buys the produce at a pre-agreed price (poultry, vegetables, medicinal plants).
- **FPO-based marketing:** collective production, grading and sale.
- **Dairy cooperatives:** village milk collection and processing, as in the Anand pattern.
- **Integrated agro-processing:** production plus on-site processing.
- **Agri-export clusters:** production for export markets.

25.4 Role of ICAR

The **Indian Council of Agricultural Research (ICAR)** is the apex body for agricultural research and education in India.

- Develops improved crop varieties and technologies.
- Runs a network of research institutes (IARI, IIHR, NDRI, CIFA, etc.).
- Coordinates the **Krishi Vigyan Kendra (KVK)** network for technology transfer.
- Promotes integrated farming systems and climate-resilient agriculture.
- Provides the scientific base for extension and policy.

25.5 Role of NABARD

The **National Bank for Agriculture and Rural Development (NABARD)** is the apex development bank for agriculture and rural development.

- Provides refinance to banks for agriculture and rural lending.
- Promotes and supports **Self-Help Groups (SHGs)** and microfinance.
- Promotes **Farmer Producer Organizations (FPOs)**.
- Supports rural infrastructure through the **Rural Infrastructure Development Fund (RIDF)**.
- Runs the **Agri-Clinics and Agri-Business Centres (ACABC)** scheme.
- Supports watershed development, tribal development and rural innovation.

25.6 ICAR and NABARD - role comparison table

Basis	ICAR	NABARD
Type	Research and education council	Development bank
Main role	Technology and knowledge	Credit and institutional support
Key instruments	Varieties, KVKs, institutes	Refinance, SHGs, FPOs, RIDF
Focus	Production technology	Finance and rural livelihoods
Tamil Nadu link	TNAU, KVKs	SHG and FPO funding

25.7 Worked example - a commercial FPO model

A banana FPO in Theni district aggregates produce from 300 member farmers.

Item	Detail
Members	300 small farmers
Annual turnover	About Rs. 5 crore (illustrative)
Functions	Input supply, grading, ripening, marketing
Benefit	Members get a higher price and steady buyers
Support	NABARD and SFAC grant, bank credit

By aggregating, the FPO negotiates better prices, buys inputs in bulk at lower cost and owns a ripening chamber, capturing value that individual farmers could not.

Business models and farmer collectives

Commercial farming works through several business models.

- **Individual commercial farm:** one household operating at scale.
- **FPO model:** farmers aggregate, market and sometimes process collectively.
- **Cooperative model:** members own the processing and marketing, as in dairy.
- **Contract farming model:** a company links farmers to a guaranteed buyer.
- **Cluster model:** a group of farms in one area specialises in one product for a buyer or export.

Choosing a model: depends on the crop, the market, the farmers' capital and their willingness to work together. The cooperative and FPO models are especially suited to smallholders because they create scale without the need to enlarge individual holdings.

25.8 Exam focus

Key points

- Commercial models produce for the market with modern inputs and management.
- Examples are contract farming, FPO marketing, dairy cooperatives and agro-processing.
- ICAR is the apex research and education body and runs the KVK network.
- NABARD is the apex development bank for agriculture and rural development.
- NABARD supports SHGs, FPOs, RIDF and the ACABC scheme.
- Collective models help smallholders reach commercial scale.

Short-answer questions with model answers

Q1. What is a commercial farming-based livelihood model? Model answer: It is an organised system in which farming is carried out primarily for the market, using modern inputs, technology and management, to generate profit and employment.

Q2. What is the role of ICAR? Model answer: ICAR is the apex body for agricultural research and education; it develops improved technologies, runs research institutes and coordinates the KVK network for technology transfer.

Q3. Name two functions of NABARD. Model answer: (i) Providing refinance to banks for agriculture and rural lending; (ii) promoting SHGs and FPOs (also RIDF and the ACABC scheme).

Long-answer questions with model answer outlines

Q1. Explain commercial farming-based livelihood models in India with examples. Outline: (i) Definition and scope; (ii) features; (iii) examples - contract farming, FPO, dairy cooperative, agro-processing; (iv) role of ICAR and NABARD; (v) worked FPO example; (vi) conclusion.

Q2. Discuss the role of ICAR and NABARD in promoting farming-based livelihoods. Outline: (i) Introduction; (ii) ICAR - research, education, KVK; (iii) NABARD - refinance, SHG, FPO, RIDF, ACABC; (iv) comparison table; (v) examples from Tamil Nadu; (vi) conclusion.

CHAPTER

26. NABARD's Initiatives in Rural Livelihoods and Agriculture

26.1 Introduction

NABARD, established in 1982, is the apex development bank for agriculture and rural development in India. Its mission is to promote sustainable and equitable agriculture and rural prosperity through credit and institutional development.

26.2 Self-Help Groups (SHGs)

- A **Self-Help Group** is a small group of 10-20 members, usually women, who save regularly and lend to each other.
- **SHG-Bank Linkage Programme:** after a period of successful saving, the group is linked to a bank for credit.
- **Benefits:** thrift habit, small loans without collateral, women's empowerment, group solidarity.
- **NABARD's role:** promotion, training, refinance and linkage support.
- Tamil Nadu has a large number of women's SHGs supported under this programme.

SHG working cycle:

```
Form group (10-20) -> Save regularly -> Lend internally
      -> Build record -> Bank linkage -> Larger loans
      -> Start enterprise -> Repay -> Grow
```

26.3 Farmer Producer Organizations (FPOs)

- An **FPO** is a collective of farmers, usually a producer company or cooperative.
- Helps smallholders aggregate produce, buy inputs in bulk and market better.
- NABARD provides credit, grant support and capacity building.
- The Government's **10,000 FPOs scheme** (implemented by NABARD, SFAC and others) aims to create and support new FPOs.
- Tamil Nadu has FPOs in banana, mango, coconut, millets and vegetables.

26.4 Agri-Clinics and Agri-Business Centres (ACABC)

- A scheme to help agriculture graduates set up their own agri-ventures.
- Provides **training and subsidy** to graduates to start agri-clinics (advisory) and agri-business centres (input supply, services).
- Creates rural employment and brings technical advice to farmers.
- Combines NABARD support with training by designated institutions.

26.5 Other NABARD initiatives

- **Rural Infrastructure Development Fund (RIDF):** finances rural roads, irrigation, warehouses and markets.
- **Watershed development:** soil and water conservation in rainfed areas.
- **Tribal Development Fund:** for tribal livelihoods.
- **Micro-enterprise development** and skill training.
- **Producer Organization Development Fund.**
- **Farm sector refinance** for crop loans, dairy, fisheries and agro-processing.

26.6 NABARD initiatives table

Initiative	Purpose	Benefit
SHG-Bank Linkage	Credit to the poor through groups	Thrift, small loans, empowerment
FPO promotion	Collective farming and marketing	Scale, better prices
ACABC	Agri-ventures by graduates	Employment, advisory services

Initiative	Purpose	Benefit
RIDF	Rural infrastructure	Roads, irrigation, storage
Watershed development	Soil and water conservation	Higher rainfed productivity
Refinance	Funding for farm and allied loans	Cheaper credit

26.7 Worked example - SHG enterprise with bank linkage

A women's SHG of 15 members saves Rs. 200 per month each for a year and receives a bank loan.

Item	Amount (Rs.)
Annual group savings (15 x 200 x 12)	36,000
Bank loan (about 4 times savings)	1,44,000
Total fund	**1,80,000**
Enterprise chosen	Dairy (4 cows)
Annual net income from dairy	60,000
Loan repayment over 3 years	48,000 per year

The group repays the loan from the dairy income and retains the balance as a group fund, which is used for further enterprises. This is the standard SHG model that NABARD promotes.

Microfinance and financial inclusion

Access to finance is the foundation of enterprise growth.

- **Micro-credit:** small loans to SHG members without collateral.
- **SHG-Bank Linkage:** group savings convert into bank credit.
- **Joint liability:** group members guarantee each other's loans, reducing default.
- **Financial literacy:** training in budgeting, saving and borrowing.
- **Digital finance:** bank accounts, UPI and mobile banking improve access.

Impact: microfinance funds small enterprises - dairy, poultry, mushroom, petty trade - and builds a saving habit. Its limitations are small loan size, limited suitability for large investments, and the risk of over-borrowing if multiple groups lend to the same household. Responsible lending and financial literacy are essential.

26.8 Exam focus

Key points

- NABARD is the apex development bank for agriculture and rural development.
- The SHG-Bank Linkage Programme gives group-based credit to the poor.
- FPOs help farmers aggregate and market collectively.
- The ACABC scheme supports agri-ventures by agriculture graduates.
- RIDF finances rural infrastructure; watershed development conserves soil and water.
- The 10,000 FPOs scheme is a major current initiative.

Short-answer questions with model answers

Q1. What is an SHG? Model answer: A Self-Help Group is a small group of 10-20 members, usually women, who save regularly, lend to each other and, after building a record, are linked to a bank for credit.

Q2. What is the ACABC scheme? Model answer: The Agri-Clinics and Agri-Business Centres scheme gives training and subsidy to agriculture graduates to set up agri-clinics for advisory services and agri-business centres for input supply and services.

Q3. Name two NABARD initiatives for rural livelihoods. Model answer: The SHG-Bank Linkage Programme and FPO promotion (also RIDF, watershed development and ACABC).

Long-answer questions with model answer outlines

Q1. Explain NABARD's initiatives in rural livelihoods and agriculture. Outline: (i) Introduction and mandate; (ii) SHG-Bank Linkage; (iii) FPOs and the 10,000 FPOs scheme; (iv) ACABC; (v) RIDF and watershed development; (vi) initiatives table; (vii) worked SHG example; (viii) conclusion.

Q2. Discuss how SHGs and FPOs promote farming-based livelihoods. Outline: (i) Concept of collective institutions; (ii) SHG model and bank linkage; (iii) FPO model; (iv) benefits - credit, scale, prices, empowerment; (v) worked example; (vi) challenges; (vii) conclusion.

CHAPTER

27. Climate-Resilient Farming, NGOs and Government Programmes

27.1 Climate-resilient farming practices under NICRA

NICRA - the **National Innovations in Climate Resilient Agriculture** - is a programme of the Indian Council of Agricultural Research (ICAR) launched to make Indian agriculture resilient to climate change. It works through a network of KVKs and research institutes.

Four pillars of NICRA:

1. **Natural resource management:** water harvesting, soil conservation, agroforestry.
2. **Climate-resilient crop and livestock practices:** tolerant varieties, contingency planning.
3. **Village-level demonstrations:** proven technologies shown on farmers' fields.
4. **Strategic research:** studying climate impact and adaptation.

Key climate-resilient practices:

- **Drought-tolerant and short-duration varieties** of paddy, millets and pulses.
- **Water harvesting structures:** farm ponds and check dams for protective irrigation.
- **Micro-irrigation:** drip and sprinkler to save water.
- **Conservation agriculture:** minimum tillage and residue retention.
- **Agroforestry and shelterbelts** to reduce heat and wind stress.
- **Crop diversification and intercropping** for risk spreading.
- **Climate-resilient livestock:** hardy breeds, fodder banks, shelter sheds.
- **Weather-based advisories and index insurance.**

27.2 NGO-led livelihood enterprises - BAIF Development Research Foundation

BAIF Development Research Foundation is a leading non-governmental organisation working on rural livelihoods, livestock development and natural resource management, mainly in western and central India, with programmes in several states.

Major approaches of BAIF:

- **Cattle and livestock development:** artificial insemination, improved breeds, fodder development to raise milk income.
- **Watershed development:** soil and water conservation to raise rainfed productivity.
- **Wadi programme:** orchard-based tribal development (mango, cashew) on degraded land, giving long-term income.
- **Women's self-help groups** and micro-enterprise promotion.
- **Sustainable agriculture** and agroforestry.

Lessons from NGO models: local participation, long-term engagement, blending technical support with social mobilisation, and linking farmers to markets are the keys to success. NGOs often fill gaps that government extension cannot reach.

27.3 Government programmes - RKVY

Rashtriya Krishi Vikas Yojana (RKVY) is a central scheme that gives grants to states for agriculture and allied development.

- States prepare agriculture development plans and receive funds to implement them.
- **RKVY-RAFTAAR** supports agri-entrepreneurship and start-ups through incubation.
- Focus on infrastructure, value addition, and innovation.
- It encourages states to increase their own spending on agriculture.

27.4 Government programmes - eNAM

eNAM - the **National Agriculture Market** - is an electronic trading platform for agricultural commodities.

- Links wholesale markets (mandis) across states on one online platform.
- Farmers and traders can trade transparently, improving price discovery.
- Reduces the role of intermediaries and improves the farmer's share.
- Provides online payment and quality assaying.
- Tamil Nadu has regulated markets linked to the platform.

27.5 Programme summary table

Programme	Type	Purpose	Benefit to farmers
NICRA	ICAR programme	Climate resilience	Tolerant varieties, water harvesting
BAIF	NGO	Livestock and watershed livelihoods	Milk income, orchards, SHGs
RKVY	Central scheme	State agri-development grants	Infrastructure, value addition
RKVY-RAFTAAR	Central scheme	Agri start-up incubation	Entrepreneurship support
eNAM	Central platform	Online market trading	Transparent prices

27.6 Worked example - impact of a climate-resilient practice

A rainfed farmer in a drought-prone block adopts a short-duration millet variety and a farm pond.

Situation	Yield (kg/ha)	Price (Rs./kg)	Gross income (Rs.)
Before (long-duration variety, no pond)	900	20	18,000
After (short-duration variety + pond)	1,400	22	30,800
Gain	**500**	**2**	**12,800**

The short-duration variety escapes terminal drought and the farm pond provides protective irrigation at flowering, raising both yield and income. The gain of about 71 per cent shows how a modest climate-resilient practice can stabilise a rainfed livelihood.

Contingency crop planning

When the monsoon fails or is delayed, a contingency plan protects income.

- **Delayed sowing:** switch to a shorter-duration variety of the same crop.
- **Early-season drought:** replace with a drought-tolerant crop such as a millet or a short pulse.
- **Mid-season drought:** reduce plant population and shift to protective irrigation.
- **Excess rain or flood:** choose submergence-tolerant varieties and improve drainage.
- **Terminal drought:** harvest for fodder if grain yield fails.
- **Post-failure:** use the land for fodder, a catch crop or a fodder-livestock enterprise.

Institutional support: NICRA and KVKs publish district-level contingency plans that list the crop and variety to sow for each rainfall situation. A household with a contingency plan loses far less in a bad year.

27.7 Exam focus

Key points

- NICRA is an ICAR programme for climate-resilient agriculture.
- NICRA works through KVKs and has four pillars, including demonstrations and research.

- Climate-resilient practices include tolerant varieties, water harvesting and conservation agriculture.
- BAIF is a leading NGO working on livestock, watershed and tribal livelihoods.
- RKVY gives development grants to states; RKVY-RAFTAAR supports agri start-ups.
- eNAM is an electronic national market platform for transparent trading.

Short-answer questions with model answers

Q1. What is NICRA? Model answer: NICRA (National Innovations in Climate Resilient Agriculture) is an ICAR programme that develops and demonstrates climate-resilient technologies, such as tolerant varieties and water harvesting, through KVKs to make agriculture resilient to climate change.

Q2. Name two approaches of BAIF. Model answer: (i) Cattle and livestock development to raise milk income; (ii) the Wadi programme of orchard-based tribal development (also watershed development and SHG promotion).

Q3. What is eNAM? Model answer: eNAM is the National Agriculture Market, an electronic trading platform that links wholesale markets across states for transparent price discovery and online trade in agricultural commodities.

Long-answer questions with model answer outlines

Q1. Explain climate-resilient farming practices under NICRA. Outline: (i) Meaning of climate resilience; (ii) NICRA and its four pillars; (iii) key practices; (iv) role of KVKs; (v) worked example; (vi) conclusion.

Q2. Discuss the role of NGOs and government programmes in promoting climate-resilient livelihoods. Outline: (i) Introduction; (ii) NGO model - BAIF; (iii) RKVY and RKVY-RAFTAAR; (iv) eNAM; (v) programme table; (vi) worked example; (vii) conclusion.

CHAPTER

28. Risks in Farming and Success Factors

28.1 Meaning of risk in farming

Risk is the possibility of loss or reduced returns from a farming activity because of uncertain events. Farming is inherently risky because it depends on weather, biology, markets and policy, all of which are uncertain.

28.2 Types of risk

Economic risk

- Price volatility of inputs and outputs.
- Cost escalation and inflation.
- Market demand changes and competition.
- Credit and interest-rate risk.

Environmental risk

- Drought, flood, cyclone and heat waves.
- Pest and disease outbreaks.
- Soil degradation and water scarcity.
- Climate change.

Technological risk

- Failure of a new variety or technology.
- Breakdown of machinery or equipment.
- Poor adaptation of technology to local conditions.
- Obsolescence.

Social risk

- Labour shortage and wage disputes.
- Social conflict and theft.
- Health and family problems.
- Policy and regulatory changes.

28.3 Risk table with management strategies

Risk type	Examples	Management strategies
Economic	Price crash, cost rise	Diversification, MSP, contract farming, futures, storage
Environmental	Drought, flood, pest	Insurance, tolerant varieties, irrigation, IPM, agroforestry
Technological	Machine failure, variety failure	Training, maintenance, proven technology, backups
Social	Labour shortage, conflict	Labour-saving tools, community action, timely wages
Institutional	Policy change, credit denial	FPO membership, formal credit, awareness

28.4 Risk management tools

- **Diversification** of crops and enterprises.
- **Crop and livestock insurance** (PMFBY, livestock insurance).
- **Irrigation** to reduce drought risk.
- **Contract farming** for assured price.
- **Storage and warehousing** to sell at better prices.
- **Savings and formal credit** to absorb shocks.

- **Weather advisories** for timely action.
- **FPO membership** for collective risk-sharing.

28.5 Success factors in farming-based livelihood enterprises

- **Resource management:** using land, water, labour and capital efficiently.
- **Innovation:** adopting new technology and practices.
- **Market access:** reliable buyers and better prices.
- **Community support:** SHGs, FPOs and cooperation.
- **Skill and knowledge:** training and advisory.
- **Financial discipline:** timely credit and repayment.
- **Diversification:** several income sources.
- **Persistence and planning:** long-term commitment.

28.6 Success factors table

Factor	What it means	Example
Resource management	Efficient use of inputs	IFS recycling, water budgeting
Innovation	Adopting new technology	Drip, improved breeds
Market access	Reliable buyer and price	FPO contract with retailer
Community support	Collective action	SHG, FPO
Skill	Technical and managerial ability	KVK training
Financial discipline	Credit and savings management	KCC, timely repayment

28.7 Worked example - risk exposure scoring

A farmer assesses the risk exposure of three enterprises on a scale of 1 (low) to 5 (high).

Enterprise	Economic	Environmental	Technological	Total (max 15)
Paddy	3	4	2	9
Dairy	2	3	3	8
Vegetables	4	3	2	9
Backyard poultry	2	2	2	6

Interpretation: backyard poultry has the lowest combined risk and paddy and vegetables the highest. A farmer seeking stability would emphasise poultry and dairy, while a farmer seeking high returns might accept the higher risk of vegetables, using insurance and diversification to manage it.

Insurance products for farming

Insurance converts a large, uncertain loss into a small, certain premium.

- **Crop insurance (PMFBY):** covers yield loss from drought, flood, pest and disease.
- **Weather-based insurance:** pays automatically when rainfall or temperature crosses a trigger.
- **Livestock insurance:** covers death of cattle, buffalo and other animals.
- **Poultry insurance:** covers bird loss from disease and disaster.
- **Crop loan insurance:** protects the borrower against crop failure.

Key ideas: premium is shared between farmer and government; claims should be settled quickly; index products avoid the cost of individual loss assessment. Insurance is most useful when combined with risk-reducing practices such as irrigation and diversification, since insurance compensates loss but does not prevent it.

28.8 Exam focus

Key points

- Risk is the possibility of loss from uncertain events in farming.
- The four types are economic, environmental, technological and social.
- Risk management tools include diversification, insurance, irrigation and contract farming.
- Success factors are resource management, innovation, market access, community support, skill and financial discipline.
- Risk scoring helps in enterprise selection.
- Diversification is the most universal risk-reduction tool.

Short-answer questions with model answers

Q1. What is risk in farming and name its types. Model answer: Risk is the possibility of loss or reduced returns from uncertain events. Its types are economic, environmental, technological and social risk.

Q2. Name three tools of risk management. Model answer: Crop insurance, diversification of enterprises and irrigation (also contract farming, storage and savings).

Q3. State two success factors of a farming enterprise. Model answer: (i) Market access with reliable buyers; (ii) resource management, using land, water, labour and capital efficiently.

Long-answer questions with model answer outlines

Q1. Explain the types of risk in farming and their management strategies. Outline: (i) Meaning of risk; (ii) economic risk; (iii) environmental risk; (iv) technological risk; (v) social risk; (vi) management tools; (vii) risk table; (viii) conclusion.

Q2. Discuss the success factors of farming-based livelihood enterprises. Outline: (i) Meaning of success factors; (ii) resource management; (iii) innovation; (iv) market access; (v) community support; (vi) skill and financial discipline; (vii) worked example; (viii) conclusion.

CHAPTER

29. Central and State Government Schemes for Farming-Based Livelihoods

29.1 Role of government schemes

Government schemes provide income support, credit, insurance, irrigation, market access and risk protection to farmers. They are designed at the central and state levels and implemented through departments, banks, KVKs and local bodies.

29.2 Central government schemes

Scheme	Purpose
PM-KISAN	Direct income support of Rs. 6,000 per year to eligible farmers
PMFBY	Crop insurance against yield loss from natural calamities
Kisan Credit Card	Concessional credit for crop and allied needs
PM Krishi Sinchayee Yojana	Irrigation - "Har Khet Ko Pani" and "More Crop Per Drop"
Soil Health Card	Soil-test based fertiliser recommendations
e-NAM	Online national agricultural market
National Food Security Mission	Raise production of rice, wheat and pulses
National Mission on Sustainable Agriculture	Climate-resilient and sustainable practices
Paramparagat Krishi Vikas Yojana	Organic and natural farming support
Agriculture Infrastructure Fund	Post-harvest and community asset financing
National Beekeeping and Honey Mission	Support for beekeeping
PM Kisan Maan Dhan Yojana	Old-age pension for small and marginal farmers
Formation of 10,000 FPOs	Support for farmer producer organizations

29.3 State government schemes (Tamil Nadu examples)

Scheme / Programme	Purpose
Uzhavan App	Mobile app for farm advisory, subsidies and weather
Free crop insurance support	State support to widen insurance cover
Micro-irrigation subsidy	Subsidy for drip and sprinkler systems
Farm mechanisation subsidy	Support for tractors, machinery and custom hiring
Millet promotion mission	Revival of millets for nutrition and income
Collective farming / FPO support	Group farming and marketing
Horticulture development schemes	Support for fruits, vegetables and flowers
Integrated farming system promotion	Support for IFS adoption
Kalaignar All Village Integrated Agricultural Development Programme	Village-level agricultural development

Note on figures: subsidy percentages and eligibility conditions change with each state budget. In an examination, describe the purpose of a scheme and its general form of support rather than quoting a specific percentage that may be outdated.

29.4 Central versus state schemes

Basis	Central schemes	State schemes
Funding	Union government	State government (with central share)
Coverage	All India	State-specific
Examples	PM-KISAN, PMFBY, e-NAM	Uzhavan, state IFS and millet missions
Focus	Broad national priorities	Local crops and needs
Implementation	Through states and agencies	Through state departments and KVKs

29.5 Worked example - a farmer's scheme portfolio

A small farmer in Tamil Nadu with one hectare and two cows accesses several schemes.

Scheme	Benefit to the farmer
PM-KISAN	Rs. 6,000 per year income support
Kisan Credit Card	Timely credit for inputs at low interest
PMFBY	Insurance against crop loss
Micro-irrigation subsidy	Drip system at reduced cost
Uzhavan App	Weather and advisory information
Dairy support	Cattle and fodder assistance

Together these schemes reduce cost, provide credit, insure against loss and give information - improving both income and stability. Using schemes in combination, rather than one at a time, gives the best result.

Scheme access and digital governance

Getting the benefit of a scheme often depends on knowing about it and applying correctly.

- **Awareness:** through KVKs, panchayats, SHGs and mobile advisories.
- **Registration:** Aadhaar-linked enrolment for direct benefit transfer.
- **Direct Benefit Transfer (DBT):** money credited straight to the bank account.
- **Digital records:** land records and soil health cards available online.
- **Grievance redressal:** helplines and portals for complaints.

Tamil Nadu practice: the Uzhavan app and the e-Sevai centres help farmers access subsidy and scheme information. Digital governance reduces leakage and delay, but requires connectivity, digital literacy and accurate land records to work fairly for all farmers, including small and marginal households.

29.6 Exam focus

Key points

- Government schemes give income support, credit, insurance, irrigation and market access.
- PM-KISAN gives direct income support of Rs. 6,000 per year.
- PMFBY insures crops against natural calamities.
- e-NAM is the national online market platform.
- Tamil Nadu has state schemes such as Uzhavan, millet promotion and IFS support.
- Combining several schemes improves income and stability.

Short-answer questions with model answers

Q1. What is PMFBY? Model answer: PMFBY (Pradhan Mantri Fasal Bima Yojana) is a crop insurance scheme that compensates farmers for yield loss caused by natural calamities such as drought, flood, pests and disease.

Q2. Name two Tamil Nadu state schemes for farmers. Model answer: The Uzhavan App for advisory and subsidies, and the millet promotion mission for nutrition and income (also micro-irrigation subsidy and IFS

support).

Q3. How does the Kisan Credit Card help farmers? Model answer: It provides timely institutional credit at concessional interest for crop and allied needs, reducing dependence on costly informal loans.

Long-answer questions with model answer outlines

Q1. Describe the central and state government schemes for promoting farming-based livelihoods. Outline: (i) Role of schemes; (ii) central schemes with purposes; (iii) Tamil Nadu state schemes; (iv) central versus state comparison; (v) worked scheme portfolio; (vi) conclusion.

Q2. Explain how government schemes support farm income and risk management. Outline: (i) Income support - PM-KISAN; (ii) credit - KCC; (iii) insurance - PMFBY; (iv) irrigation and inputs; (v) market - e-NAM; (vi) worked example; (vii) conclusion.

CHAPTER

30. Role of Public and Private Organizations

30.1 Public organizations

Public organizations are government bodies that provide research, extension, credit, regulation and welfare to agriculture.

- **ICAR and agricultural universities (TNAU):** research and education.
- **Krishi Vigyan Kendras (KVKs):** technology transfer and training.
- **State Departments of Agriculture:** implementation of schemes and extension.
- **NABARD:** development finance and institutional support.
- **Commodity boards:** (Coffee Board, Spices Board, Coconut Development Board).
- **Cooperatives:** milk, credit and marketing cooperatives.
- **Regulated markets and eNAM:** marketing infrastructure.

30.2 Private organizations

Private organizations are firms, companies and individuals that participate in agriculture for profit or service.

- **Input companies:** seed, fertiliser, pesticide and machinery firms.
- **Agro-processing companies:** buyers and processors of farm produce.
- **Contract farming firms:** supply inputs and buy output.
- **Retailers and e-commerce platforms:** sell farm produce to consumers.
- **Banks and microfinance institutions:** private credit.
- **Agri-tech start-ups:** digital advisory, drones, sensors and market platforms.
- **NGOs:** social mobilisation, training and livelihood projects.

30.3 Public and private role comparison table

Function	Public organizations	Private organizations
Research	ICAR, TNAU	Private R&D in seeds and agri-tech
Extension	KVK, departments	Input dealers, agri-tech apps
Credit	NABARD, cooperatives	Banks, microfinance
Inputs	Public seed farms	Seed, fertiliser, machinery firms
Marketing	Regulated markets, eNAM	Retailers, e-commerce, processors
Welfare	Schemes and subsidies	CSR and NGO projects

30.4 Public-private partnership (PPP) in agriculture

- **Contract farming:** a firm supplies inputs and buys produce at a fixed price.
- **Agri-clinics and agri-business centres:** graduates provide private advisory.
- **Custom hiring centres:** private machinery services.
- **e-commerce linkage:** FPOs selling on private platforms.
- **Joint research:** public institutes with private companies.

PPP combines the public sector's reach and research with the private sector's efficiency and market access.

30.5 Worked example - a public-private support chain for a farmer

Stage	Organization	Support given
Advice	KVK / Uzhavan App	Crop and weather advisory

Stage	Organization	Support given
Inputs	Private dealer + subsidy	Seed and fertiliser
Credit	Cooperative bank / KCC	Crop loan
Insurance	PMFBY	Crop insurance
Machinery	Custom hiring centre	Tractor and harvester
Marketing	FPO + e-NAM	Grading and better price
Processing	Private processor	Value addition

No single organization meets all a farmer's needs. The farmer's livelihood is strongest when public and private support are combined.

Extension and advisory services

Extension is the bridge between research and the farmer's field.

- **Public extension:** state departments, KVKs and agricultural universities.
- **Private extension:** input dealers, agri-tech apps and companies.
- **Collective extension:** FPOs and SHGs sharing knowledge.
- **Digital extension:** mobile apps, call centres and video advisories.

Approaches: individual farm visits, group meetings, demonstrations, farmer field schools and field days. **Qualities of good extension:** it is timely, local, in the farmer's language and linked to input and market support. In Tamil Nadu, KVKs and the Uzhavan app together form the core of the extension system for smallholders.

30.6 Exam focus

Key points

- Public organizations provide research, extension, credit, regulation and welfare.
- Private organizations provide inputs, processing, retail, credit and agri-tech.
- ICAR and TNAU do research; KVKs do extension; NABARD does development finance.
- Private firms add efficiency and market access.
- Public-private partnership combines the strengths of both.
- A farmer benefits most from a combined support chain.

Short-answer questions with model answers

Q1. Name two public organizations that support farmers. Model answer: Krishi Vigyan Kendras (KVKs) for extension and training, and NABARD for development finance (also ICAR, TNAU and state departments).

Q2. What role do private organizations play in agriculture? Model answer: Private organizations supply inputs, buy and process produce, provide credit and agri-tech services, run contract farming and sell produce to consumers, adding efficiency and market access.

Q3. What is a public-private partnership in agriculture? Model answer: It is a collaboration in which public and private organizations work together - for example, contract farming, custom hiring centres or joint research - combining public reach with private efficiency.

Long-answer questions with model answer outlines

Q1. Discuss the role of public and private organizations in promoting farming-based livelihoods. Outline: (i) Introduction; (ii) public organizations and their functions; (iii) private organizations and their functions; (iv) comparison table; (v) public-private partnership; (vi) worked support chain; (vii) conclusion.

Q2. Explain how public and private support together improve farm livelihoods. Outline: (i) Need for multiple actors; (ii) public role - research, extension, credit, welfare; (iii) private role - inputs, market, technology; (iv) PPP examples; (v) worked chain; (vi) conclusion.

CHAPTER

31. Circular Economy and Zero-Waste Farming Systems

31.1 Concept of circular economy

The **circular economy** is an economic model in which resources are kept in use for as long as possible, the maximum value is extracted from them while in use, and products and materials are recovered and regenerated at the end of each use. It contrasts with the **linear economy** of "take, make, use, dispose".

Three principles of the circular economy:

1. **Reduce:** use fewer raw materials and less energy.
2. **Reuse:** use products and resources again instead of discarding them.
3. **Recycle:** recover materials and turn them back into useful inputs.

31.2 Linear versus circular economy

Basis	Linear economy	Circular economy
Flow	Take - make - dispose	Make - use - recover - reuse
Waste	Waste is a problem	Waste is a resource
Resources	Consumed and lost	Kept in use
Cost	Rising with scarcity	Lower through recycling
Environment	Polluting	Regenerative

31.3 Sustainability and waste reduction in farming

Agriculture is naturally suited to the circular economy because farm waste is a resource:

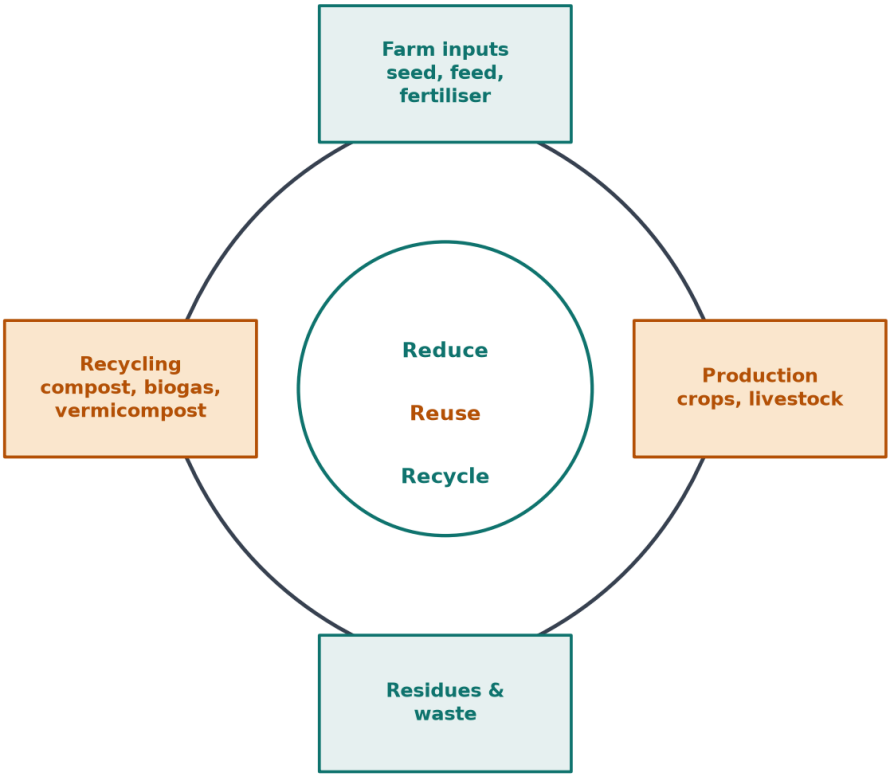
- **Crop residues** feed livestock, are composted, or are used as mulch.
- **Animal dung** becomes manure, biogas and vermicompost.
- **Kitchen and market waste** becomes compost or animal feed.
- **Water** is reused through farm ponds and recycling.

31.4 Zero-waste farming systems

A **zero-waste farming system** aims to use every output of one enterprise as an input to another, so that little is wasted.

Practices:

- **Recycling organic waste:** crop residues, dung and kitchen waste returned to the soil.
- **Composting:** aerobic decomposition of organic waste into manure.
- **Vermicomposting:** using earthworms to convert waste into nutrient-rich compost.
- **Biogas:** converting dung and waste into cooking gas and enriched slurry.
- **Integrated waste management:** separating, treating and reusing different waste streams.
- **Mulching:** using residues to conserve moisture and suppress weeds.

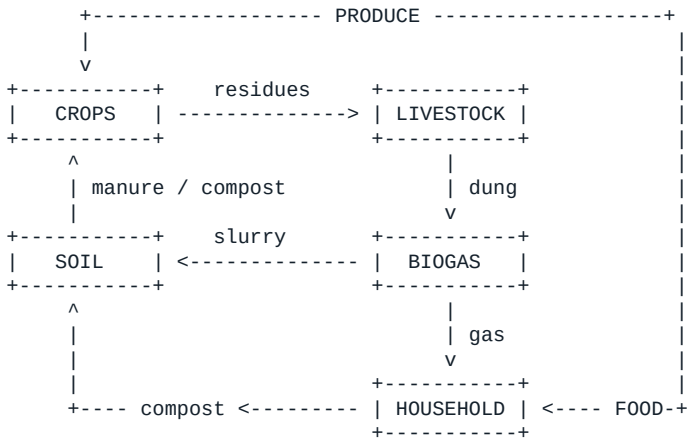


Linear "take–make–dispose" becomes a loop: every output becomes an input, so little is wasted.

Figure 31.1 Circular economy loop

The circular economy loop — farm inputs, production, residues and recycling keep materials in use so that little is wasted.

31.5 Circular economy loop (text diagram)



Every arrow is a reuse. The loop keeps nutrients, energy and money inside the farm.

31.6 Composting methods table

Method	Raw material	Product and use
Farmyard manure	Dung, bedding	Manure for crops
Vermicompost	Dung, crop waste	High-quality compost
NADEP compost	Crop residue, dung	Compost for fields
Biogas slurry	Dung in biogas plant	Enriched manure and gas
Green manure	Legume crops	Nitrogen for the next crop
Mulch	Straw, leaves	Moisture and weed control

31.7 Worked example - savings from a zero-waste system

A farm household with two cows and one hectare recycles waste instead of buying fertiliser.

Item	Before (Rs./year)	After (Rs./year)
Purchased fertiliser	12,000	4,000
Cooking fuel (LPG)	9,000	3,000
Livestock feed (partly home-grown)	60,000	54,000
Total cost	**81,000**	**61,000**
Annual saving	-	**20,000**

By converting dung to biogas and compost, using crop residues as feed and mulch, and returning slurry to the field, the household saves about Rs. 20,000 a year - a substantial gain for a small farm. The saving is a direct result of closing the loop.

Waste-to-wealth enterprises

Recycling farm and rural waste can itself become a business.

- **Vermicompost unit:** sells organic manure to other farmers and nurseries.
- **Biogas plant:** supplies cooking gas and enriched slurry.
- **Coir unit:** converts coconut husk into rope, mats and pith.
- **Mushroom from straw:** turns paddy straw into a high-value food.
- **Briquette and fuel unit:** converts crop residue into fuel.
- **Organic input shop:** sells compost, bio-fertiliser and bio-pesticide.

Benefit: the household earns from what was once waste, cuts its own input cost and reduces pollution. Waste-to-wealth enterprises are well suited to SHGs and FPOs because they need modest capital and use locally available material such as dung, straw and husk.

31.8 Exam focus

Key points

- The circular economy keeps resources in use and treats waste as a resource.
- Its three principles are reduce, reuse and recycle.
- Zero-waste farming uses every output as an input to another enterprise.
- Composting, vermicomposting, biogas and mulching are key practices.
- Circular farming reduces input cost and pollution.
- Agriculture is naturally suited to the circular economy.

Short-answer questions with model answers

Q1. Define circular economy. Model answer: The circular economy is an economic model in which resources are kept in use for as long as possible, maximum value is extracted while in use, and materials are recovered and regenerated, in contrast to the take-make-dispose linear economy.

Q2. What is a zero-waste farming system? Model answer: It is a farming system that uses every output of one enterprise as an input to another, so that little is wasted, through practices such as composting, vermicomposting and biogas.

Q3. Name three composting methods. Model answer: Farmyard manure, vermicompost and NADEP compost (also biogas slurry and green manure).

Long-answer questions with model answer outlines

Q1. Explain the circular economy concept and zero-waste farming systems. Outline: (i) Meaning of circular economy; (ii) linear versus circular; (iii) principles; (iv) zero-waste practices; (v) circular loop diagram; (vi) worked savings example; (vii) conclusion.

Q2. Discuss how organic waste can be recycled in a farming-based livelihood system. Outline: (i) Types of farm waste; (ii) composting methods; (iii) biogas; (iv) mulching; (v) integrated waste management; (vi) economic and environmental benefits; (vii) conclusion.

CHAPTER

32. Green Economy and Sustainable Agriculture

32.1 Definition of green economy

A **green economy** is one that results in improved human well-being and social equity while significantly reducing environmental risks and ecological scarcities. In a green economy, growth in income and employment is driven by public and private investment that reduces carbon emissions and pollution, enhances energy and resource efficiency, and prevents the loss of biodiversity.

In agriculture, a green economy means producing food and income while protecting soil, water, biodiversity and climate.

32.2 Principles of the green economy

- **Low carbon:** reduce greenhouse gas emissions.
- **Resource efficient:** use less to produce more.
- **Socially inclusive:** benefits shared fairly.
- **Biodiverse:** protect ecosystems and species.
- **Sustainable:** meet present needs without harming the future.

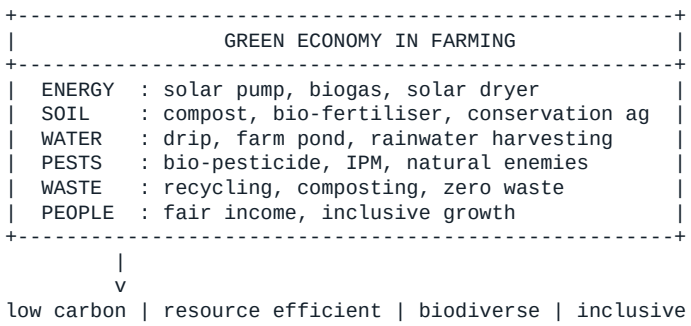
32.3 Sustainable agriculture practices

- **Organic and natural farming:** no synthetic chemicals, on-farm inputs.
- **Integrated nutrient management:** balanced organic and inorganic sources.
- **Integrated pest management:** biological, cultural, mechanical and need-based chemical control.
- **Conservation agriculture:** minimum tillage, residue retention, rotation.
- **Agroforestry:** trees with crops for carbon and soil benefits.
- **Micro-irrigation:** water efficiency.
- **Crop diversification:** resilience and soil health.
- **Renewable energy:** solar pumps, biogas, solar dryers.

32.4 Eco-friendly technologies in farming

Technology	Function	Benefit
Solar pump	Irrigation	Saves diesel, cuts emissions
Biogas plant	Cooking gas and slurry	Saves fuel and fertiliser
Vermicompost	Organic manure	Cuts chemical fertiliser
Drip irrigation	Water delivery	Saves water and labour
Solar dryer	Drying produce	Reduces waste, adds value
Bio-pesticides	Pest control	Safe, low residue
Azolla and blue-green algae	Nitrogen fixation	Free nitrogen for paddy
Bio-fertilisers	Nutrient supply	Reduces fertiliser need

32.5 Green economy in farming (text diagram)



32.6 Worked example - solar pump economics

A farmer replaces a diesel pump with a solar pump.

Item	Diesel pump	Solar pump
Capital cost (Rs.)	40,000	2,00,000 (with subsidy, lower)
Annual running cost (Rs.)	30,000	3,000
Life (years)	10	20
Annual saving (Rs.)	-	27,000

The solar pump costs more initially but saves about Rs. 27,000 a year in fuel, recovering the extra cost in a few years with subsidy, and it cuts carbon emissions. This is the green economy in practice: higher initial investment, lower running cost and lower pollution.

Carbon farming and ecosystem services

Farming can be paid for environmental services, not only for produce.

- **Carbon sequestration:** agroforestry, conservation agriculture and residue retention store carbon in soil and trees.
- **Carbon credits:** farmers may earn from verified emission reductions.
- **Water conservation:** recharge structures provide a service to the community.
- **Biodiversity:** maintaining diverse crops and habitats supports pollinators and natural enemies.
- **Pollination services:** beekeeping supports crop yields across the neighbourhood.

Potential and caution: carbon markets are developing and rules vary; income is uncertain. The reliable benefits today are the on-farm gains - better soil, water retention and yields - that come from the same practices that sequester carbon. Green practices pay first through lower cost and higher yield, and only later, if at all, through carbon payments.

32.7 Exam focus

Key points

- A green economy improves well-being and equity while cutting environmental risk.
- Its principles are low carbon, resource efficiency, inclusion, biodiversity and sustainability.
- Sustainable practices include organic farming, INM, IPM, conservation agriculture and agroforestry.
- Eco-friendly technologies include solar pumps, biogas, drip and bio-fertilisers.
- Green technologies raise initial cost but lower running cost and pollution.
- Agriculture is central to the green economy.

Short-answer questions with model answers

Q1. Define green economy. Model answer: A green economy is one that improves human well-being and social equity while significantly reducing environmental risks and ecological scarcities, with growth driven by investment that reduces pollution and enhances resource efficiency.

Q2. Name three sustainable agriculture practices. Model answer: Organic farming, integrated pest management and conservation agriculture (also INM, agroforestry and micro-irrigation).

Q3. Give two examples of eco-friendly technologies in farming. Model answer: Solar irrigation pumps and biogas plants (also drip irrigation, solar dryers and bio-fertilisers).

Long-answer questions with model answer outlines

Q1. Explain the green economy and sustainable agriculture practices. Outline: (i) Definition of green economy; (ii) principles; (iii) sustainable agriculture practices; (iv) eco-friendly technologies; (v) worked example; (vi) conclusion.

Q2. Discuss the role of eco-friendly technologies in sustainable farming livelihoods. Outline: (i) Meaning of eco-friendly technology; (ii) energy technologies; (iii) soil and water technologies; (iv) pest and waste technologies; (v) economics; (vi) conclusion.

CHAPTER

33. Digitalization and Its Impact on Farming-Based Livelihood Enterprises

33.1 Meaning of digitalization in agriculture

Digitalization is the use of digital technologies - mobile devices, sensors, drones, satellites, data analytics and online platforms - to improve farming decisions, operations and marketing. It is often called **digital agriculture** or **smart farming**.

33.2 Digital tools and technologies

Drones

- Aerial spraying of pesticides and nutrients.
- Crop health monitoring through aerial images.
- Mapping and surveying of fields.
- Faster and more uniform coverage than manual spraying.

Sensors and IoT

- Soil moisture sensors for irrigation scheduling.
- Weather stations for micro-climate data.
- Livestock sensors for health and estrus detection.
- Internet of Things (IoT) links sensors to decision systems.

Data analytics

- Yield prediction and crop planning.
- Precision input recommendation (variable-rate fertiliser).
- Disease and pest forecasting.
- Price trend analysis.

33.3 Digital platforms for market access

- **eNAM:** national online market platform for transparent trading.
- **E-commerce:** selling produce directly to consumers online.
- **Direct-to-consumer (D2C) sales:** farm-to-home delivery and subscription boxes.
- **Mobile apps:** price information, advisory and buying.
- **Digital payments:** UPI and bank transfers for instant, safe payment.

33.4 Financial technologies and digital services

- **eNAM** for market price discovery and online payment.
- **Digital seed systems:** online seed ordering and traceability.
- **Mobile apps:** Uzhavan in Tamil Nadu for advisory, subsidy and weather.
- **Kisan Credit Card and digital lending:** faster credit access.
- **Weather-based insurance:** digital claims using weather data.

33.5 Digital platforms table

Platform / tool	Function	Benefit to farmer
eNAM	Online market trading	Transparent price, wider buyer base
Uzhavan App (TN)	Advisory, subsidy, weather	Timely information
E-commerce	Direct selling	Better price, reach

Platform / tool	Function	Benefit to farmer
Drones	Spraying and monitoring	Saves labour and input
Soil sensors	Irrigation scheduling	Saves water
Digital payments (UPI)	Money transfer	Instant, safe payment
Weather-based insurance	Risk cover	Quick claims

33.6 Digitalization flow (text diagram)

```
SENSE           DECIDE            ACT              SELL
------          ------            ---              ----
soil sensor --> data analytics --> drip irrigation --> e-market
weather app --> advisory       --> timely sowing  --> e-commerce
drone image --> crop diagnosis --> precision spray-> D2C delivery
price app   --> selling plan   --> storage/sale   --> digital payment
     |              |                 |                  |
     +---- data connects every stage from field to market ----+
```

33.7 Impact on livelihood enterprises

- **Higher income:** better prices through e-markets and direct selling.
- **Lower cost:** precision input use and labour saving.
- **Reduced risk:** weather advisories and insurance.
- **Better decisions:** data-driven planning.
- **New enterprises:** agri-tech services, drone services, digital marketing.
- **Inclusion:** mobile access reaches remote villages.

Challenges: digital divide, poor connectivity, low digital literacy, cost of devices, data privacy and the need for local-language content.

33.8 Worked example - gain from digital market access

A farmer sells 100 quintals of paddy through the local trader and then through an e-market platform.

Channel	Price per quintal (Rs.)	Gross return (Rs.)
Local trader	1,900	1,90,000
e-NAM platform	2,080	2,08,000
Gain	**180**	**18,000**

A price gain of Rs. 180 per quintal on 100 quintals adds Rs. 18,000 - about 9.5 per cent more - with no extra yield. Digital market access converts the same produce into higher income.

Data ownership and digital literacy

Digitalization raises new issues beyond technology.

- **Data ownership:** who owns farm data collected by apps and sensors?
- **Privacy:** use of personal and farm data must be transparent and consent-based.
- **Digital literacy:** training in using apps, payments and platforms.
- **Language:** content in Tamil and other local languages improves adoption.
- **Connectivity:** rural internet coverage is essential for equity.
- **Trust:** farmers adopt digital tools when they see clear benefit and fair use of data.

Balanced view: digital tools offer real gains in advisory, market access and finance, but their benefits are uneven if connectivity, literacy and data rules are weak. Inclusive digitalization means affordable devices, local-language content and clear data governance, so that smallholders are not left behind.

33.9 Exam focus

Key points

- Digitalization is the use of digital technologies to improve farming.
- Key tools are drones, sensors, IoT and data analytics.
- Digital platforms include eNAM, e-commerce and D2C sales.
- Fintech includes digital payments, digital lending and weather-based insurance.
- Digitalization raises income, lowers cost and reduces risk.
- Challenges include the digital divide, connectivity and digital literacy.

Short-answer questions with model answers

Q1. What is digital agriculture? Model answer: Digital agriculture is the use of digital technologies such as mobile devices, sensors, drones, data analytics and online platforms to improve farming decisions, operations and marketing.

Q2. How do drones help farmers? Model answer: Drones are used for aerial spraying of pesticides and nutrients and for crop health monitoring and mapping, giving faster, more uniform coverage and saving labour and inputs.

Q3. Name two digital platforms for market access. Model answer: eNAM and e-commerce platforms (also direct-to-consumer apps and digital payment systems).

Long-answer questions with model answer outlines

Q1. Explain digitalization in agriculture and its impact on farming-based livelihood enterprises. Outline: (i) Meaning; (ii) digital tools - drones, sensors, analytics; (iii) market platforms; (iv) fintech; (v) impact on income, cost and risk; (vi) worked example; (vii) challenges; (viii) conclusion.

Q2. Discuss digital platforms and financial technologies for farm market access in India and Tamil Nadu. Outline: (i) Market access problem; (ii) eNAM; (iii) e-commerce and D2C; (iv) Uzhavan and mobile apps; (v) digital payments and insurance; (vi) worked example; (vii) conclusion.

CHAPTER

34. Changing Lifestyles and Their Impact on Farming-Based Livelihood Enterprises

34.1 Changing lifestyles

Indian lifestyles are changing with urbanisation, rising incomes, nuclear families, dual-income households, health awareness and digital connectivity. These changes reshape what people eat, where they buy, and how they spend leisure time - and therefore reshape farm livelihoods.

34.2 Impacts on farming-based livelihood enterprises

- **Demand for processed and convenience food:** ready mixes, millet health foods, cut and packed vegetables.
- **Demand for safe and organic food:** premium for chemical-free produce.
- **Demand for quality and branded products:** grading, packaging and branding.
- **Growth of online buying:** e-commerce and direct-to-consumer sales.
- **Demand for experience:** agri-tourism, farm stays and farm visits.
- **Health and nutrition awareness:** millets, fruits, vegetables and nutraceuticals.
- **Convenience:** home delivery and subscription of fresh produce.

34.3 Diversification of income through tourism

Eco-tourism

- Nature-based tourism that conserves the environment and supports local communities.
- In Tamil Nadu, hill areas such as the Nilgiris and forest fringe areas have eco-tourism potential.
- Farmers can provide guided walks, local food and accommodation.

Agri-tourism

- Visiting a working farm for education, recreation and experience.
- Activities: fruit picking, milking, planting, bullock-cart rides, farm meals.
- Income from entry fees, food, produce sales and farm stays.
- Educates urban consumers and builds a direct market for farm products.

Farm visits

- School and college educational visits.
- Corporate and family day visits.
- Farmer-to-farmer learning visits.

34.4 Lifestyle-driven enterprise table

Lifestyle change	New enterprise opportunity	Example
Health awareness	Millet and health foods	Millet ready-mix unit
Convenience	Cut and packed vegetables	Fresh-cut vegetable unit
Organic preference	Organic farming and delivery	Organic vegetable box
Online buying	E-commerce and D2C	Farm-to-home app
Experience demand	Agri-tourism	Farm stay and farm tour
Branding preference	Grading and packaging	Branded banana or honey

34.5 Agri-tourism income model (worked example)

A farm near a city offers farm visits and a small farm stay.

Income source	Annual amount (Rs.)
Entry fees (5,000 visitors at Rs. 50)	2,50,000
Farm meals and refreshments	1,50,000
Sale of farm produce to visitors	1,00,000
Farm stay (2 rooms, 100 nights at Rs. 1,500)	1,50,000
Total gross income	**6,50,000**
Operating cost	3,00,000
Net income	**3,50,000**

Agri-tourism can add a large income stream to a farm, using the same land, with modest extra investment in facilities, food and marketing. It also sells farm produce directly at retail prices.

34.6 Balancing opportunity and caution

- Agri-tourism needs location advantage (near a city or tourist route), investment in facilities and hospitality skills.
- It must not disturb the main farm operations.
- Diversification into tourism should complement, not replace, crop and livestock income.
- Market research and promotion are essential for success.

Consumer trends and direct marketing

Changing consumer behaviour is reshaping how farmers sell.

- **Freshness and traceability:** consumers want to know where food comes from.
- **Health and safety:** demand for low-residue and organic produce.
- **Convenience:** home delivery, subscription boxes and cut produce.
- **Local and seasonal:** preference for locally grown, seasonal food.
- **Digital buying:** apps and social media as sales channels.

Direct marketing channels: farm-gate sales, farmers' markets, community-supported agriculture, online stores and social-media selling. Direct marketing raises the farmer's share of the consumer rupee, builds customer loyalty and gives feedback on quality. It requires reliable production, packaging, delivery and communication - new skills that farmers and FPOs are increasingly developing.

34.7 Exam focus

Key points

- Changing lifestyles are driven by urbanisation, income, health awareness and digital access.
- They raise demand for processed, safe, branded and convenient food.
- Online buying creates e-commerce and direct-to-consumer opportunities.
- Agri-tourism and eco-tourism add income from farm visits and stays.
- Lifestyle-driven enterprises include millet foods, fresh-cut vegetables and organic delivery.
- Tourism should complement, not replace, core farm income.

Short-answer questions with model answers

Q1. How do changing lifestyles affect farming-based enterprises? Model answer: Changing lifestyles raise demand for processed, safe, branded and convenient food, encourage online buying, and create demand for farm experiences such as agri-tourism, opening new income opportunities for farmers.

Q2. What is agri-tourism? Model answer: Agri-tourism is visiting a working farm for education, recreation and experience, generating income for the farmer from entry fees, food, produce sales and farm stays.

Q3. Name two lifestyle-driven enterprise opportunities. Model answer: A millet health-food unit and fresh-cut and packed vegetables (also organic delivery and agri-tourism).

Long-answer questions with model answer outlines

Q1. Discuss how changing lifestyles influence farming-based livelihood enterprises. Outline: (i) Drivers of lifestyle change; (ii) impact on food demand; (iii) online buying and D2C; (iv) agri-tourism and eco-tourism; (v) lifestyle-driven enterprises table; (vi) worked example; (vii) conclusion.

Q2. Explain income diversification through eco-tourism, agri-tourism and farm visits. Outline: (i) Meaning of diversification through tourism; (ii) eco-tourism; (iii) agri-tourism; (iv) farm visits; (v) income model example; (vi) cautions; (vii) conclusion.

CHAPTER

Glossary of Key Terms

This glossary defines the important terms used across the 34 lectures. Each definition is written in simple exam-oriented language.

A

Agri-clinic: An advisory service set up by an agriculture graduate to provide technical advice to farmers, supported under the ACABC scheme.

Agricultural livelihood system (ALS): The whole set of farm and farm-related activities, resources and relationships through which a farming household earns its living; its components are land, labour, capital and knowledge.

Agro-climatic zone: A region with a distinct combination of climate, soil, rainfall, topography and cropping pattern; India has 15 such zones and Tamil Nadu has 7.

Agroforestry: A land-use system in which woody perennials are deliberately grown on the same land as crops or livestock.

Agro-processing: The conversion of raw farm produce into processed products such as flour, oil, pulp or packaged foods to add value.

Agro-tourism (agri-tourism): Visiting a working farm for education, recreation and experience, generating income for the farmer.

Apiculture: Beekeeping; the rearing of honey bees for honey, beeswax and pollination.

Aquaculture: Fish farming; the rearing of fish in ponds and tanks.

B

BAIF: A non-governmental organisation working on livestock development, watershed management and tribal livelihoods.

Balanced ration: A feed mixture supplying all nutrients in the correct proportion for maintenance and production of livestock.

Benefit-cost ratio (B:C ratio): The ratio of total returns to total cost; a value above 1 indicates profit.

Bio-fertiliser: A preparation containing living micro-organisms that supply nutrients to crops, such as Rhizobium or Azotobacter.

Biogas: A combustible gas produced from the anaerobic digestion of dung and organic waste, used as cooking fuel; the slurry is a rich manure.

C

Capital (in ALS): The physical, financial and working resources a farm household can use to produce income.

Cauvery delta: The rice-growing tract of Tamil Nadu irrigated by the Cauvery, covering Thanjavur, Thiruvarur, Nagapattinam and nearby districts.

Circular economy: An economic model in which resources are kept in use as long as possible and materials are recovered and regenerated, in contrast to the linear take-make-dispose model.

Climate-smart agriculture (CSA): Farming that sustainably increases productivity and income, builds resilience to climate change and reduces greenhouse gas emissions where possible.

Commercial farming: Farming carried out mainly for sale and profit, using modern inputs and technology.

Composite fish culture: The practice of stocking several fish species that feed at different levels of a pond to maximise total yield.

Conservation agriculture: A farming approach based on minimum soil disturbance, retention of crop residues and crop rotation.

Contract farming: An arrangement in which a farmer grows produce for a company that supplies inputs and buys the output at a pre-agreed price.

Cropping intensity: (Gross cropped area / net sown area) x 100; it shows how many crops are grown per year on the same land.

Cropping pattern: The proportion of area under different crops at a point of time.

Cropping system: The pattern, sequence and management of crops on a piece of land over time.

Crop rotation: Growing different crops in a planned sequence on the same land.

D

Dalwai Committee: The Committee on Doubling Farmers' Income (2016) that framed the strategy to raise farm income.

Digital agriculture: The use of digital technologies such as mobile devices, sensors, drones, data analytics and online platforms to improve farming.

Doubling farmers' income: A policy goal to raise real farm income substantially, requiring productivity gains, better prices and lower costs.

Drip irrigation: A micro-irrigation method that delivers water directly to the plant root zone, saving water.

Dryland: Rainfed land with low and uncertain rainfall, suited to millets, pulses, goat rearing and agroforestry.

Duck-cum-fish: An integrated model in which ducks are housed beside a pond, their droppings fertilising the pond for fish.

E

eNAM: The National Agriculture Market, an electronic platform that links wholesale markets across states for transparent trading.

Eco-tourism: Nature-based tourism that conserves the environment and supports local communities.

Enterprise (farm-based): Any income-generating activity carried out on or around the farm.

Extension: The process of transferring agricultural knowledge and technology from research to farmers.

F

Farm typology: The grouping of farms into types that share similar resources, enterprises and constraints.

Farming system: The combination of farm enterprises together with the resources and management practices used to produce food and income.

Farming-based livelihood system: A farming system viewed as a means of living, focusing on food, income, employment and security.

Farmer Producer Organization (FPO): A collective of farmers, usually a producer company or cooperative, that aggregates produce, buys inputs and markets in bulk.

Feasibility: Whether a farming system can be successfully established and sustained in a given agro-climatic zone.

Floriculture: The cultivation of flowers and ornamental plants.

Fodder: Feed for livestock, including green fodder and dry fodder.

G

Garden land: Irrigated upland suited to crops, horticulture, dairy and poultry.

Global GAP: A widely recognised standard for good agricultural practices in food production.

Green economy: An economy that improves human well-being and social equity while reducing environmental risks and ecological scarcities.

Green manure: A crop, usually a legume, grown and ploughed back into the soil to add nutrients.

Gross cropped area: The total area sown with crops in a year, counting each crop season separately.

Groundwater: Water stored below the ground in aquifers, tapped by wells and borewells.

H

Horticulture: The science and art of cultivating fruits, vegetables, flowers, spices and medicinal and aromatic plants.

Hydroponics: Growing plants in a nutrient solution without soil, often in vertical farming.

I

ICAR: The Indian Council of Agricultural Research, the apex body for agricultural research and education in India.

Integrated Farming System (IFS): A combination of two or more enterprises on the same farm so that the waste or by-product of one becomes the input of another.

Integrated nutrient management (INM): The balanced use of organic and inorganic nutrient sources.

Integrated pest management (IPM): The combined use of biological, cultural, mechanical and need-based chemical methods to control pests.

Intercropping: Growing two or more crops in definite rows on the same field at the same time.

K

Kisan Credit Card (KCC): A scheme providing concessional institutional credit for crop and allied needs.

Krishi Vigyan Kendra (KVK): A district-level farm science centre for technology transfer, training and demonstrations.

L

Landholding: The area of land owned or operated by a household; classified as marginal (below 1 ha), small (1-2 ha), semi-medium (2-4 ha), medium (4-10 ha) and large (above 10 ha).

Limiting factor: The scarcest component of a system, which controls its overall performance.

Livelihood: The capabilities, assets and activities required for a means of living.

Livelihood indicator: A measurable variable used to assess the level and sustainability of a household's living.

Linear economy: The take-make-dispose economic model, in contrast to the circular economy.

M

Mixed cropping: Growing two or more crops together in the same field without a definite row arrangement.

Mixed farming: Farming that combines crop production with livestock.

Monocropping: Growing a single crop on the same land year after year.

MSP (Minimum Support Price): The assured price at which the government buys notified crops; recommended by the CACP.

Multiple cropping: Growing more than one crop on the same land in a year.

N

NABARD: The National Bank for Agriculture and Rural Development, the apex development bank for agriculture and rural development.

Net sown area: The area sown with crops once in a year, counted once.

NICRA: The National Innovations in Climate Resilient Agriculture programme of ICAR for climate-resilient farming.

O

Olericulture: The cultivation of vegetables.

Organic farming: Farming without synthetic chemicals, using on-farm organic inputs.

P

PMFBY: Pradhan Mantri Fasal Bima Yojana, the crop insurance scheme against yield loss from natural calamities.

PM-KISAN: A central scheme giving direct income support of Rs. 6,000 per year to eligible farmers.

Pomology: The cultivation of fruit crops.

Post-harvest loss: The loss of produce between harvest and consumption, caused by poor handling, storage and transport.

Poultry: The rearing of domesticated birds such as chicken and ducks for meat and eggs.

Precision farming: The use of GPS, sensors and data to apply inputs precisely where and when needed.

Protected cultivation: Growing crops under polyhouse or shade-net to control temperature, pests and moisture.

R

RKVY: Rashtriya Krishi Vikas Yojana, a central scheme giving development grants to states for agriculture.

RIDF: The Rural Infrastructure Development Fund of NABARD, financing rural roads, irrigation and storage.

Risk: The possibility of loss or reduced returns from uncertain events in farming.

Rural Infrastructure Development Fund: See RIDF.

S

Self-Help Group (SHG): A small group of 10-20 members, usually women, who save regularly, lend to each other and are linked to a bank for credit.

Sericulture: The rearing of silkworms for silk production.

Silviculture: The cultivation of trees; in agroforestry, trees with pasture or crops.

Soil Health Card: A card giving soil-test based fertiliser recommendations to farmers.

Subsistence farming: Farming that produces mainly for the family's own consumption.

Sustainable agriculture: Farming that meets present needs without harming the ability of future generations to meet theirs.

Sustainable Livelihood Framework: A framework linking a household's assets, the vulnerability context, institutions and policies, livelihood strategies and outcomes.

T

Thanjavur: A district in the Cauvery delta of Tamil Nadu, known as the rice bowl of the state.

Transect walk: A walk across a village to observe land use, crops and resources, used in participatory assessment.

TNAU: Tamil Nadu Agricultural University, which conducts research and education in agriculture and develops Tamil Nadu's agro-climatic zoning.

U

Uzhavan App: A Tamil Nadu government mobile app providing farm advisory, subsidy information and weather data.

V

Value chain: The full sequence of activities and actors that take a product from the farm to the final consumer, adding value at each stage.

Value addition: The increase in the value of a product at a stage of the chain, equal to output value minus input value.

Vermicompost: Compost produced by earthworms from organic waste, rich in nutrients.

Vertical farming: Growing crops in stacked layers, often with hydroponics, to save land and water.

Vulnerability context: The shocks, trends and seasonality that affect a household's livelihood.

W

Wetland: Low-lying land with assured water, suited to paddy, fish and duck.

Zero-waste farming: A farming system that uses every output of one enterprise as an input to another, so that little is wasted.

CHAPTER

Abbreviations and Acronyms

Abbreviation	Full form
ACABC	Agri-Clinics and Agri-Business Centres
ALS	Agricultural Livelihood System
BAIF	Bharatiya Agro Industries Foundation (BAIF Development Research Foundation)
B:C	Benefit-Cost ratio
CACP	Commission for Agricultural Costs and Prices
CSA	Climate-Smart Agriculture
D2C	Direct-to-Consumer
eNAM	Electronic National Agriculture Market
FPO	Farmer Producer Organization
GDP	Gross Domestic Product
GVA	Gross Value Added
GSDP	Gross State Domestic Product
ICAR	Indian Council of Agricultural Research
ICT	Information and Communication Technology
IFS	Integrated Farming System
INM	Integrated Nutrient Management
IoT	Internet of Things
IPM	Integrated Pest Management
KCC	Kisan Credit Card
KVK	Krishi Vigyan Kendra
MSP	Minimum Support Price
NABARD	National Bank for Agriculture and Rural Development
NICRA	National Innovations in Climate Resilient Agriculture
NSSO	National Sample Survey Office
PMFBY	Pradhan Mantri Fasal Bima Yojana
PM-KISAN	Pradhan Mantri Kisan Samman Nidhi
PRA	Participatory Rural Appraisal
RIDF	Rural Infrastructure Development Fund
RKVY	Rashtriya Krishi Vikas Yojana
SHG	Self-Help Group
TNAU	Tamil Nadu Agricultural University
UPI	Unified Payments Interface

Mounish Raj · MouRa · Agri AI

CHAPTER

References and Further Reading

The following standard texts, government reports and institutional publications are recommended for deeper study. Students should consult the latest editions and current official publications for updated figures.

Standard textbooks on farming systems and livelihood

1. Panda, S.C. *Farming Systems and Sustainable Agriculture*. Agrobios, Jodhpur.
2. Jayanthi, C. and others. *Integrated Farming Systems*. Tamil Nadu Agricultural University, Coimbatore.
3. Singh, S.S. *Crop Management*. Kalyani Publishers, New Delhi.
4. Balasubramanian, P. and others. *Principles of Crop Production*. Tamil Nadu Agricultural University.
5. Reddy, S.R. *Principles of Crop Production*. Kalyani Publishers.
6. Chambers, R. and Conway, G. *Sustainable Rural Livelihoods: Practical Concepts for the 21st Century*. Institute of Development Studies.
7. Scoones, I. *Sustainable Livelihoods: A Framework for Analysis*. Institute of Development Studies.
8. Ellis, F. *Rural Livelihoods and Diversity in Developing Countries*. Oxford University Press.
9. Ashby, J. and others. *Farming Systems Research*. Various editions.

Government and institutional publications

1. Government of India. *Report of the Committee on Doubling Farmers' Income* (Dalwai Committee). Ministry of Agriculture and Farmers' Welfare.
2. Government of India. *Agricultural Statistics at a Glance*. Directorate of Economics and Statistics.
3. Government of India. *Economic Survey*. Ministry of Finance, various years.
4. NABARD. *Annual Report* and *SHG-Bank Linkage Programme: Status of Microfinance in India*. NABARD, Mumbai.
5. NABARD. *Farmer Producer Organizations - Status and Progress*. NABARD.
6. ICAR. *Vision 2050* and *NICRA* programme documents. Indian Council of Agricultural Research.
7. NITI Aayog. *Doubling Farmers' Income - Rationale, Strategy and Action Plan*.

ICAR e-course material

1. ICAR. *e-Course on Farming Systems and Sustainable Agriculture*.
2. ICAR. *e-Course on Agricultural Economics*.
3. ICAR. *e-Course on Livestock and Poultry Production*.
4. ICAR. *e-Course on Horticulture and Agroforestry*.
5. ICAR. *e-Course on Agricultural Extension*.

TNAU material

1. Tamil Nadu Agricultural University. *Crop Production Guide*. Directorate of Agriculture, Government of Tamil Nadu.
2. Tamil Nadu Agricultural University. *Agro-climatic Zones of Tamil Nadu*.
3. Tamil Nadu Agricultural University. *Integrated Farming System - Research and Extension Bulletins*.
4. Tamil Nadu Agricultural University. *TNAU AgriTech Portal and Uzhavan App* resources.

Reports, journals and online sources

1. Ministry of Agriculture and Farmers' Welfare. Scheme guidelines for PM-KISAN, PMFBY, eNAM and RKVY.
2. Department of Agriculture, Government of Tamil Nadu. *Policy Notes* and state scheme guidelines.
3. *Indian Journal of Agricultural Economics*, *Indian Journal of Agronomy* and *Indian Journal of Extension Education*.

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4 FAO. *Sustainable Livelihoods and Climate-Smart Agriculture* documents.

5 Reserve Bank of India. *Report on Trend and Progress of Banking in India*.

Note on sources: Where an exact figure is uncertain or changes with each survey or budget, the material in this compilation gives the standard range and advises verification against the latest official source. Students should quote the most recent published figure available at the time of the examination and cite the source.

CHAPTER

Appendix A - Quick Revision Tables

This appendix consolidates the most important comparison tables in one place for last-minute revision.

A.1 Comparison of farming systems

Basis	Subsistence	Commercial	Mixed	Organic	Agroforestry
Purpose	Home use	Market profit	Both	Market, premium	Both
Farm size	Small	Medium-large	Small-medium	Varies	Varies
Inputs	Low	High	Moderate	Organic only	Low-moderate
Technology	Traditional	Modern	Mixed	Traditional-modern	Traditional
Main risk	Weather	Price	Moderate	Yield, certification	Slow returns
Example	Rainfed paddy	Sugarcane, poultry	Crop + dairy	Certified vegetables	Coconut + banana

A.2 Livelihood indicators

Dimension	Main indicators	Measure
Economic	Income, employment, productivity, B:C	Rupees, person-days, ratio
Social	Education, health, gender equity, food security	Percentage, months, score
Environmental	Soil carbon, groundwater, biodiversity, recycling	Percentage, level, count

A.3 Components of farming-based livelihood systems

Component	Includes	Livelihood role
Cropping systems	Cereals, pulses, oilseeds, horticulture	Food and cash
Dairy and livestock	Cattle, buffalo, goat, pig, poultry, duck	Daily income, manure
Horticulture	Fruits, vegetables, flowers, spices	High-value income
Agroforestry	Trees with crops and animals	Long-term income, soil
Fisheries	Fish in ponds and tanks	Protein and income
Secondary enterprises	Processing, handicrafts	Value addition, jobs

A.4 IFS models by agro-climatic zone

Feature	Wetland IFS	Garden land IFS	Dryland IFS
Water source	Canal, assured	Borewell	Rainfed, farm pond
Main crop	Paddy	Maize, cotton, turmeric	Millets, pulses
Livestock	Dairy, duck	Dairy, poultry	Goat, backyard poultry
Fisheries	Farm pond fish	Limited	Small pond
Trees	Coconut on bunds	Mango, sapota	Border agroforestry
Key stabiliser	Farm pond	Borewell	Farm pond and goat
Main risk	Flood	Water scarcity	Drought

A.5 Value chain stages

Stage	Activity	Value added
Input supply	Seed, feed, credit, advice	Foundation for production
Production	Crop and livestock	Raw produce
Primary processing	Cleaning, grading, drying	Quality and price
Secondary processing	Milling, pulping, packaging	Higher value product
Logistics	Storage, cold chain, transport	Reduced loss
Marketing	Wholesale, retail, e-commerce	Consumer access
Consumption	Final consumer	-

A.6 Enterprises by size

Size	Examples	Investment	Suitability
Small	Beekeeping, mushroom, sericulture	Low	Marginal, women, landless
Medium	Agro-processing, nursery, dairy	Moderate	Small, semi-medium
Large	Commercial dairy, poultry, food plant	High	Medium, large, companies

A.7 Risks in farming and management

Risk type	Examples	Management
Economic	Price crash, cost rise	Diversification, MSP, contract farming, storage
Environmental	Drought, flood, pest	Insurance, tolerant varieties, irrigation, IPM
Technological	Machine or variety failure	Training, maintenance, proven technology
Social	Labour shortage, conflict	Labour-saving tools, community action
Institutional	Policy change, credit denial	FPO membership, formal credit

A.8 Digital platforms in agriculture

Platform / tool	Function	Benefit
eNAM	Online market trading	Transparent price
Uzhavan App	Advisory and subsidy (TN)	Timely information
E-commerce and D2C	Direct selling	Better price, reach
Drones	Spraying and monitoring	Saves labour
Soil sensors and IoT	Irrigation scheduling	Saves water
Digital payments	Money transfer	Instant, safe

A.9 Central and state schemes

Scheme	Purpose
PM-KISAN	Rs. 6,000 per year income support
PMFBY	Crop insurance
Kisan Credit Card	Concessional credit
PM Krishi Sinchayee Yojana	Irrigation
Soil Health Card	Soil-based fertiliser advice

Scheme	Purpose
e-NAM	Online market
Paramparagat Krishi Vikas Yojana	Organic farming support
10,000 FPOs scheme	Farmer producer organizations
Uzhavan App (TN)	Advisory and subsidy
Millet mission (TN)	Millets for nutrition and income

A.10 Key formulas

Concept	Formula
Cropping intensity	(Gross cropped area / net sown area) x 100
Benefit-cost ratio	Total returns / total cost
Net income	Gross income - total cost
Value added at a stage	Output value - input value
Farmer's share of consumer price	(Farm-gate price / consumer price) x 100
Weighted indicator score	Sum of (indicator score x weight)

CHAPTER

Appendix B - Model Question Papers

B.1 Model Question Paper 1 (Full syllabus)

Time: 3 hours. Maximum marks: 100.

Section A - Answer all (10 x 2 = 20 marks).

1 Define agro-climatic zone and state the number of zones in India.
2 What is MSP and who recommends it?
3 Define livelihood.
4 Name the four major components of an agricultural livelihood system.
5 What is cropping intensity?
6 Define composite fish culture.
7 What is a value chain?
8 What is NICRA?
9 Define green economy.
10 What is agri-tourism?

Section B - Answer any five (5 x 6 = 30 marks).

1 Explain the role of agriculture in India's economy.
2 Describe the types of farming systems with examples.
3 Explain the integration of livestock with crop farming.
4 Discuss the factors affecting integration of enterprises.
5 Explain the circular economy concept in farming.
6 Describe digital platforms for agricultural market access.

Section C - Answer any five (5 x 10 = 50 marks).

1 Discuss the challenges and opportunities for enhancing farmers' income, with a worked example.
2 Explain the sustainable livelihood framework with a diagram.
3 Describe the IFS models for wetland, garden land and dryland with budgets.
4 Explain the types of risk in farming and their management strategies.
5 Discuss NABARD's initiatives in rural livelihoods.
6 Explain value chains and the steps in value chain creation with a margin example.

B.2 Model Question Paper 2 (Full syllabus)

Time: 3 hours. Maximum marks: 100.

Section A - Answer all (10 x 2 = 20 marks).

1 State agriculture's approximate share in India's GVA and employment.
2 What is PM-KISAN?
3 Name the five capitals of the sustainable livelihood framework.
4 Define farm typology.
5 Differentiate intercropping from mixed cropping.
6 Name two integrated fish farming models.
7 What is an FPO?
8 Define climate-smart agriculture.
9 What is a zero-waste farming system?
10 Define digital agriculture.

Section B - Answer any five (5 x 6 = 30 marks).

1 Explain the livelihood indicators with examples.
2 Describe agroforestry and its benefits.
3 Explain the horticulture-based livelihood system.
4 Discuss the role of ICAR and NABARD.
5 Explain the types of small farm-based enterprises.
6 Describe the impact of changing lifestyles on farm enterprises.

Section C - Answer any five (5 x 10 = 50 marks).

1 Explain the status of agriculture in India and Tamil Nadu.
2 Discuss the central and state government schemes for farming-based livelihoods.
3 Explain the factors affecting the integration of enterprises with a worked example.
4 Describe the circular economy and green economy concepts in farming.
5 Explain the role of public and private organizations in agriculture.
6 Discuss the challenges and future prospects of farming-based livelihood systems.

B.3 Answer key - selected short answers

Agriculture's share: about 16-20 per cent of GVA and 42-48 per cent of employment.

PM-KISAN: direct income support of Rs. 6,000 per year in three instalments to eligible farmers.

Five capitals: natural, physical, human, financial and social.

Farm typology: grouping of farms by similar resources, enterprises and constraints.

Intercropping versus mixed cropping: intercropping uses definite rows; mixed cropping does not.

Integrated fish farming models: duck-cum-fish and dairy-cum-fish (also poultry-cum-fish and piggery-cum-fish).

FPO: a collective of farmers, usually a producer company or cooperative, for aggregation and marketing.

Climate-smart agriculture: farming that raises productivity, builds resilience and cuts emissions.

Zero-waste farming: uses every output as an input to another enterprise.

Digital agriculture: use of digital technologies to improve farming decisions and marketing.

CHAPTER

Appendix C - Practice Numericals with Solutions

C.1 Cropping intensity

Problem: A farm has a net sown area of 2 hectares. In one year it grows paddy on 2 ha, a pulse on 1.5 ha and a vegetable on 0.5 ha. Find the cropping intensity.

Solution: Gross cropped area = 2.0 + 1.5 + 0.5 = 4.0 ha. Cropping intensity = (4.0 / 2.0) x 100 = **200 per cent**.

C.2 Benefit-cost ratio of a dairy unit

Problem: A dairy unit has total returns of Rs. 1,58,000 and total cost of Rs. 1,20,000. Find the net income and the B:C ratio.

Solution: Net income = 1,58,000 - 1,20,000 = **Rs. 38,000**. B:C ratio = 1,58,000 / 1,20,000 = **1.32**.

C.3 Benefit-cost ratio of a poultry unit

Problem: A poultry unit has total cost of Rs. 30,000 and total returns of Rs. 43,700. Find the net income and the B:C ratio.

Solution: Net income = 43,700 - 30,000 = **Rs. 13,700**. B:C ratio = 43,700 / 30,000 = **1.46**.

C.4 Value-chain margin and farmer's share

Problem: The farm-gate price of a crop is Rs. 1,200 per quintal and the consumer price is Rs. 2,400 per quintal. Find the farmer's share and the total value added after the farm gate.

Solution: Farmer's share = (1,200 / 2,400) x 100 = **50 per cent**. Value added after the farm gate = 2,400 - 1,200 = **Rs. 1,200 per quintal**.

C.5 Weighted livelihood indicator score

Problem: A household scores net income 4 (weight 0.4) and environment 3 (weight 0.6). Find the weighted score.

Solution: Weighted score = (4 x 0.4) + (3 x 0.6) = 1.6 + 1.8 = **3.4 out of 5**.

C.6 Integrated farm income

Problem: A one-hectare wetland IFS has enterprise net incomes of Rs. 36,000 (paddy), Rs. 38,000 (dairy), Rs. 14,000 (coconut), Rs. 8,000 (vegetables), Rs. 7,000 (duck) and Rs. 5,000 (fish). Find the total net income and the increase over paddy alone.

Solution: Total net income = 36,000 + 38,000 + 14,000 + 8,000 + 7,000 + 5,000 = **Rs. 1,08,000**. Increase over paddy alone = (1,08,000 - 36,000) / 36,000 x 100 = **200 per cent** (using paddy-alone net income of Rs. 36,000 from the same model).

C.7 Digital market gain

Problem: A farmer sells 100 quintals of paddy at Rs. 1,900 per quintal through a trader and at Rs. 2,080 per quintal through e-NAM. Find the gain.

Solution: Gain per quintal = 2,080 - 1,900 = Rs. 180. Total gain = 180 x 100 = **Rs. 18,000**, which is about **9.5 per cent** of the trader-channel return.

C.8 Zero-waste system saving

Problem: Before recycling, a farm spends Rs. 12,000 on fertiliser and Rs. 9,000 on cooking fuel. After recycling, it spends Rs. 4,000 and Rs. 3,000 respectively. Find the annual saving.

Solution: Saving on fertiliser = 12,000 - 4,000 = Rs. 8,000. Saving on fuel = 9,000 - 3,000 = Rs. 6,000. Total annual saving from these two items = **Rs. 14,000** (part of the total Rs. 20,000 saving when feed savings are also included).

C.9 Fish pond B:C ratio

Problem: A one-hectare fish pond has total cost of Rs. 1,60,000 and returns of Rs. 2,60,000. Find the net return and the B:C ratio.

Solution: Net return = 2,60,000 - 1,60,000 = **Rs. 1,00,000**. B:C ratio = 2,60,000 / 1,60,000 = **1.63**.

C.10 Diversification gain

Problem: A farmer earns Rs. 45,000 net from one hectare of paddy. After diversifying to paddy, vegetables and fruit, the net income is Rs. 78,000. Find the percentage gain.

Solution: Gain = (78,000 - 45,000) / 45,000 x 100 = **73.3 per cent**.

CHAPTER

Appendix D - Summary of Schemes and Institutions at a Glance

D.1 Central schemes - quick reference

Scheme	Support type	Main beneficiary
PM-KISAN	Direct income support	All eligible landholding farmers
PMFBY	Crop insurance	Farmers growing notified crops
Kisan Credit Card	Institutional credit	Farmers and allied borrowers
PM Krishi Sinchayee Yojana	Irrigation support	Farmers in irrigation projects
Soil Health Card	Advisory	All farmers
e-NAM	Market platform	Farmers and traders
Paramparagat Krishi Vikas Yojana	Organic support	Organic farmers and groups
Agriculture Infrastructure Fund	Post-harvest finance	FPOs, entrepreneurs
10,000 FPOs scheme	Institutional support	Farmer collectives
National Beekeeping and Honey Mission	Enterprise support	Beekeepers

D.2 Tamil Nadu state schemes - quick reference

Scheme / programme	Support type	Purpose
Uzhavan App	Digital advisory	Information and subsidy access
Micro-irrigation subsidy	Input support	Drip and sprinkler
Farm mechanisation subsidy	Machinery support	Custom hiring, equipment
Millet promotion mission	Crop promotion	Nutrition and income
IFS promotion	System support	Integrated farming adoption
Horticulture schemes	Crop support	Fruits, vegetables, flowers
Collective farming support	Institutional	Group farming and marketing

D.3 Institutions and their roles

Institution	Type	Main role
ICAR	Research council	Technology and research
TNAU	Agricultural university	Research, education, zoning
KVK	Extension centre	Technology transfer, training
NABARD	Development bank	Credit, SHG, FPO, RIDF
SFAC	Central agency	FPO promotion
State Department of Agriculture	Government	Scheme implementation
FPO	Farmer collective	Aggregation and marketing
SHG	Community group	Savings and micro-credit

D.4 Final revision checklist

1 Be able to define every key term in the glossary in one or two lines.

2 Be able to draw the farming systems classification tree, the sustainable livelihood framework and the circular economy loop.

3 Be able to reproduce the comparison tables for farming systems, IFS models, risks and enterprises.

4 Be able to solve numericals on cropping intensity, B:C ratio, value chain margin and weighted scoring.

5 Be able to give at least one Tamil Nadu example for every major topic - Thanjavur and the Cauvery delta, the Uzhavan app, FPOs in banana and mango, KVK demonstrations and TN IFS models.

6 Be able to state the purpose of every major central and state scheme in one line.

7 Be able to explain any concept with a definition, a structure (headings or bullets) and an example.

8 Keep all figures as standard ranges and cite the latest official source where required.

CHAPTER

Appendix E - Quick Reference Tables for Revision

E.1 Agro-climatic zones of India and representative crops

Zone	Representative region	Main crops
Western Himalayan	J&K, Himachal, Uttarakhand hills	Wheat, maize, apple, potato
Eastern Himalayan	North Bengal, Sikkim, NE hills	Rice, maize, tea, horticulture
Lower Gangetic Plains	West Bengal	Rice, jute, potato
Middle Gangetic Plains	Eastern UP, Bihar	Rice, wheat, sugarcane
Upper Gangetic Plains	Western UP, Uttarakhand plains	Wheat, rice, sugarcane
Trans-Gangetic Plains	Punjab, Haryana, Delhi	Wheat, rice, cotton
Eastern Plateau and Hills	Chhattisgarh, Odisha, Jharkhand	Rice, pulses, oilseeds
Central Plateau and Hills	Madhya Pradesh, Rajasthan (east)	Wheat, soybean, pulses
Western Plateau and Hills	Maharashtra, MP (west)	Cotton, sorghum, sugarcane
Southern Plateau and Hills	Karnataka, AP (interior), TN (upland)	Millets, pulses, oilseeds
East Coast Plains and Hills	Coastal AP, Odisha, TN coast	Rice, groundnut, coconut
West Coast Plains and Ghats	Kerala, coastal Karnataka	Rice, coconut, spices
Gujarat Plains and Hills	Gujarat	Groundnut, cotton, wheat
Western Dry Region	Rajasthan (west)	Bajra, pulses, livestock
Islands	Andaman and Nicobar, Lakshadweep	Rice, coconut, spices

E.2 Agro-climatic zones of Tamil Nadu

Zone	Representative districts	Main crops
North Eastern	Vellore, Tiruvannamalai, Kanchipuram	Rice, groundnut, sugarcane
North Western	Dharmapuri, Krishnagiri, Salem	Millets, cotton, mango
Western	Coimbatore, Erode, Tiruppur	Cotton, sugarcane, turmeric, coconut
Cauvery Delta	Thanjavur, Thiruvarur, Nagapattinam	Rice, pulses, sugarcane
Southern	Madurai, Virudhunagar, Sivaganga	Millets, cotton, pulses, banana
High Rainfall	Kanyakumari, Tirunelveli (parts)	Rice, coconut, rubber, spices
Hill and High Altitude	Nilgiris, Kodaikanal	Tea, potato, vegetables, flowers

E.3 Tamil Nadu crop calendar (season-wise)

Season	Months	Main crops	Notes
Samba	June-July to Dec-Jan	Rice (delta)	Main delta season
Thaladi	Aug-Sept to Jan-Feb	Rice (delta)	Follows kuruvai
Kuruvai	April-May to Aug	Short-duration rice	Needs early water
Navarai	Dec-Jan to April	Rice (limited area)	Requires irrigation
Summer	Feb-May	Pulses, vegetables	Short-duration crops
Rainfed (Adi)	June-July sowing	Millets, pulses, cotton	Rainfed drylands

E.4 Livestock and poultry breeds

Category	Breeds	Notes
Indigenous cattle	Sahiwal, Gir, Red Sindhi, Kangayam, Umblachery	Hardy, low yield, disease tolerant
Crossbred cattle	HF and Jersey crosses	Higher milk yield, need better feed
Buffalo	Murrah, Surti, Jaffarabadi, Mehsana	High fat milk
Goat	Tellicherry, Jamunapari, Sirohi, Boer	Meat and milk; dryland suited
Sheep	Madras Red, Mecheri, Nellore	Wool and meat; Tamil Nadu
Pig	Large White Yorkshire, Landrace, Ghungroo	Meat; NE India and Kerala
Poultry	Aseel, Kadaknath, Vanaraja, Gramapriya	Backyard and improved types
Duck	Khaki Campbell, indigenous	Wetland and water bodies

E.5 Major soil types and suitable crops

Soil type	Where found	Suitable crops
Alluvial	Indo-Gangetic plains, delta	Rice, wheat, sugarcane, pulses
Black (regur)	Deccan, Maharashtra, MP	Cotton, soybean, sorghum
Red	Tamil Nadu uplands, Karnataka	Millets, pulses, groundnut
Laterite	Kerala, Karnataka coast, hills	Tea, coffee, cashew, rubber
Sandy and arid	Rajasthan, TN coast	Bajra, pulses, groundnut
Peaty and saline	Coastal tracts, deltas	Salt-tolerant rice, coconut

E.6 Horticultural crop reference

Crop group	Examples	Tamil Nadu note
Fruits	Mango, banana, coconut, sapota, guava	Banana and coconut lead
Vegetables	Tomato, brinjal, onion, tapioca	Peri-urban belts
Flowers	Jasmine, marigold, rose, crossandra	Madurai, Dindigul jasmine
Spices	Chilli, turmeric, pepper, cardamom	Erode turmeric
Plantation	Coconut, cashew, tea, coffee	Coconut is widespread
Medicinal	Coleus, senna, aloe, glory lily	Dryland and export crops

E.7 Fisheries reference

Category	Examples	Notes
Indian major carps	Rohu, catla, mrigal	Composite culture
Exotic carps	Grass carp, silver carp, common carp	Different feeding niches
Integrated models	Duck, poultry, dairy, piggery cum fish	Waste feeds the pond
Rice-fish	Fish in paddy fields	Weed and pest control
Brackish water	Prawn, milkfish	Coastal Tamil Nadu

CHAPTER

Appendix F - Case Studies, Exam Tips and Syllabus Map

F.1 Case study summaries

Case	Location / theme	Key lesson
Wetland IFS	Cauvery delta, Thanjavur	Paddy, dairy, duck and pond give higher, steadier income
Dryland IFS	Ramanathapuram, TN	Millets, goat, pond and agroforestry stabilise rainfed income
NICRA demonstration	KVK network across India	Climate-resilient practices cut risk and raise yield
BAIF	Livestock and watershed, western India	Technical support with social mobilisation works
FPO marketing	Theni banana FPO, TN	Aggregation raises prices and market power
SHG enterprise	Women's groups across TN	Savings plus bank linkage fund small enterprises
Agri-tourism	Peri-urban farms	Farm visits and stays add a large income stream

F.2 Common examination mistakes to avoid

- Quoting a single unverified figure instead of a standard range with a source.
- Describing only one enterprise when the question asks about a livelihood system.
- Forgetting to define terms at the start of an answer.
- Writing long paragraphs without headings, bullets or tables.
- Omitting the worked steps and units in numerical answers.
- Giving no Indian or Tamil Nadu example.
- Confusing MSP with market price, and CACP with procurement agencies.
- Mixing up the five capitals or the three dimensions of livelihood indicators.
- Drawing no diagram where one is expected (framework, classification tree, cycle).
- Failing to conclude the answer with a one-line summary.

F.3 Syllabus map - all 34 lectures at a glance

Lecture	Core theme
1	Status of agriculture in India and Tamil Nadu
2	Farmers' income, MSP and policy initiatives
3	Challenges and opportunities for income
4	Livelihood - definition, concept and patterns
5	Livelihood indicators for analysis
6	Agricultural livelihood systems - definition and components
7	Approaches to ALS - traditional, modern, IFS, framework
8	Challenges and future perspectives in ALS
9	Farming systems and farming-based livelihood systems
10	Types of farming systems and farm typology
11	Prevalent farming systems in India
12	Component - cropping systems
13	Component - dairy and small livestock
14	Component - horticulture
15	Component - agroforestry

Lecture	Core theme
16	Component - fish farming and integrated models
17	Mid semester examination and revision
18	Small, medium and large farm-based enterprises
19	Value chains in farming-based livelihood systems
20	Secondary enterprises and their role
21	Challenges and future prospects of FBLS
22	Factors affecting integration of enterprises
23	Feasibility of farming systems across agro-climatic zones
24	Future directions in integrated farming systems
25	Commercial farming-based livelihood models - ICAR and NABARD
26	NABARD's initiatives - SHGs, FPOs, ACABC
27	NICRA, NGO enterprises and RKVY and eNAM
28	Risks in farming and success factors
29	Central and state government schemes
30	Role of public and private organizations
31	Circular economy and zero-waste farming
32	Green economy and sustainable agriculture
33	Digitalization and its impact
34	Changing lifestyles and their impact

F.4 One-page formula and figure card

Formulas

- Cropping intensity = (Gross cropped area / Net sown area) x 100
- Net income = Gross income - Total cost
- Benefit-cost ratio = Total returns / Total cost
- Value added = Output value - Input value
- Farmer's share = (Farm-gate price / Consumer price) x 100
- Weighted score = Sum of (score x weight)

Key standard ranges

- Agriculture's share in India's GVA: 16-20 per cent.
- Agriculture's share in employment: 42-48 per cent.
- Small and marginal holdings: about 86 per cent.
- Average holding size: about 1.0-1.1 hectares.
- Cropping intensity: about 140-145 per cent.
- Agro-climatic zones: India 15, Tamil Nadu 7.
- PM-KISAN support: Rs. 6,000 per year.
- MSP basis: 1.5 times A2+FL for many crops.

Key Tamil Nadu references

- Cauvery delta and Thanjavur - rice bowl, Grand Anicut.
- Uzhavan app - advisory and subsidy.
- FPOs - banana, mango, coconut, millets.
- KVK network - technology demonstrations.

- TNAU - research, education and agro-climatic zoning.
- TN IFS models - wetland, garden land and dryland.

Final note: this compilation follows the AGR 101 syllabus in exact lecture order and gives standard ranges wherever an exact figure is uncertain. Students should verify current figures against the latest official publication and cite the source in the examination.